Module 3: Free Electron Theory

Student Copy — Comprehensive Lecture & Tutorial Guide

PH7302: Condensed Matter Physics / Electronic Structure of Materials

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Abstract

This document constitutes the unified, self-contained student edition for **Module 3: Free Electron Theory and the Origin of Energy Bands** in PH7302 Condensed Matter Physics. It synthesizes classical Drude transport theory, quantum Sommerfeld Fermi gas statistics, density of states across dimensions, the quantum Boltzmann Transport Equation (BTE), electronic and thermal conductivity, the Wiedemann–Franz law, thermoelectric phenomena (Seebeck, Peltier, Thomson, and Mott formula), failure modes of the free electron approximation, Bloch's theorem, nearly free electron band gap formation, and Fermi surface experimental probes (de Haas–van Alphen quantum oscillations). The guide features an upfront assignment problem set of 26 curated problems, 18 step-by-step solved in-text numerical and conceptual examples, targeted student learning boxes, and interactive Python simulation scripts.

How to Use This Student Edition

This student edition has been structured to guide you from foundational physical concepts to advanced mathematical mastery of electronic transport and band formation. To maximize your learning efficiency, consider the following study roadmap:

1. **Work the Upfront Assignment Continuously**: The section immediately following the Table of Contents contains the complete module assignment problem set (26 problems across 5 parts). Each problem provides exact parameters and concise final numerical answers or hints in brackets (`[Ans: ...]`). Attempt each problem as soon as you finish the corresponding lecture chapter. Intermediate derivations are intentionally omitted to encourage active problem-solving.
2. **Study the In-Text Worked Examples**: Embedded within Chapters 1–10 are 18 fully solved worked examples. Each example breaks down the physical setup, algebraic substitutions in SI units, unit conversions, and physical interpretation. Work through these derivations with pen and paper before attempting the assignment exercises.
3. **Engage with the Conceptual Learning Boxes**:
    - **Conceptual Checkpoints**: Probe deeper into physical principles (e.g., screening, quasiparticles, and wave interference).
    - **Study Takeaways**: Highlight overarching physical takeaways to commit to memory.
    - **Problem-Solving Strategies**: Provide systematic calculation recipes for exams and research.
    - **Common Misconceptions & Pitfalls**: Flag algebraic traps, unit inconsistencies, and physical fallacies.
4. **Run the Computational Demonstrations**: Appendix A includes 5 complete Python scripts illustrating key quantum phenomena: Fermi energy calculation, thermal smearing of the Fermi edge, low-temperature specific heat fitting, zone-boundary band gap splitting, and de Haas–van Alphen oscillations. Executing and modifying these scripts will build concrete numerical intuition.

1 Module Overview and Syllabus Alignment

The content of Module 3 spans classical to quantum models of conduction electrons, culminating in the necessity of energy band theory:

Chapter	Core Topic	Syllabus Learning Outcomes & Competencies
Chapter 1	Classical Drude Theory	Equations of motion, drift velocity, DC/AC conductivity, mobility, velocity hierarchy, mean free path paradox, Drude Lorenz number, Hall effect.
Chapter 2	Sommerfeld Free Electron Gas	1D/3D wells, periodic boundary conditions, k-space quantization, Fermi sphere, k_F, p_F, E_F, v_F, T_F, empirical electron density formula and metal data.
Chapter 3	Density of States & Ground State	Phase-space cell quantization, PBC method, 1D/2D/3D comparison, ground state energy $U_0 = \frac{3}{5}NE_F$, Fermi pressure P_0, electronic bulk modulus B.
Chapter 4	Quantum Statistics & Specific Heat	Fermi–Dirac distribution, chemical potential $\mu(T)$, Sommerfeld integral expansion, linear $C_{el} = \gamma T$, Debye crossover $C/T = \gamma + \beta T^2$, Pauli paramagnetism.
Chapter 5	Electrical & Thermal Transport	Boltzmann Transport Equation in RTA, linearized distribution, Fermi surface integration, recovery of σ, Matthiessen's rule, quantum thermal conductivity, Wiedemann–Franz law ($L = \frac{\pi^2}{3}(k_B/e)^2$).
Chapter 6	Thermoelectric Phenomena	Seebeck effect, Peltier effect, Thomson effect, Kelvin relations, microscopic Peltier flux derivation, Mott formula for thermopower $S \approx -\frac{\pi^2 k_B}{2e}(T/T_F)$.
Chapter 7	Model Limitations & Band Origin	Positive Hall coefficients, divalent metals, magnetoresistance, distinction between conductors/insulators, periodic potential necessity, coupled-oscillator band formation.
Chapter 8	Periodic Potential & Bloch Theorem	Translation operators, Bloch's theorem proof, crystal momentum $\hbar\mathbf{k}$, Born–von Karman boundary, extended, reduced, and periodic zone schemes.
Chapter 9	Nearly Free Electron Model	Weak potential perturbation, zone-boundary degeneracy, 2-state secular equation, band gap $\Delta E = 2\|V_{\mathbf{G}}\|$, standing waves $\psi_\pm$ and electrostatic energy splitting.
Chapter 10	Fermi Surfaces & Probes	Fermi surface construction, Landau levels, cyclotron resonance, semiclassical Onsager relation, de Haas–van Alphen oscillations, extremal cross-sections.
Chapter 11	Capstone Problem Solving	Comprehensive multi-concept tutorial problems integrating thermodynamics, transport, band gaps, and quantum oscillations.

Table 1: Structure, syllabus coverage, and learning outcomes for Module 3.

Contents

How to Use This Student Edition 1

1 Module Overview and Syllabus Alignment 2

2 Physical Constants and Useful Identities 6

Assignment: Module 3 — Practice Problems Set 7

3 Classical Drude Theory of Metals 13
- 3.1 Historical Foundations and the Electron Gas Concept . . . 13
- 3.2 Core Assumptions of the Drude Model . . . 13
- 3.3 Microscopic Equation of Motion and Collisional Damping . . . 13
- 3.4 Steady-State Drift Velocity, Mobility, and DC Conductivity . . . 14
- 3.5 AC Electrical Conductivity and Plasma Frequency . . . 15
- 3.6 Hierarchy of Velocities and the Mean Free Path Paradox . . . 15
- 3.7 Classical Thermal Conductivity and the Wiedemann–Franz Law . . . 16
- 3.8 Galvanomagnetic Transport: The Classical Hall Effect . . . 17

4 Quantum Sommerfeld Free Electron Gas 20
- 4.1 Quantum Foundations and the 1D Potential Well . . . 20
- 4.2 The 3D Box and Periodic Boundary Conditions (Born–von Karman) . . . 20
- 4.3 The Fermi Sphere at Absolute Zero . . . 21
- 4.4 Fundamental Fermi Parameters . . . 21
- 4.5 Empirical Electron Density Formula and Reference Metal Data . . . 22

5 Density of States and Ground State Properties 26
- 5.1 Definition and Significance of the Density of States . . . 26
- 5.2 Dual Derivations of the 3D Density of States . . . 26
  - 5.2.1 Method 1: Phase-Space Cell Quantization (Heisenberg Cells) . . . 26
  - 5.2.2 Method 2: Periodic Boundary Conditions in k-Space . . . 27
- 5.3 Dimensionality Comparison: 1D, 2D, and 3D Systems . . . 27
- 5.4 Volumetric Density of States and Values at the Fermi Level . . . 28
- 5.5 Ground-State Internal Energy at Absolute Zero . . . 28
- 5.6 Thermodynamic Degeneracy Pressure and Electronic Bulk Modulus . . . 29

6 Quantum Statistics and Electronic Heat Capacity 32
- 6.1 The Fermi–Dirac Distribution Function . . . 32
- 6.2 Temperature Dependence of the Chemical Potential . . . 32
- 6.3 The Classical Electronic Heat Capacity Failure . . . 33
- 6.4 Qualitative Quantum Resolution via the Pauli Principle . . . 33
- 6.5 Rigorous Sommerfeld Expansion and Electronic Heat Capacity . . . 34
- 6.6 Separation of Electronic and Phonon Heat Capacities . . . 35
- 6.7 Pauli Spin Paramagnetism of the Free Electron Gas . . . 37

7 Electrical and Thermal Transport in Metals 41
- 7.1 Semiclassical Dynamics of Conduction Electrons . . . 41
- 7.2 The Boltzmann Transport Equation in the Relaxation Time Approximation . . . 41
- 7.3 Linearization of the Distribution Function . . . 41
- 7.4 Rigorous Evaluation of Current Density and Conductivity Recovery . . . 42
- 7.5 Microscopic Scattering Mechanisms and Matthiessen's Rule . . . 43
- 7.6 Quantum Thermal Conductivity and the Wiedemann–Franz Law . . . 44

8 Thermoelectric Phenomena in Metals **47**
8.1 Coupled Transport of Charge and Heat . . . 47
8.2 The Three Fundamental Thermoelectric Effects . . . 47
8.2.1 1. The Seebeck Effect . . . 47
8.2.2 2. The Peltier Effect . . . 47
8.2.3 3. The Thomson Effect . . . 47
8.3 Kelvin (Onsager) Thermodynamic Relations . . . 48
8.4 Microscopic Derivation of the Peltier Heat Flux . . . 48
8.5 The Mott Formula for Thermopower . . . 48

9 Limitations of Free Electron Theory and Transition to Band Theory **53**
9.1 Scorecard of the Free Electron Model . . . 53
9.1.1 Six Major Successes of the Free Electron Model . . . 53
9.1.2 Six Critical Failures Demanding a Periodic Potential . . . 53
9.2 The Physical Necessity of a Periodic Lattice Potential . . . 54
9.3 Intuitive Model of Band Formation: Coupled Oscillators . . . 54

10 Electrons in a Periodic Potential and Bloch's Theorem **57**
10.1 The Periodic Crystal Potential and Translation Operators . . . 57
10.2 Rigorous Proof of Bloch's Theorem . . . 57
10.3 Physical Nature of Crystal Momentum $\hbar\mathbf{k}$ . . . 58
10.4 Born–von Karman Boundary Conditions and State Counting . . . 59
10.5 The Three Zone Schemes . . . 59

11 Nearly Free Electron Model and Energy Gaps **62**
11.1 Perturbation Theory for a Weak Periodic Potential . . . 62
11.2 Matrix Elements and Selection Rules . . . 62
11.3 Non-Degenerate Perturbation Theory and its Breakdown . . . 62
11.4 Degenerate Perturbation Theory and the 2×2 Secular Equation . . . 63
11.5 Energy Band Gap at the Brillouin Zone Boundary . . . 63
11.6 Physical Origin of the Band Gap: Standing Wave Electrostatics . . . 64

12 Fermi Surfaces and Experimental Probes **68**
12.1 Geometry of the Fermi Surface in Brillouin Zones . . . 68
12.2 Semiclassical Electron Dynamics in a Magnetic Field . . . 69
12.3 Landau Level Quantization . . . 69
12.4 The de Haas–van Alphen (dHvA) Effect . . . 70

13 Comprehensive Solved Problems and Tutorial Solutions **74**
13.1 Problem 11.1: Complete Electronic Ground State of Monovalent Noble Metals . . . 74
13.2 Problem 11.2: Quantum Degeneracy Pressure and White Dwarf Collapse . . . 75
13.3 Problem 11.3: Determination of Fermi Energy from Specific Heat . . . 76
13.4 Problem 11.4: Band Gaps in a 1D Superlattice Potential . . . 77
13.5 Problem 11.5: Multi-Orbit Fermi Surface Reconstruction in Divalent Metals . . . 78

A Appendix A: Python Demonstration Scripts **79**
A.1 Fermi Energy Calculator . . . 79
A.2 Fermi–Dirac Distribution Visualization . . . 79
A.3 Electronic and Phonon Specific Heat . . . 80
A.4 Nearly Free Electron Gap Illustration . . . 80
A.5 de Haas–van Alphen Oscillation Model . . . 81

B Appendix B: Master Formula Reference Table **82**

Module 3 Summary: Key Takeaways & Bridge to Band Theory **84**

References and Recommended Reading **85**

2 Physical Constants and Useful Identities

The following standard physical constants and unit conversions are used consistently throughout this module:

Quantity	Symbol	SI Value	Alternate / Practical Value
Planck constant	h	6.6261×10^{-34} J · s	4.1357×10^{-15} eV · s
Reduced Planck constant	$\hbar = h/(2\pi)$	1.0546×10^{-34} J · s	6.5821×10^{-16} eV · s
Electron rest mass	m_e	9.1094×10^{-31} kg	0.5110 MeV/c^2
Elementary charge	e	1.6022×10^{-19} C	—
Boltzmann constant	k_B	1.3807×10^{-23} J/K	8.6173×10^{-5} eV/K
Avogadro constant	N_A	6.0221×10^{23} mol^{-1}	—
Bohr magneton	$\mu_B = e\hbar/(2m_e)$	9.2740×10^{-24} J/T	5.7884×10^{-5} eV/T
Permeability of vacuum	μ_0	$4\pi \times 10^{-7}$ H/m	1.2566×10^{-6} N/A^2
Permittivity of vacuum	ϵ_0	8.8542×10^{-12} F/m	—
Universal gas constant	$R = N_A k_B$	8.3145 J/(mol · K)	—
Sommerfeld Lorenz number	$L = \frac{\pi^2}{3}(k_B/e)^2$	2.4428×10^{-8} W Ω K^{-2}	—

Table 2: Fundamental physical constants adopted in this text.

Useful Numerical Conversions and Benchmarks

- 1 eV $= 1.6022 \times 10^{-19}$ J.
- Thermal energy scale at $T = 300$ K: $k_B T = 0.02585$ eV $\approx \frac{1}{40}$ eV.
- Free electron coefficient: $\frac{\hbar^2}{2m_e} \approx 3.81$ eV · Å^2 $= 6.10 \times 10^{-39}$ J · m^2.
- Wavevector to energy conversion: $E(\text{eV}) \approx 3.81 \times [k(\text{Å}^{-1})]^2$.
- 1 atm $= 1.01325 \times 10^5$ Pa $= 1.01325 \times 10^5$ N/m^2.

Assignment: Module 3 — Free Electron Theory and Band Origin

Instructions for Students: This assignment covers the theoretical derivations, physical concepts, and quantitative problem-solving techniques developed across Module 3. Problems are organized into five thematic parts corresponding to lecture chapters. Final answers and essential hints are provided in brackets (`[Ans: ...]`) to facilitate self-evaluation. Full step-by-step model derivations for representative problem archetypes are provided as in-text worked examples within the respective lecture chapters.

Part I: Classical Drude Theory & Electrical Transport

1. Aluminum Relaxation Time and Drift Kinetics:
Metallic aluminum has an electrical conductivity $\sigma = 3.5 \times 10^7\ \Omega^{-1}\text{m}^{-1}$ at room temperature (300 K) and a conduction electron concentration $n = 1.8 \times 10^{29}\ \text{m}^{-3}$.

(a) Calculate the momentum relaxation time τ using the Drude model.

(b) If an electric field of $E = 0.50$ V/m is applied across an aluminum wire, determine the steady-state electron drift velocity v_d.

(c) Compute the net distance traversed by an electron in 1.0 second of steady drift.

[**Ans:** (a) $\tau \approx 6.9 \times 10^{-15}$ s $= 6.9$ fs; (b) $v_d \approx 6.07 \times 10^{-4}$ m/s $= 0.61$ mm/s; (c) $d \approx 0.61$ mm]

2. Residual Resistivity Ratio (RRR) of Copper:
A high-purity copper specimen exhibits an electrical resistivity of $\rho(300\text{ K}) = 1.70 \times 10^{-8}\ \Omega\cdot\text{m}$ at room temperature, while at liquid helium temperature (4.2 K) its resistivity falls to $\rho(4.2\text{ K}) = 1.00 \times 10^{-10}\ \Omega \cdot \text{m}$.

(a) Calculate the ratio of momentum relaxation times $\tau(4.2\text{ K})/\tau(300\text{ K})$.

(b) Identify the dominant microscopic scattering mechanism at 300 K and at 4.2 K, and calculate the percentage of room-temperature resistance attributable to thermal phonons.

[**Ans:** (a) $\tau(4.2\text{ K})/\tau(300\text{ K}) = 170$; (b) Phonon scattering accounts for 99.4% of room-temperature resistivity; at 4.2 K, static impurity and defect scattering dominates exclusively]

3. Comprehensive Transport Analysis of Metallic Copper:
Consider monovalent metallic copper with mass density $\rho_m = 8.96\ \text{g/cm}^3$, atomic mass $A = 63.55$ g/mol, and measured electrical resistivity $\rho = 1.70 \times 10^{-8}\ \Omega \cdot \text{m}$ at 300 K. Under an applied field $E = 1.0$ V/m:

(a) Determine the conduction electron density n.

(b) Evaluate the Fermi wavevector k_F, Fermi energy E_F, and Fermi velocity v_F.

(c) Calculate the momentum relaxation time τ and the quantum mean free path $l = v_F\tau$.

(d) Calculate the average drift velocity v_d.

(e) Determine the dimensionless ratio v_d/v_F.

[**Ans:** (a) $n \approx 8.49 \times 10^{28}\ \text{m}^{-3}$; (b) $k_F \approx 1.36 \times 10^{10}\ \text{m}^{-1}$, $E_F \approx 7.04$ eV, $v_F \approx 1.57 \times 10^6$ m/s; (c) $\tau \approx 2.46 \times 10^{-14}$ s, $l \approx 387$ Å; (d) $v_d \approx 4.32 \times 10^{-3}$ m/s $= 4.32$ mm/s; (e) $v_d/v_F \approx 2.75 \times 10^{-9}$]

4. Matthiessen's Rule and Low-Temperature Mean Free Path:
Applying Matthiessen's rule $\rho(T) = \rho_0 + \rho_{\text{ph}}(T)$ to the copper sample from Assignment 2:

(a) Determine the residual resistivity ρ_0 and the phonon-induced resistivity $\rho_{\text{ph}}(300\text{ K})$.

(b) Given that the room-temperature mean free path is $l(300\text{ K}) \approx 400$ Å, calculate the electron mean free path at liquid helium temperature $l(4.2\text{ K})$.

[**Ans:** (a) $\rho_0 = 1.00 \times 10^{-10}\ \Omega\cdot\text{m}$, $\rho_{\text{ph}}(300\ \text{K}) = 1.69 \times 10^{-8}\ \Omega\cdot\text{m}$; (b) $l(4.2\ \text{K}) = 170 \times 400\ \text{Å} = 68{,}000\ \text{Å} = 6.8\ \mu\text{m}$]

5. **Defect Scattering in Cold-Worked and Annealed Copper**:
A heavily cold-worked copper wire has an electrical resistivity of $\rho = 2.00 \times 10^{-8}\ \Omega\cdot\text{m}$ at 300 K. After thorough high-temperature annealing (which removes dislocations and internal grain boundaries), its resistivity decreases to $1.70 \times 10^{-8}\ \Omega\cdot\text{m}$. What percentage of the initial room-temperature resistivity was caused by work-induced lattice defects? [**Ans:** Defect percentage $= (\Delta\rho/\rho_{\text{initial}}) \times 100\% = (0.30/2.00) \times 100\% = 15.0\%$]

Part II: Quantum Free Electron Gas & Fermi–Dirac Statistics

6. **Fermi Energy and Characteristic Temperatures of Sodium**:
Metallic sodium crystallizes in a BCC structure with one valence electron per atom, yielding an electron density $n = 2.65 \times 10^{28}\ \text{m}^{-3}$.

   (a) Calculate the Fermi wavevector k_F and Fermi momentum p_F.

   (b) Calculate the Fermi energy E_F in Joules and in electron-volts.

   (c) Determine the Fermi temperature T_F and Fermi velocity v_F.

   [**Ans:** (a) $k_F \approx 9.22 \times 10^9\ \text{m}^{-1}$, $p_F \approx 9.72 \times 10^{-25}\ \text{kg}\cdot\text{m/s}$; (b) $E_F \approx 5.19 \times 10^{-19}\ \text{J} \approx 3.24\ \text{eV}$; (c) $T_F \approx 3.76 \times 10^4\ \text{K}$, $v_F \approx 1.07 \times 10^6\ \text{m/s}$]

7. **Quantum Degeneracy Parameter of Sodium at Room Temperature**:
Using the Fermi temperature of sodium from Assignment 6, compute the degeneracy ratio T/T_F at room temperature ($T = 300$ K). Discuss why the electron gas remains deeply in the extreme quantum regime despite the ambient temperature being high by human standards. [**Ans:** $T/T_F \approx 300/37{,}600 \approx 0.0080 = 0.8\% \ll 1$. Since $k_B T \ll E_F$, thermal energy cannot excite electrons lying deeper than $\approx k_B T$ below the Fermi level; the system is frozen into a degenerate Fermi sea]

8. **Dirac Delta Property of the Fermi–Dirac Derivative**:
Demonstrate mathematically that in the zero-temperature limit:

$$\lim_{T\to 0}\left(-\frac{\partial f_0}{\partial E}\right) = \delta(E - E_F)$$

   where $f_0(E) = [e^{(E-\mu)/k_B T} + 1]^{-1}$. Explain why this property ensures that charge transport at low temperatures is strictly a Fermi-surface phenomenon. [**Hint:** Integrate $-\frac{\partial f_0}{\partial E}$ from $-\infty$ to $+\infty$ and show the integral equals 1 for all $T > 0$. Then show that for any fixed $E \neq \mu$, the derivative vanishes exponentially as $T \to 0$]

9. **Density of States and Conductivity in a 2D Electron Gas**:
Consider a two-dimensional electron gas (2DEG) confined to an area $A = L^2$ in the xy-plane:

   (a) Show that the 2D density of states per unit area (including electron spin) is strictly independent of energy: $g_{2\text{D}}(E) = \frac{m}{\pi\hbar^2} = \text{constant}$.

   (b) Express the 2D areal carrier density $n_{2\text{D}}$ in terms of the Fermi wavevector k_F.

   (c) Verify that the Drude-type conductivity formula $\sigma_{2\text{D}} = \frac{n_{2\text{D}} e^2 \tau}{m}$ holds identically in two dimensions.

   [**Ans:** (a) $g_{2\text{D}}(E) = \frac{dN}{dE} = 2 \times \frac{2\pi k\, dk}{(2\pi/L)^2 dE} = \frac{Am}{\pi\hbar^2}$; (b) $n_{2\text{D}} = \frac{k_F^2}{2\pi}$; (c) $\sigma_{2\text{D}} = \frac{n_{2\text{D}} e^2 \tau}{m}$]

10. Copper Fermi Temperature Benchmark:
Given that the Fermi energy of metallic copper is $E_F \approx 7.00$ eV, verify quantitatively that its Fermi temperature is $T_F \approx 81{,}200$ K. Contrast this with the melting temperature of copper ($T_{\text{melt}} = 1358$ K) and explain why copper vaporizes long before its electron gas could ever behave classically. [**Ans:** $T_F = E_F/k_B = 7.00 \text{ eV}/(8.6173 \times 10^{-5} \text{ eV/K}) \approx 81{,}230 \text{ K} \approx 81{,}200$ K. Since $T_{\text{melt}} \ll T_F$, solid copper is always degenerate]

11. Quantum Degeneracy Pressure in Copper:
Conduction electrons in metallic copper have density $n = 8.5 \times 10^{28}$ m^{-3} and Fermi energy $E_F = 7.0$ eV.

(a) Calculate the electron degeneracy pressure $P_0 = \frac{2}{5} n E_F$ in Pascals (N/m^2).

(b) Express this pressure in atmospheres (1 atm $\approx 1.013 \times 10^5$ Pa).

(c) Compute the electronic contribution to the bulk modulus $B = \frac{5}{3} P_0 = \frac{2}{3} n E_F$.

[**Ans:** (a) $P_0 \approx 3.81 \times 10^{10}$ Pa; (b) $P_0 \approx 3.76 \times 10^5$ atm $\approx 380{,}000$ atm; (c) $B \approx 6.35 \times 10^{10}$ Pa]

12. Chemical Potential Deviation Threshold:
The low-temperature Sommerfeld expansion for the chemical potential of a 3D electron gas is:

$$\mu(T) \approx E_F \left[1 - \frac{\pi^2}{12} \left(\frac{T}{T_F} \right)^2 \right]$$

Determine the temperature (expressed as a fraction of T_F) at which $\mu(T)$ drops by exactly 1.0% below E_F. Evaluate this temperature numerically for copper ($T_F \approx 81{,}200$ K). [**Ans:** $\frac{\pi^2}{12}(T/T_F)^2 = 0.01 \implies T/T_F = \sqrt{0.12/\pi^2} \approx 0.110 \implies T \approx 0.11\, T_F$; for copper, $T \approx 8{,}950$ K]

Part III: Thermal Transport, Wiedemann–Franz Law & Thermoelectricity

13. Mott Formula and Thermopower of Metallic Sodium:
State the Mott formula for electronic thermopower:

$$S = -\frac{\pi^2 k_B^2 T}{3e} \left[\frac{d \ln \sigma(E)}{dE} \right]_{E_F}$$

Assuming a 3D parabolic energy band with energy-independent scattering relaxation time ($\sigma(E) \propto D(E) v^2(E) \propto E^{3/2}$):

(a) Derive the simplified expression $S = -\frac{\pi^2 k_B^2 T}{2e E_F} = -\frac{\pi^2 k_B}{2e} \left(\frac{T}{T_F} \right)$.

(b) Calculate the numerical Seebeck coefficient S for sodium ($E_F = 3.15$ eV) at $T = 300$ K.

(c) Explain the physical significance of the negative algebraic sign.

[**Ans:** (a) $\frac{d \ln \sigma}{dE} = \frac{3}{2E_F} \implies S = -\frac{\pi^2 k_B^2 T}{2e E_F}$; (b) $S \approx -3.49$ μV/K; (c) The negative sign indicates that charge is transported by negative electron carriers diffusing from hot to cold]

14. Extracting Fermi Energy from Calorimetric Heat Capacity:
Calorimetric measurements on a monovalent metal at liquid helium temperatures yield an electronic specific heat coefficient $C_{el} = 2.50 \times 10^{-4} T$ J $\cdot$ mol^{-1} $\cdot$ K^{-1}. Assuming the Sommerfeld free electron relation $\gamma = \frac{\pi^2}{2} \frac{R k_B}{E_F}$:

(a) Determine the Fermi energy E_F in Joules and in electron-volts.

(b) Calculate the Fermi temperature T_F.

[**Ans:** (a) $\gamma = 2.50 \times 10^{-4}$ J/(mol $\cdot$ K^2) $\implies$ $E_F = \frac{\pi^2 R k_B}{2\gamma} \approx 2.27 \times 10^{-18}$ J ≈ 14.1 eV; (b) $T_F \approx 1.64 \times 10^5$ K]

15. Temperature Independence of Pauli Paramagnetism:
Starting from the spin-split density of states $g(E \pm \mu_B B)$ in a uniform magnetic field B:

(a) Show that the zero-temperature Pauli paramagnetic volume susceptibility is $\chi_{\text{Pauli}} = \mu_0 \mu_B^2 g(E_F) = \frac{3\mu_0 n \mu_B^2}{2E_F}$.

(b) Using the Sommerfeld expansion, prove that thermal corrections to χ_{Pauli} enter only at order $(T/T_F)^2$, explaining why Pauli paramagnetism is experimentally temperature-independent up to melting temperatures.

[**Ans:** (a) $\Delta N \approx 2 \times \frac{1}{2} g(E_F) \mu_B B \implies M = \mu_B^2 g(E_F) B \implies \chi = \mu_0 \mu_B^2 g(E_F)$; (b) $\chi(T) = \chi(0)[1 - \frac{\pi^2}{12}(T/T_F)^2]$; since $T/T_F \sim 10^{-2}$, corrections are $< 0.01\%$]

Part IV: Periodic Potential, Bloch's Theorem, Band Gaps & Quantum Oscillations

16. Successes and Classical Failures of Free Electron Theory:
Enumerate three fundamental experimental observations accurately explained by the Sommerfeld free electron theory, and three distinct phenomena that decisively contradict free electron theory and necessitate the introduction of a periodic lattice potential. [**Ans:** Successes: (1) Linear electronic specific heat $C_{el} = \gamma T$; (2) Temperature-independent Pauli paramagnetism; (3) Wiedemann–Franz law and universal Lorenz number. Failures: (1) Inability to distinguish metals, semiconductors, and insulators; (2) Positive Hall coefficients ($R_H > 0$ in Al, Zn, Cd); (3) Anisotropic non-spherical Fermi surfaces and giant magnetoresistance]

17. Origin of Positive Hall Coefficient in Aluminum:
Explain how energy band curvature $\frac{\partial^2 E}{\partial k^2} < 0$ produces negative effective mass states ($m^* < 0$) near the top of an energy band, and describe how hole-like pockets in the second Brillouin zone of aluminum give rise to a measured Hall coefficient of $R_H > 0$. [**Ans:** When a band is nearly full, electron vacancies near the band maximum accelerate in the direction of the applied electric field rather than opposite to it. Semiclassically, these vacant states behave as positive quasiparticles (holes) with charge $+e$ and positive effective mass $|m^*|$, dominating the Hall deflection and yielding $R_H > 0$]

18. Band Formation and Overlap in Metallic Sodium:
Describe how the discrete $3s$ and $3p$ atomic levels of isolated sodium atoms broaden into continuous energy bands as the interatomic spacing is reduced to the equilibrium lattice parameter $a = 4.23$ Å. Explain why the half-filled $3s$ band and its energetic overlap with the unoccupied $3p$ band guarantee metallic conduction. [**Ans:** Overlap of atomic orbitals splits N degenerate levels into a band of N orbital states ($2N$ states with spin). Sodium has 1 valence electron per atom, filling the $3s$ band to exactly 50%. Furthermore, at the metallic lattice spacing, the bottom of the $3p$ band drops below the $3s$ band top, providing continuous unoccupied states adjacent to E_F]

19. Time-Reversal Symmetry and Complex Conjugate Bloch States:
If $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r})$ is an eigenstate of the crystal Hamiltonian $\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})$ with eigenvalue $E(\mathbf{k})$ in a real periodic potential $V(\mathbf{r}) = V^*(\mathbf{r})$:

(a) Prove that $\psi_{\mathbf{k}}^*(\mathbf{r})$ is also a Bloch eigenstate of $\hat{H}$ with identical energy $E(\mathbf{k})$.

(b) Determine the crystal momentum of $\psi_{\mathbf{k}}^*(\mathbf{r})$ and deduce the fundamental dispersion symmetry $E(-\mathbf{k}) = E(\mathbf{k})$.

[**Ans:** (a) Taking the complex conjugate of $\hat{H}\psi = E\psi$ gives $\hat{H}\psi^* = E\psi^*$ since $\hat{H}$ is purely real; (b) $\psi^*_{\mathbf{k}}(\mathbf{r}) = e^{-i\mathbf{k}\cdot\mathbf{r}}u^*_{\mathbf{k}}(\mathbf{r})$ has crystal momentum $-\hbar\mathbf{k}$, proving time-reversal invariance $E(-\mathbf{k}) = E(\mathbf{k})$]

20. State Capacity of the First Brillouin Zone:
A macroscopic crystal contains $N_c = 1.0 \times 10^{23}$ primitive unit cells:

(a) How many distinct orbital wavevector states $\mathbf{k}$ exist within the first Brillouin zone?

(b) What is the maximum number of electrons that can be accommodated within a single non-degenerate energy band including electron spin?

(c) What can be concluded about the electrical conductivity of an insulator having an even number of valence electrons per primitive unit cell?

[**Ans:** (a) Exactly $N_c = 1.0 \times 10^{23}$ orbital states; (b) $2N_c = 2.0 \times 10^{23}$ electrons per band; (c) If each unit cell has an even number of valence electrons, bands can be completely filled, leaving an energy band gap between the highest occupied band (valence band) and lowest empty band (conduction band)]

21. Band Gap for a 1D Sinusoidal Potential:
A one-dimensional crystal potential is given by $V(x) = V_0\cos(2\pi x/a)$.

(a) Express $V(x)$ as a complex Fourier series $V(x) = \sum_G V_G e^{iGx}$ and determine the non-zero Fourier coefficients V_G.

(b) Using the nearly free electron secular equation, calculate the energy band gap opened at the first Brillouin zone boundary $k = \pm\pi/a$.

[**Ans:** (a) $V(x) = \frac{V_0}{2}e^{i(2\pi/a)x} + \frac{V_0}{2}e^{-i(2\pi/a)x} \implies V_{2\pi/a} = V_{-2\pi/a} = V_0/2$; (b) Band gap $\Delta E = 2|V_G| = 2|V_0/2| = |V_0|$]

22. Fermi Surface Geometry of a 2D Square Monovalent Lattice:
Consider a 2D square lattice with lattice constant a and 1 valence electron per unit cell:

(a) Show that the free-electron Fermi wavevector is $k_F = \frac{\sqrt{2\pi}}{a} \approx 0.798\left(\frac{\pi}{a}\right)$.

(b) Does the free-electron Fermi circle intersect the first Brillouin zone boundary? What fraction of the first Brillouin zone area is occupied by the Fermi sea?

[**Ans:** (a) $n_{2D} = 1/a^2$, $k_F = \sqrt{2\pi n_{2D}} = \sqrt{2\pi}/a \approx 0.798(\pi/a)$; (b) Since $k_F = 0.798(\pi/a) < \pi/a$, the Fermi circle lies entirely inside the first BZ and does not touch the boundary. It fills exactly 50% of the first zone area]

23. Experimental Criteria for Observing Quantum Oscillations:
State the two fundamental physical criteria that must be satisfied simultaneously to observe de Haas–van Alphen oscillations in metals. Why are high magnetic fields ($B \geq 10$ T) and cryogenic temperatures ($T \leq 4.2$ K) indispensable? [**Ans:** (1) $k_BT \ll \hbar\omega_c$ (thermal energy must be much smaller than the Landau level spacing to prevent thermal smearing of the oscillatory peaks); (2) $\omega_c\tau \gg 1$ (electrons must execute at least one full cyclotron orbit before undergoing collisional scattering)]

24. Band Gap Magnitude from Negative Fourier Amplitude:
A periodic potential has a Fourier component $V_G = -0.50$ eV for reciprocal lattice vector G. Determine the energy band gap opened at the corresponding Brillouin zone boundary. [**Ans:** $\Delta E = 2|V_G| = 2|-0.50\text{ eV}| = 1.00$ eV]

Part V: Comprehensive Synoptic & Conceptual Exercises

25. Five Conceptual Transport & Fermi Surface Checkpoints:
Provide concise physical explanations (2–3 sentences each) for the following core phenomena:

(a) Why the electronic specific heat of metals at room temperature is suppressed by two orders of magnitude relative to the classical Dulong–Petit equipartition value.

(b) Why electrons in a mathematically ideal, static periodic crystal experience zero electrical resistance.

(c) Why average drift velocity ($v_d \sim 10^{-4}$ m/s) is eight to nine orders of magnitude smaller than Fermi velocity ($v_F \sim 10^6$ m/s).

(d) Why electrical conduction at low temperatures occurs strictly in an active shell near the Fermi surface.

(e) The physical meaning of the relaxation time approximation (RTA) and the physical condition under which it breaks down.

[**Ans:** (a) Pauli exclusion principle forbids 99% of electrons deep in the Fermi sea from absorbing thermal energy; (b) Bloch waves are stationary eigenstates of a periodic potential that propagate without scattering in the absence of broken symmetry; (c) Conduction is an imperceptible net drift superimposed on a frenzy of quantum carriers; (d) Symmetrical states below E_F cannot scatter because adjacent momentum states are completely full; (e) RTA assumes collisions instantly restore equilibrium independently of previous scattering states, breaking down for large-angle inelastic electron-phonon collisions at intermediate temperatures]

26. Derivation of Electrical Conductivity from the Boltzmann Equation:
Starting from the steady-state Boltzmann Transport Equation in the relaxation time approximation:

$$-\frac{e}{\hbar}\mathbf{E}\cdot\nabla_{\mathbf{k}}f = -\frac{f-f_0}{\tau}$$

(a) Linearize the distribution function to obtain $f_1(\mathbf{k}) = e\tau(\mathbf{v}\cdot\mathbf{E})\left(-\frac{\partial f_0}{\partial E}\right)$.

(b) Express the current density $\mathbf{j}$ as an integral over k-space and prove that the equilibrium distribution f_0 carries zero net current.

(c) Transform the integral to iso-energetic coordinates $d^3k = \frac{dS_E dE}{\hbar v}$ and perform the angular and energy integrations to prove $\sigma = \frac{ne^2\tau}{m}$.

[**Ans:** (a) $f \approx f_0 + f_1 \implies \nabla_{\mathbf{k}} f \approx \hbar\mathbf{v}\frac{\partial f_0}{\partial E} \implies f_1 = e\tau(\mathbf{v}\cdot\mathbf{E})(-\partial f_0/\partial E)$; (b) Equilibrium integral vanishes identically by parity (v_x is odd, $f_0(E)$ is even); (c) Using $\langle v_x^2\rangle = \frac{1}{3}v_F^2$, $-\partial f_0/\partial E \to \delta(E-E_F)$, and $S_F = 4\pi k_F^2$ yields $j_x = \frac{e^2\tau E_x}{12\pi^3\hbar}v_F(4\pi k_F^2) = \frac{ne^2\tau}{m}E_x \implies \sigma = \frac{ne^2\tau}{m}$]

3 Classical Drude Theory of Metals

3.1 Historical Foundations and the Electron Gas Concept

Shortly after J. J. Thomson's discovery of the electron in 1897, Paul Drude (1900) proposed the first microscopic physical model of electrical and thermal conduction in metals. Drude applied the newly developed kinetic theory of neutral dilute gases (pioneered by Maxwell and Boltzmann) to the conduction electrons in a solid.

In an isolated neutral metallic atom, such as sodium ($Z = 11$, configuration $1s^2 2s^2 2p^6 3s^1$), the core electrons remain strongly localized in closed shells around the nucleus. When N_c atoms condense into a solid crystal, the outermost valence orbitals overlap extensively. Because the ionization potential of valence electrons in metals is relatively low, these valence electrons detach from their parent atoms and become delocalized over the entire volume V of the specimen. The crystal is thus conceptualized as an array of heavy, immobile positive ion cores immersed in a sea of mobile conduction electrons (often referred to as the *electron gas* or *Fermi sea*).

3.2 Core Assumptions of the Drude Model

The classical Drude model rests upon four fundamental postulates:

1. **Free Electron Approximation**: Between collisions, the interactions of electrons with the positive ion cores are neglected. The crystal potential is assumed to be spatially flat ($V(\mathbf{r}) = 0$) throughout the interior of the metal.

2. **Independent Electron Approximation**: Electron–electron Coulomb repulsions are completely neglected. Each electron moves independently of all other conduction electrons.

3. **Relaxation Time Approximation (Collisions)**: Electrons undergo instantaneous, stochastic scattering events with a characteristic mean collision time (relaxation time) τ. The probability that an electron suffers a collision in an infinitesimal time interval dt is given by dt/τ, independent of its position or velocity.

4. **Thermal Equilibrium Post-Collision**: Collisions are assumed to completely randomize the electron velocity. Immediately after a collision, an electron emerges with a velocity that is completely uncorrelated with its pre-collision velocity, distributed according to the Maxwell–Boltzmann distribution at local temperature T.

Conceptual Checkpoint: Why Does the Independent-Electron Approximation Work?

At first glance, neglecting Coulomb repulsion between electrons appears physically unjustifiable: the average electron density in metals is $n \sim 10^{28}$–10^{29} m^{-3}, corresponding to an average inter-electron separation of only $r_s \sim 2$–3 Å. At this distance, the bare Coulomb repulsion energy $e^2/(4\pi\epsilon_0 r_s) \approx 5$ eV is comparable to the total kinetic energy!
The resolution lies in **dielectric screening**: mobile conduction electrons rapidly rearrange themselves in response to any local charge imbalance, screening the Coulomb potential from a long-range $1/r$ dependence into an exponentially decaying Yukawa-type potential $V(r) \sim \frac{e^{-k_s r}}{r}$ (Thomas–Fermi screening). Furthermore, as shown decades later by Lev Landau in his Fermi Liquid Theory, Pauli exclusion drastically restricts the phase space available for electron–electron scattering, turning strongly interacting electrons into weakly interacting *quasiparticles* with long lifetimes near the Fermi surface.

3.3 Microscopic Equation of Motion and Collisional Damping

Consider an ensemble of n electrons per unit volume subject to a macroscopic, uniform external electric field $\mathbf{E}$. In the absence of collisions, each electron accelerates according to Newton's second law: $m\frac{d\mathbf{v}}{dt} = -e\mathbf{E}$.

Collisions act as an effective dynamical frictional drag force opposing the net motion. In a time interval dt, a fraction dt/τ of the electrons undergoes a collision and has their directional momentum completely randomized to zero mean value, while the remaining fraction $(1 - dt/\tau)$ continues to accelerate freely. The change in the average electron velocity $\langle \mathbf{v} \rangle$ over time dt is:

$$\langle \mathbf{v}(t + dt) \rangle = \left(1 - \frac{dt}{\tau}\right) \left[\langle \mathbf{v}(t) \rangle - \frac{e\mathbf{E}}{m} dt\right] + \frac{dt}{\tau} \cdot \mathbf{0} \tag{1}$$

Expanding to first order in dt and taking the limit $dt \to 0$ yields the fundamental **Drude equation of motion**:

$$\frac{d\langle \mathbf{v} \rangle}{dt} + \frac{\langle \mathbf{v} \rangle}{\tau} = -\frac{e\mathbf{E}}{m} \tag{2}$$

The term $-m\langle \mathbf{v} \rangle/\tau$ represents an effective linear damping force arising from microscopic collisions.

3.4 Steady-State Drift Velocity, Mobility, and DC Conductivity

Under a steady (DC) electric field $\mathbf{E}$, the electron ensemble reaches a dynamical steady state where the acceleration imparted by the field is exactly balanced by collisional momentum loss: $\frac{d\langle \mathbf{v} \rangle}{dt} = 0$. Equation (2) immediately yields the **steady-state drift velocity** $\mathbf{v}_d \equiv \langle \mathbf{v} \rangle$:

$$\mathbf{v}_d = -\frac{e\tau}{m}\mathbf{E} \tag{3}$$

The proportionality constant between drift speed and electric field magnitude is defined as the **electron mobility** μ:

$$\mu \equiv \frac{|\mathbf{v}_d|}{E} = \frac{e\tau}{m} \tag{4}$$

The macroscopic electrical current density $\mathbf{j}$ is the net charge flowing per unit area per unit time:

$$\mathbf{j} = n(-e)\mathbf{v}_d = -ne\left(-\frac{e\tau}{m}\mathbf{E}\right) = \left(\frac{ne^2\tau}{m}\right)\mathbf{E} \tag{5}$$

Comparing this result directly with the macroscopic Ohm's law $\mathbf{j} = \sigma_0 \mathbf{E}$, we obtain the celebrated **Drude DC electrical conductivity**:

$$\boxed{\sigma_0 = \frac{ne^2\tau}{m}} \tag{6}$$

The corresponding electrical resistivity ρ_0 is simply the reciprocal of conductivity:

$$\boxed{\rho_0 = \frac{1}{\sigma_0} = \frac{m}{ne^2\tau}} \tag{7}$$

Common Misconception & Pitfall: Drift Velocity vs. Carrier Speed

A persistent source of confusion for students is conflating the average drift velocity $\mathbf{v}_d$ with the instantaneous speed of individual electrons.

- **Drift velocity** $v_d \sim 10^{-4}$–10^{-3} m/s (fractions of a millimeter per second) is a sluggish directional drift imposed on the electron cloud.
- **Thermal speed** $v_{th} \sim 10^5$ m/s (classical) or **Fermi velocity** $v_F \sim 10^6$ m/s (quantum) represents the lightning-fast, chaotic motion of individual carriers.

An electric current in a wire is not a high-speed river of charge, but a microscopic bias superimposed on an ultra-high-speed isotropic swarm.

3.5 AC Electrical Conductivity and Plasma Frequency

When an alternating (AC) electric field $\mathbf{E}(t) = \mathbf{E}_0 e^{-i\omega t}$ is applied to a metal (e.g., from an electromagnetic wave), we seek steady-state harmonic solutions of the form $\mathbf{v}(t) = \mathbf{v}_0 e^{-i\omega t}$. Substituting into the equation of motion (2):

$$-i\omega \mathbf{v}_0 + \frac{\mathbf{v}_0}{\tau} = -\frac{e\mathbf{E}_0}{m} \implies \mathbf{v}_0 = -\frac{e\tau/m}{1 - i\omega\tau}\mathbf{E}_0 \tag{8}$$

The resulting AC current density $\mathbf{j}(\omega) = \sigma(\omega)\mathbf{E}(\omega)$ defines the **frequency-dependent optical conductivity**:

$$\boxed{\sigma(\omega) = \frac{\sigma_0}{1 - i\omega\tau} = \sigma_0 \left(\frac{1 + i\omega\tau}{1 + \omega^2\tau^2} \right)} \tag{9}$$

At optical frequencies where $\omega\tau \gg 1$ (typically $\tau \sim 10^{-14}$ s, so $\omega \gg 10^{14}$ rad/s), the conductivity becomes purely imaginary: $\sigma(\omega) \approx i\frac{\sigma_0}{\omega\tau} = i\frac{ne^2}{m\omega}$.

Substituting this into Maxwell's curl equation $\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \mu_0\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = -i\omega\mu_0 \left(\epsilon_0 + \frac{i\sigma}{\omega}\right) \mathbf{E}$, we obtain the **complex dielectric function**:

$$\epsilon(\omega) = 1 + \frac{i\sigma(\omega)}{\epsilon_0 \omega} \approx 1 - \frac{ne^2}{\epsilon_0 m\omega^2} = 1 - \frac{\omega_p^2}{\omega^2} \tag{10}$$

where the fundamental **electron plasma frequency** ω_p is defined as:

$$\boxed{\omega_p = \sqrt{\frac{ne^2}{\epsilon_0 m}}} \tag{11}$$

For frequencies below ω_p, $\epsilon(\omega) < 0$, electromagnetic waves cannot propagate into the metal, and light is almost totally reflected (explaining the shiny, lustrous appearance of polished metals). For ultraviolet frequencies $\omega > \omega_p$, $\epsilon(\omega) > 0$, and metals become transparent (*ultraviolet transparency*).

3.6 Hierarchy of Velocities and the Mean Free Path Paradox

In the classical Drude model, the average thermal speed of electrons was evaluated using Maxwell–Boltzmann equipartition:

$$\frac{1}{2}mv_{th}^2 = \frac{3}{2}k_B T \implies v_{th} = \sqrt{\frac{3k_B T}{m}} \tag{12}$$

At room temperature ($T = 300$ K):

$$v_{th} = \sqrt{\frac{3(1.38 \times 10^{-23}\ \text{J/K})(300\ \text{K})}{9.11 \times 10^{-31}\ \text{kg}}} \approx 1.17 \times 10^5\ \text{m/s} \tag{13}$$

Using a typical experimental relaxation time $\tau \approx 2.5 \times 10^{-14}$ s for copper, Drude deduced the classical **mean free path**:

$$l_{\text{classical}} = v_{th}\tau \approx (1.17 \times 10^5\ \text{m/s})(2.5 \times 10^{-14}\ \text{s}) \approx 2.9 \times 10^{-10}\ \text{m} \approx 3\ \text{Å} \tag{14}$$

This distance (3 Å) is precisely of the order of the interatomic lattice spacing in solid copper ($d \approx 2.5$–3 Å). This numerical coincidence led Drude to conclude that electrons simply bounce off adjacent hard-sphere ion cores like billiard balls!

The Catastrophic Breakdown at Low Temperatures: Experimentally, when pure copper is cooled to liquid helium temperature (4.2 K), its electrical conductivity increases by a factor of 10^3 to 10^5 (as seen in Assignment 2). Under classical physics, $v_{th} \propto \sqrt{T}$ decreases to $\sim 10^4$ m/s, requiring τ to increase by 10^5, which would predict a cryogenic mean free path exceeding:

$$l_{\text{cryogenic}} > 10^{-4} \text{ m} = 0.1 \text{ mm} = 10^6 \text{ Å} \tag{15}$$

Classical physics cannot explain how an electron could glide past *millions of solid ion cores* without colliding with a single one! This glaring paradox was resolved only with quantum mechanics: electrons are de Broglie wavepackets that propagate through a perfectly periodic lattice without any resistance whatsoever; scattering occurs strictly when the crystal periodicity is disrupted by thermal lattice vibrations (phonons) or static defects.

3.7 Classical Thermal Conductivity and the Wiedemann–Franz Law

In 1853, Gustav Wiedemann and Rudolf Franz empirically discovered that the ratio of thermal conductivity (K) to electrical conductivity (σ) for a wide range of metals at a fixed temperature is nearly constant, and is directly proportional to absolute temperature:

$$\frac{K}{\sigma} = LT \tag{16}$$

where L is a universal constant known as the **Lorenz number**.

Drude applied classical kinetic gas theory to calculate the electronic thermal conductivity:

$$K = \frac{1}{3} C_V v l \tag{17}$$

Using the classical monatomic gas heat capacity per unit volume $C_V = \frac{3}{2} n k_B$, and substituting $l = v_{th}\tau$ and $\frac{1}{2}mv_{th}^2 = \frac{3}{2}k_B T$:

$$K_{\text{Drude}} = \frac{1}{3}\left(\frac{3}{2}nk_B\right) v_{th}(v_{th}\tau) = \frac{1}{2}nk_B\tau v_{th}^2 = \frac{1}{2}nk_B\tau\left(\frac{3k_BT}{m}\right) = \frac{3}{2}\frac{nk_B^2\tau T}{m} \tag{18}$$

Dividing by the Drude electrical conductivity $\sigma_0 = \frac{ne^2\tau}{m}$:

$$\frac{K}{\sigma T} = \frac{\frac{3}{2}\frac{nk_B^2\tau T}{m}}{\left(\frac{ne^2\tau}{m}\right)T} = \frac{3}{2}\left(\frac{k_B}{e}\right)^2 \equiv L_{\text{Drude}} \tag{19}$$

Evaluating this numerically:

$$L_{\text{Drude}} = \frac{3}{2}\left(\frac{1.3807 \times 10^{-23} \text{ J/K}}{1.6022 \times 10^{-19} \text{ C}}\right)^2 \approx 1.11 \times 10^{-8} \text{ W}\,\Omega\,\text{K}^{-2} \tag{20}$$

Experimental values for typical metals lie in the range $(2.2\text{–}2.5) \times 10^{-8}$ W Ω K^{-2}. Drude's classical calculation was off by only a factor of 2!

This apparent triumph was later revealed to be the result of a **fortuitous cancellation of two massive errors**:

1. Drude *overestimated* the electronic specific heat C_V by a factor of ≈ 100, because classical equipartition wrongly assumed all n electrons absorb thermal energy (in reality, Pauli exclusion restricts thermal excitation to a fraction $\sim T/T_F \approx 10^{-2}$).

2. Drude *underestimated* the mean square velocity by an identical factor of ≈ 100, because he used $v_{th}^2 \propto T$ instead of the true quantum Fermi velocity $v_F^2 \approx 100\, v_{th}^2$.

Because $K \propto C_V v^2$, the factor of 100 in C_V canceled the factor of $1/100$ in v^2! Sommerfeld's quantum theory later resolved both errors simultaneously, yielding the exact universal value $L = \frac{\pi^2}{3}(k_B/e)^2 \approx 2.44 \times 10^{-8}$ W Ω K^{-2}.

3.8 Galvanomagnetic Transport: The Classical Hall Effect

When an electric current flows through a conductor in the presence of a perpendicular magnetic field $\mathbf{B} = B\hat{z}$, the moving electrons experience a transverse magnetic Lorentz force $\mathbf{F}_B = -e(\mathbf{v} \times \mathbf{B})$, deflecting them toward one side of the specimen. This creates a transverse charge accumulation and a resulting transverse electric field $\mathbf{E}_H$ known as the **Hall field**.

In the Drude model, the equation of motion in combined fields $\mathbf{E} = (E_x, E_y, 0)$ and $\mathbf{B} = (0, 0, B)$ is:

$$m\left(\frac{d\mathbf{v}}{dt} + \frac{\mathbf{v}}{\tau}\right) = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{21}$$

In steady state ($d\mathbf{v}/dt = 0$), writing out the Cartesian components:

$$m\frac{v_x}{\tau} = -eE_x - ev_yB \implies v_x = -\frac{e\tau}{m}E_x - \omega_c\tau v_y \tag{22}$$

$$m\frac{v_y}{\tau} = -eE_y + ev_xB \implies v_y = -\frac{e\tau}{m}E_y + \omega_c\tau v_x \tag{23}$$

where $\omega_c = \frac{eB}{m}$ is the **cyclotron frequency**.

In standard Hall geometry, current is restricted to flow strictly along the x-direction ($j_y = 0 \implies v_y = 0$). Setting $v_y = 0$ in equation (23):

$$0 = -\frac{e\tau}{m}E_y + \omega_c\tau v_x \implies E_y = \frac{m}{e}\omega_c v_x = \frac{m}{e}\left(\frac{eB}{m}\right)v_x = v_xB \tag{24}$$

Since $j_x = -nev_x \implies v_x = -\frac{j_x}{ne}$, the transverse Hall field is:

$$E_y = -\frac{1}{ne}j_xB \tag{25}$$

The **Hall coefficient** R_H is defined as the transverse field per unit current density and unit applied magnetic field:

$$\boxed{R_H \equiv \frac{E_y}{j_xB} = -\frac{1}{ne}} \tag{26}$$

Drude's classical result predicts two striking features:

1. R_H is strictly negative (for negative electron charges $q = -e$).
2. R_H depends inversely on carrier concentration n, providing a direct experimental method to measure the conduction electron density of any metallic specimen.

While this prediction is well verified for alkali metals (Na, K) and noble metals (Cu, Ag, Au), it fails spectacularly for metals such as aluminum ($R_H > 0$), zinc ($R_H > 0$), and beryllium ($R_H > 0$), where the measured Hall coefficient is positive! This major limitation provides direct evidence for the quantum band structure of solids, where holes with negative effective masses dominate transport (as explored in Chapters 7 and 8).

Worked Example: Copper Electrical Transport and Velocity Hierarchy

Problem Statement: A pure copper wire at room temperature ($T = 300$ K) carries a current driven by a steady electric field $E = 1.00$ V/m. The measured parameters of copper are:

- Conduction electron concentration: $n = 8.50 \times 10^{28}$ m^{-3}
- Electrical conductivity: $\sigma = 5.96 \times 10^{7}$ Ω^{-1}m^{-1}
- Quantum Fermi velocity: $v_F = 1.57 \times 10^{6}$ m/s

Calculate: (a) the momentum relaxation time τ, (b) the steady-state electron drift velocity v_d, (c) the classical thermal speed v_{th} at 300 K, and (d) the dimensionless ratios v_d/v_{th} and v_d/v_F.

Step-by-Step Solution:

(a) Momentum relaxation time τ: From the Drude conductivity relation $\sigma = \frac{ne^2\tau}{m}$, we solve for τ:

$$\tau = \frac{m\sigma}{ne^2} = \frac{(9.109 \times 10^{-31}\ \text{kg})(5.96 \times 10^{7}\ \Omega^{-1}\text{m}^{-1})}{(8.50 \times 10^{28}\ \text{m}^{-3})(1.6022 \times 10^{-19}\ \text{C})^2}$$
$$= \frac{5.429 \times 10^{-23}}{8.50 \times 10^{28} \times 2.567 \times 10^{-38}} = \frac{5.429 \times 10^{-23}}{2.182 \times 10^{-9}} \approx \mathbf{2.49 \times 10^{-14}}\ \text{s} \quad (\approx 25\ \text{fs})$$

(b) Drift velocity v_d: Using equation (3) under $E = 1.00$ V/m:

$$v_d = \frac{e\tau E}{m} = \frac{(1.6022 \times 10^{-19}\ \text{C})(2.49 \times 10^{-14}\ \text{s})(1.00\ \text{V/m})}{9.109 \times 10^{-31}\ \text{kg}}$$
$$= \frac{3.989 \times 10^{-33}}{9.109 \times 10^{-31}} \approx \mathbf{4.38 \times 10^{-3}}\ \text{m/s} = \mathbf{4.38}\ \text{mm/s}$$

(c) Classical thermal speed v_{th} at 300 K:

$$v_{th} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3(1.3807 \times 10^{-23}\ \text{J/K})(300\ \text{K})}{9.109 \times 10^{-31}\ \text{kg}}}$$
$$= \sqrt{\frac{1.243 \times 10^{-20}}{9.109 \times 10^{-31}}} = \sqrt{1.364 \times 10^{10}} \approx \mathbf{1.17 \times 10^{5}}\ \text{m/s}$$

(d) Comparison of velocity scales:

$$\frac{v_d}{v_{th}} = \frac{4.38 \times 10^{-3}\ \text{m/s}}{1.17 \times 10^{5}\ \text{m/s}} \approx \mathbf{3.74 \times 10^{-8}}$$
$$\frac{v_d}{v_F} = \frac{4.38 \times 10^{-3}\ \text{m/s}}{1.57 \times 10^{6}\ \text{m/s}} \approx \mathbf{2.79 \times 10^{-9}}$$

Physical Discussion: The average drift speed (4.4 mm/s) is nearly nine orders of magnitude smaller than the quantum Fermi velocity (1.57×10^6 m/s). Conduction in copper does not involve a rapid movement of the entire Fermi sea, but an almost infinitesimal directional displacement of quantum states near the Fermi boundary.

Worked Example: Resolution of the Drude vs. Quantum Velocity Paradox

Problem Statement: In classical Drude theory, the DC electrical conductivity is written as $\sigma = \frac{ne^2\tau}{m}$. In quantum Sommerfeld theory (Chapter 5), current is shown to be carried exclusively by electrons near the Fermi surface with speed v_F, yet the resulting formula for σ is mathematically identical: $\sigma = \frac{ne^2\tau}{m}$. Explain physically why the classical carrier concentration n appears in the quantum conductivity formula even though 99% of the electrons are prevented by the Pauli principle from carrying net transport.

Step-by-Step Resolution:

1. **Classical Perspective**: Every single one of the n conduction electrons is assumed to drift

with average speed $v_d = \frac{e\tau E}{m}$. The total current density is:

$$j_{\text{classical}} = nev_d = ne\left(\frac{e\tau E}{m}\right) = \frac{ne^2\tau}{m}E$$

2. **Quantum Perspective**: Electrons occupy states inside the Fermi sphere up to energy E_F. Electrons deep below E_F cannot be accelerated into forward momentum states because all adjacent states are completely full (Pauli exclusion principle). Only electrons in a narrow surface shell of thickness $\Delta E \approx k_B T$ around E_F can undergo transitions.

   The active fraction of carriers is approximately $f_{\text{active}} \approx \frac{k_B T}{E_F} = \frac{T}{T_F} \sim 10^{-2}$.

3. **The Quantum Velocity Enhancement**: However, unlike classical electrons crawling at drift speed v_d, the uncompensated electrons at the Fermi surface move with the immense Fermi velocity $v_F = \frac{\hbar k_F}{m} \approx 10^6$ m/s. When an electric field E is applied, the entire Fermi sphere shifts in k-space by $\Delta k = \frac{eE\tau}{\hbar}$.

   The volume of uncompensated momentum states produced at the Fermi surface when integrated over the spherical boundary yields an effective charge displacement proportional to $k_F^3 \propto n$. As derived rigorously via the Boltzmann Transport Equation in Chapter 5:

$$j_{\text{quantum}} = \frac{e^2\tau E}{12\pi^3\hbar} v_F (4\pi k_F^2) = \frac{e^2\tau k_F^3}{3\pi^2 m}E$$

   Using $k_F^3 = 3\pi^2 n$, this simplifies exactly to:

$$j_{\text{quantum}} = \left(\frac{ne^2\tau}{m}\right)E \implies \sigma = \frac{ne^2\tau}{m}$$

4. **Conclusion**: The classical Drude formula remains valid in quantum mechanics because the suppression factor (T/T_F) in active carrier density is exactly compensated by the quantum velocity enhancement factor (v_F/v_d) at the Fermi surface!

4 Quantum Sommerfeld Free Electron Gas

4.1 Quantum Foundations and the 1D Potential Well

In 1928, Arnold Sommerfeld revolutionized metallic theory by combining Drude's free-electron concept with quantum wave mechanics and the Pauli exclusion principle. By treating conduction electrons as non-interacting quantum particles obeying the Fermi–Dirac distribution rather than the classical Boltzmann distribution, Sommerfeld resolved the classical specific heat and transport paradoxes without abandoning the free-electron picture.

To understand quantum wavevector quantization, consider first a single electron of mass m_e confined to a one-dimensional box of length L with impenetrable boundaries:

$$V(x) = \begin{cases} 0, & 0 < x < L \\ \infty, & \text{otherwise} \end{cases} \tag{27}$$

Inside the well, the time-independent Schrödinger equation is:

$$-\frac{\hbar^2}{2m_e}\frac{d^2\psi}{dx^2} = E\psi \tag{28}$$

Applying the Dirichlet boundary conditions $\psi(0) = \psi(L) = 0$ yields the familiar standing-wave solutions:

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, 3, \dots \tag{29}$$

with quantized energy eigenvalues:

$$E_n = \frac{\hbar^2 k_n^2}{2m_e} = \frac{\hbar^2\pi^2 n^2}{2m_e L^2}, \quad k_n = \frac{n\pi}{L} \tag{30}$$

4.2 The 3D Box and Periodic Boundary Conditions (Born–von Karman)

In macroscopic crystals, stationary standing waves are inconvenient for modeling traveling electric currents. Instead, we place the electron gas in a three-dimensional cube of volume $V = L^3$ and impose **Born–von Karman periodic boundary conditions**:

$$\psi(x + L, y, z) = \psi(x, y + L, z) = \psi(x, y, z + L) = \psi(x, y, z) \tag{31}$$

Under these boundary conditions, the Schrödinger eigenfunctions are traveling plane waves:

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{V}}e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{\sqrt{V}}e^{i(k_x x + k_y y + k_z z)} \tag{32}$$

with parabolic energy dispersion:

$$E(\mathbf{k}) = \frac{\hbar^2 k^2}{2m_e} = \frac{\hbar^2}{2m_e}(k_x^2 + k_y^2 + k_z^2) \tag{33}$$

Applying the periodic boundary conditions (31) demands:

$$e^{ik_x L} = 1, \quad e^{ik_y L} = 1, \quad e^{ik_z L} = 1 \tag{34}$$

This restricts the allowed wavevectors to a discrete cubic lattice in reciprocal k-space:

$$\mathbf{k} = \left(\frac{2\pi n_x}{L}, \frac{2\pi n_y}{L}, \frac{2\pi n_z}{L}\right), \quad n_x, n_y, n_z \in \{0, \pm 1, \pm 2, \dots\} \tag{35}$$

Each distinct allowed wavevector state occupies a volume in k-space given by:

$$\Delta V_k = \Delta k_x \Delta k_y \Delta k_z = \left(\frac{2\pi}{L}\right)^3 = \frac{8\pi^3}{V} \tag{36}$$

Consequently, the density of allowed orbital wavevector states in k-space is:

$$\rho_k = \frac{1}{\Delta V_k} = \frac{V}{8\pi^3} = \frac{V}{(2\pi)^3} \tag{37}$$

4.3 The Fermi Sphere at Absolute Zero

According to the Pauli exclusion principle, each orbital state labeled by wavevector $\mathbf{k}$ can accommodate at most **two electrons** of opposite spin ($m_s = \pm 1/2$).

At absolute zero ($T = 0$ K), N electrons in the crystal minimize the total energy of the system by filling the lowest available single-particle energy levels up to a maximum kinetic energy E_F, known as the **Fermi energy**. In wavevector space, the occupied states form a sphere of radius k_F, termed the **Fermi sphere**:

$$|\mathbf{k}| \leq k_F \iff E(\mathbf{k}) \leq E_F \tag{38}$$

The total number of electrons N accommodated within the Fermi sphere is obtained by multiplying the sphere volume by the density of states in k-space and including the spin degeneracy factor $g_s = 2$:

$$N = 2 \times \frac{\frac{4}{3}\pi k_F^3}{\left(\frac{2\pi}{L}\right)^3} = 2 \times \frac{\frac{4}{3}\pi k_F^3}{\frac{8\pi^3}{V}} = \frac{V}{3\pi^2} k_F^3 \tag{39}$$

Dividing by crystal volume V gives the fundamental relation between conduction electron concentration $n = N/V$ and the **Fermi wavevector** k_F:

$$n = \frac{k_F^3}{3\pi^2} \iff \boxed{k_F = (3\pi^2 n)^{1/3}} \tag{40}$$

4.4 Fundamental Fermi Parameters

From the Fermi wavevector k_F, all fundamental kinematic and thermal parameters of the degenerate electron gas follow directly:

1. **Fermi Momentum** p_F: The quantum momentum of electrons situated at the Fermi boundary:

$$\boxed{p_F = \hbar k_F = \hbar (3\pi^2 n)^{1/3}} \tag{41}$$

2. **Fermi Energy** E_F: The kinetic energy of the highest occupied state at $T = 0$ K:

$$\boxed{E_F = \frac{\hbar^2 k_F^2}{2m_e} = \frac{\hbar^2}{2m_e}(3\pi^2 n)^{2/3}} \tag{42}$$

3. **Fermi Velocity** v_F: The magnitude of the group velocity of electrons on the Fermi surface:

$$\boxed{v_F = \frac{\hbar k_F}{m_e} = \sqrt{\frac{2E_F}{m_e}}} \tag{43}$$

4. **Fermi Temperature** T_F: The characteristic thermal temperature scale corresponding to E_F:

$$\boxed{T_F = \frac{E_F}{k_B}} \tag{44}$$

Study Takeaway: The Extreme Quantum Nature of Metals at Room Temperature

For typical metals, the Fermi energy is on the order of 2–12 eV. Translating this into temperature units via $T_F = E_F/k_B$ gives:

$$T_F \approx \frac{(5 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{1.381 \times 10^{-23} \text{ J/K}} \sim 50{,}000\text{–}100{,}000 \text{ K!}$$

Because ambient temperatures ($T_{\text{room}} \approx 300$ K) are orders of magnitude smaller than T_F, the

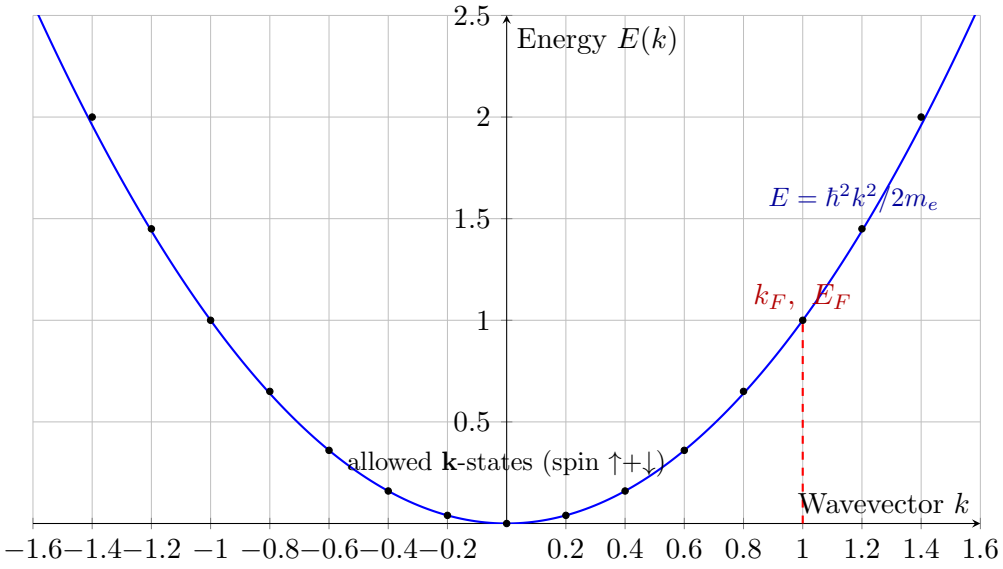


Figure 1: Free-electron dispersion $E(k) = \hbar^2k^2/2m_e$ as a function of one component of **k** (1D cut). Dots: allowed **k**-values from periodic boundary conditions on a length L. Vertical line: Fermi wavevector k_F that bounds the occupied states at $T = 0$.

ratio $T/T_F \sim 10^{-2} \ll 1$.
Consequently, conduction electrons in metals at room temperature are in the **deeply degenerate quantum regime**. Classical Boltzmann statistics is wholly inapplicable; the electron gas is frozen into its quantum ground state with only a minute thermal skin of thickness $\sim k_BT$ excited at the Fermi boundary.

4.5 Empirical Electron Density Formula and Reference Metal Data

For any elemental metal with mass density ρ_m (kg/m^3), molar mass A (kg/mol), and valence Z_v (number of conduction electrons contributed per atom), the conduction electron density n is calculated from Avogadro's number $N_A \approx 6.022 \times 10^{23}$ mol^{-1} as:

$$\boxed{n = \frac{Z_v N_A \rho_m}{A}} \tag{45}$$

Alternatively, for cubic crystals with lattice constant a and N_c atoms per primitive unit cell:

$$n = \frac{Z_v N_c}{a^3} \tag{46}$$

where $N_c = 1$ for simple cubic, $N_c = 2$ for body-centered cubic (BCC), and $N_c = 4$ for face-centered cubic (FCC) unit cells.

Problem-Solving Strategy: Calculating Conduction Electron Density from Crystallographic Data

When given crystallographic data instead of macroscopic density:

1. Identify the Bravais lattice: Simple Cubic (1 atom/cell), BCC (2 atoms/cell), FCC (4 atoms/cell).
2. Determine valence Z_v: Alkali metals (Li, Na, K) have $Z_v = 1$; Noble metals (Cu, Ag, Au) have $Z_v = 1$; Divalent metals (Be, Mg, Zn) have $Z_v = 2$; Trivalent metals (Al, Ga, In) have $Z_v = 3$.

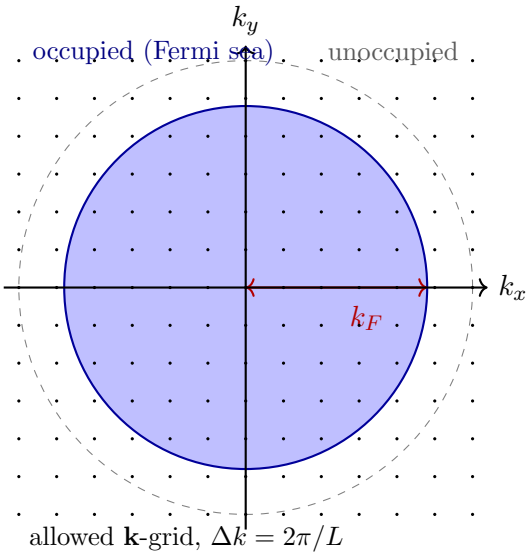


Figure 2: Filled Fermi sphere in **k**-space at $T = 0$. Each **k**-point on the cubic lattice (spacing $2\pi/L$) accommodates two electrons (spin up and down). All states inside the sphere of radius $k_F = (3\pi^2 n)^{1/3}$ are filled; all states outside are empty.

Element	**Valence** Z_v	**Density** n (10^{28} m^{-3})	k_F (10^9 m^{-1})	E_F (eV)	v_F (10^6 m/s)	T_F (10^4 K)
Lithium (Li)	1	4.70	11.2	4.74	1.29	5.51
Sodium (Na)	1	2.65	9.22	3.24	1.07	3.77
Potassium (K)	1	1.40	7.46	2.12	0.86	2.46
Copper (Cu)	1	8.47	13.6	7.00	1.57	8.16
Silver (Ag)	1	5.86	12.0	5.49	1.39	6.38
Gold (Au)	1	5.90	12.1	5.53	1.40	6.42
Aluminum (Al)	3	18.1	17.5	11.7	2.03	13.6

Table 3: Conduction electron density and calculated Sommerfeld Fermi parameters for selected elemental metals.

3. Compute unit cell volume $V_{\text{cell}} = a^3$ in m^3.
4. Compute electron density $n = \frac{Z_v N_{\text{atoms}}}{a^3}$.
5. Use $k_F = (3\pi^2 n)^{1/3}$ and $E_F = \frac{\hbar^2 k_F^2}{2m_e}$ to find all remaining parameters.

Worked Example: Fermi Energy and Electron Density of Sodium from Mass Density

Problem Statement: Metallic sodium has mass density $\rho_m = 970$ kg/m^3 and atomic weight $A = 23.0$ g/mol $= 23.0 \times 10^{-3}$ kg/mol. Sodium is monovalent ($Z_v = 1$). Calculate:

(a) The conduction electron density n.

(b) The Fermi wavevector k_F.

(c) The Fermi energy E_F in Joules and in electron-volts.

(d) The Fermi velocity v_F and Fermi temperature T_F.

Step-by-Step Solution:

(a) Conduction electron concentration n: Using equation (45):

$$n = \frac{Z_v N_A \rho_m}{A} = \frac{(1)(6.022 \times 10^{23}\ \text{mol}^{-1})(970\ \text{kg/m}^3)}{23.0 \times 10^{-3}\ \text{kg/mol}}$$
$$= \frac{5.841 \times 10^{26}}{2.30 \times 10^{-2}} \approx \mathbf{2.54 \times 10^{28}}\ \text{electrons/m}^3$$

(b) Fermi wavevector k_F:

$$k_F = (3\pi^2 n)^{1/3} = \left[3\pi^2 (2.54 \times 10^{28}\ \text{m}^{-3})\right]^{1/3}$$
$$= (7.521 \times 10^{29})^{1/3} \approx \mathbf{9.09 \times 10^{9}}\ \text{m}^{-1} = 0.909\ \text{Å}^{-1}$$

(c) Fermi energy E_F:

$$E_F = \frac{\hbar^2 k_F^2}{2m_e} = \frac{(1.0546 \times 10^{-34}\ \text{J}\cdot\text{s})^2 (9.094 \times 10^{9}\ \text{m}^{-1})^2}{2(9.109 \times 10^{-31}\ \text{kg})}$$
$$= \frac{(1.112 \times 10^{-68})(8.270 \times 10^{19})}{1.822 \times 10^{-30}} = \frac{9.196 \times 10^{-49}}{1.822 \times 10^{-30}} \approx \mathbf{5.05 \times 10^{-19}}\ \text{J}$$

Converting to electron-volts:

$$E_F = \frac{5.05 \times 10^{-19}\ \text{J}}{1.6022 \times 10^{-19}\ \text{J/eV}} \approx \mathbf{3.15}\ \text{eV}$$

(d) Fermi velocity v_F and Fermi temperature T_F:

$$v_F = \frac{\hbar k_F}{m_e} = \frac{(1.0546 \times 10^{-34}\ \text{J}\cdot\text{s})(9.094 \times 10^{9}\ \text{m}^{-1})}{9.109 \times 10^{-31}\ \text{kg}} \approx \mathbf{1.05 \times 10^{6}}\ \text{m/s}$$
$$T_F = \frac{E_F}{k_B} = \frac{5.05 \times 10^{-19}\ \text{J}}{1.3807 \times 10^{-23}\ \text{J/K}} \approx \mathbf{3.66 \times 10^{4}}\ \text{K} \approx 36{,}600\ \text{K}$$

Worked Example: Fermi Sphere Parameters of Silver from FCC Lattice Constant

Problem Statement: Silver (Ag) crystallizes in a face-centered cubic (FCC) lattice with lattice constant $a = 4.09$ Å $= 4.09 \times 10^{-10}$ m. Each silver atom contributes 1 conduction electron ($Z_v = 1$). Calculate:

(a) The conduction electron concentration n.

(b) The Fermi wavevector k_F and Fermi momentum p_F.

(c) The Fermi energy E_F in eV.

(d) The Fermi velocity v_F and Fermi temperature T_F.

Step-by-Step Solution:

(a) Electron concentration n: An FCC unit cell contains 4 atoms ($8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$). The unit cell volume is $V_{\text{cell}} = a^3$:

$$V_{\text{cell}} = (4.09 \times 10^{-10}\ \text{m})^3 \approx 6.842 \times 10^{-29}\ \text{m}^3$$
$$n = \frac{Z_v \times 4}{a^3} = \frac{4}{6.842 \times 10^{-29}\ \text{m}^3} \approx \mathbf{5.85 \times 10^{28}}\ \text{m}^{-\mathbf{3}}$$

(b) Fermi wavevector k_F and momentum p_F:

$$k_F = (3\pi^2 n)^{1/3} = [3\pi^2(5.846 \times 10^{28})]^{1/3} = (1.731 \times 10^{30})^{1/3} \approx \mathbf{1.20 \times 10^{10}}\ \text{m}^{-1} = 1.20\ \text{Å}^{-1}$$

$$p_F = \hbar k_F = (1.0546 \times 10^{-34}\ \text{J}\cdot\text{s})(1.201 \times 10^{10}\ \text{m}^{-1}) \approx \mathbf{1.27 \times 10^{-24}}\ \text{kg}\cdot\text{m/s}$$

(c) Fermi energy E_F:

$$E_F = \frac{\hbar^2 k_F^2}{2m_e} = \frac{(1.0546 \times 10^{-34})^2 (1.201 \times 10^{10})^2}{2(9.109 \times 10^{-31})} = \frac{1.604 \times 10^{-48}}{1.822 \times 10^{-30}} \approx 8.798 \times 10^{-19}\ \text{J}$$

$$E_F = \frac{8.798 \times 10^{-19}\ \text{J}}{1.6022 \times 10^{-19}\ \text{J/eV}} \approx \mathbf{5.49}\ \text{eV}$$

(d) Fermi velocity and temperature:

$$v_F = \frac{\hbar k_F}{m_e} = \frac{1.266 \times 10^{-24}\ \text{kg}\cdot\text{m/s}}{9.109 \times 10^{-31}\ \text{kg}} \approx \mathbf{1.39 \times 10^{6}}\ \text{m/s}$$

$$T_F = \frac{E_F}{k_B} = \frac{8.798 \times 10^{-19}\ \text{J}}{1.3807 \times 10^{-23}\ \text{J/K}} \approx \mathbf{6.37 \times 10^{4}}\ \text{K} \approx 63{,}700\ \text{K}$$

5 Density of States and Ground State Properties

5.1 Definition and Significance of the Density of States

In statistical mechanics and condensed matter physics, the **Density of States** (DOS), denoted $g(E)$, represents the number of available single-particle quantum states per unit energy interval around energy E:

$$g(E) \equiv \frac{dN(E)}{dE} \tag{47}$$

where $N(E)$ is the cumulative number of quantum states with energy less than or equal to E. The density of states per unit volume of the specimen is denoted $D(E) \equiv g(E)/V$.

The DOS is the foundational bridge between microscopic single-particle eigenvalues $E(\mathbf{k})$ and macroscopic thermodynamic properties. Any thermodynamic average of a physical quantity $A(E)$ can be evaluated as an energy integral:

$$\langle A \rangle = \int_0^\infty A(E) g(E) f(E)\, dE \tag{48}$$

where $f(E)$ is the Fermi–Dirac distribution function.

5.2 Dual Derivations of the 3D Density of States

We present two complementary, rigorous derivations of the density of states for a three-dimensional free electron gas.

5.2.1 Method 1: Phase-Space Cell Quantization (Heisenberg Cells)

According to the quantum correspondence principle and Heisenberg's uncertainty principle ($\Delta x \Delta p_x \geq h$), a single quantum state occupies an elementary volume of h^3 in the six-dimensional classical phase space $(\mathbf{r}, \mathbf{p})$.

For a spatial volume $V = \iiint d^3r$, the phase-space volume corresponding to electrons with momentum magnitude between p and $p + dp$ is given by the spherical shell in momentum space:

$$d\Omega_{\text{phase}} = V \times 4\pi p^2 dp \tag{49}$$

Dividing by the elementary quantum volume h^3 and multiplying by the electron spin degeneracy $g_s = 2$ ($m_s = \pm 1/2$), the number of quantum states $g(p)dp$ in the momentum interval $[p, p + dp]$ is:

$$g(p)dp = 2 \times \frac{V \cdot 4\pi p^2 dp}{h^3} = \frac{8\pi V}{h^3} p^2 dp \tag{50}$$

For non-relativistic free electrons, energy and momentum are related by $E = \frac{p^2}{2m_e} \implies p = \sqrt{2m_e E}$, with differential:

$$dp = \frac{m_e}{\sqrt{2m_e E}} dE = \sqrt{\frac{m_e}{2E}} dE \implies p^2 dp = (2m_e E)\sqrt{\frac{m_e}{2E}} dE = m_e \sqrt{2m_e E}\, dE \tag{51}$$

Substituting into equation (50) and using $h = 2\pi\hbar$:

$$g(E)dE = \frac{8\pi V}{(2\pi\hbar)^3} m_e \sqrt{2m_e E}\, dE = \frac{8\pi V}{8\pi^3\hbar^3} \sqrt{2} m_e^{3/2} E^{1/2} dE \tag{52}$$

Simplifying yields the celebrated 3D density of states:

$$\boxed{g(E) = \frac{V}{2\pi^2} \left(\frac{2m_e}{\hbar^2}\right)^{3/2} E^{1/2}} \tag{53}$$

5.2.2 Method 2: Periodic Boundary Conditions in k-Space

In reciprocal space, the total number of orbital states inside a sphere of radius k is the sphere volume divided by the cell volume per state $(2\pi/L)^3 = 8\pi^3/V$. Multiplying by spin degeneracy $g_s = 2$:

$$N(k) = 2 \times \frac{\frac{4}{3}\pi k^3}{\frac{8\pi^3}{V}} = \frac{V}{3\pi^2}k^3 \tag{54}$$

Expressing k in terms of energy $E = \frac{\hbar^2 k^2}{2m_e} \implies k = \left(\frac{2m_e E}{\hbar^2}\right)^{1/2}$:

$$N(E) = \frac{V}{3\pi^2}\left(\frac{2m_e E}{\hbar^2}\right)^{3/2} = \frac{V}{3\pi^2}\left(\frac{2m_e}{\hbar^2}\right)^{3/2} E^{3/2} \tag{55}$$

Differentiating with respect to E:

$$g(E) = \frac{dN(E)}{dE} = \frac{V}{3\pi^2}\left(\frac{2m_e}{\hbar^2}\right)^{3/2}\left(\frac{3}{2}E^{1/2}\right) = \frac{V}{2\pi^2}\left(\frac{2m_e}{\hbar^2}\right)^{3/2} E^{1/2} \tag{56}$$

Both derivations yield identically equation (53).

5.3 Dimensionality Comparison: 1D, 2D, and 3D Systems

The power-law dependence of the electronic density of states on energy depends fundamentally on the spatial dimensionality d of the conductor:

1. **One-Dimensional Conductor (Quantum Wire)**: A wire of length L with k-state spacing $\Delta k = 2\pi/L$. State count up to k: $N(k) = 2 \times \frac{2k}{2\pi/L} = \frac{2Lk}{\pi}$. Using $k = \sqrt{2m_e E}/\hbar$:

$$g_{1D}(E) = \frac{dN}{dE} = \frac{L}{\pi}\sqrt{\frac{2m_e}{\hbar^2}}E^{-1/2} \propto E^{-1/2} \tag{57}$$

   The $E^{-1/2}$ divergence as $E \to 0$ is a classic one-dimensional **Van Hove singularity**.

2. **Two-Dimensional Conductor (Quantum Well / 2DEG)**: A quantum well of area $A = L^2$ with k-space cell area $(2\pi/L)^2 = 4\pi^2/A$. State count inside a circle of radius k: $N(k) = 2 \times \frac{\pi k^2}{4\pi^2/A} = \frac{Ak^2}{2\pi}$. Using $k^2 = \frac{2m_e E}{\hbar^2}$:

$$N(E) = \frac{Am_e E}{\pi\hbar^2} \implies \boxed{g_{2D}(E) = \frac{dN}{dE} = \frac{Am_e}{\pi\hbar^2} = \text{constant}} \tag{58}$$

   The 2D density of states is strictly **energy-independent**!

3. **Three-Dimensional Conductor (Bulk Metal)**:

$$g_{3D}(E) \propto E^{1/2} \tag{59}$$

Conceptual Checkpoint: Why is the 2D Density of States Energy-Independent?

In d spatial dimensions with parabolic dispersion $E \propto k^2$, the k-space volume scales as $k^d \propto E^{d/2}$. The cumulative state count is $N(E) \propto E^{d/2}$, and differentiating gives the density of states:

$$g(E) = \frac{dN}{dE} \propto E^{\frac{d}{2}-1}$$

- For $d = 1$: $g(E) \propto E^{-1/2}$ (decreases with energy).

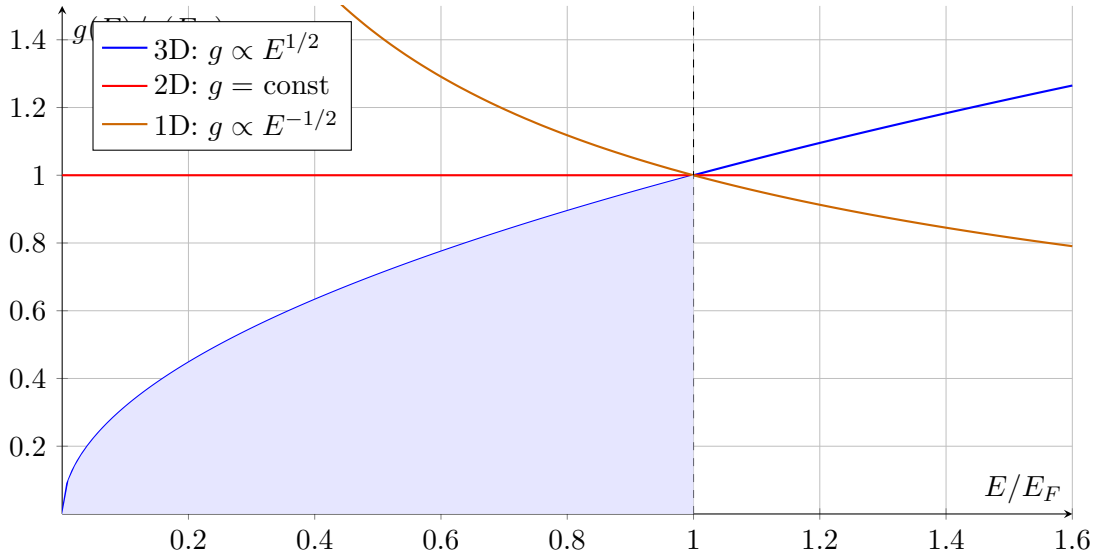


Figure 3: Comparison of the free-electron density of states in 1D, 2D, and 3D, each normalized to $g(E_F)$ at $E = E_F$. The 3D DOS grows as $E^{1/2}$, the 2D DOS is constant, and the 1D DOS diverges as $E^{-1/2}$ at the band edge. The shaded area shows the 3D occupied states at $T = 0$.

- For $d = 2$: $g(E) \propto E^0 =$ constant (flat step-like spectrum).
- For $d = 3$: $g(E) \propto E^{1/2}$ (increases monotonically with energy).

The energy-independent density of states in 2D systems is of enormous practical significance in semiconductor physics: it creates sharp, step-like absorption spectra in GaAs/AlGaAs quantum wells, enabling low-threshold semiconductor quantum-well lasers and high-frequency HEMTs.

5.4 Volumetric Density of States and Values at the Fermi Level

Dividing equation (53) by volume V gives the density of states per unit volume:

$$D(E) \equiv \frac{g(E)}{V} = \frac{1}{2\pi^2}\left(\frac{2m_e}{\hbar^2}\right)^{3/2} E^{1/2} \tag{60}$$

This can be expressed in an exceptionally compact, intuitive form using the Fermi energy $E_F = \frac{\hbar^2}{2m_e}(3\pi^2 n)^{2/3} \implies n = \frac{1}{3\pi^2}\left(\frac{2m_e E_F}{\hbar^2}\right)^{3/2}$:

$$\boxed{D(E) = \frac{3n}{2E_F}\left(\frac{E}{E_F}\right)^{1/2}} \tag{61}$$

Evaluating this directly at the Fermi level $E = E_F$:

$$\boxed{D(E_F) = \frac{3n}{2E_F}} \quad \text{and} \quad \boxed{g(E_F) = \frac{3N}{2E_F}} \tag{62}$$

This beautiful relationship states that the density of states at E_F is exactly 1.5 times the average density of states n/E_F across the entire occupied band!

5.5 Ground-State Internal Energy at Absolute Zero

At $T = 0$ K, all single-particle states up to E_F are filled with probability $f(E) = 1$, and all states above E_F are empty ($f(E) = 0$). The total internal energy U_0 of the electron gas is obtained by

integrating the energy times the density of states:

$$U_0 = \int_0^{E_F} E\, g(E)\, dE = \int_0^{E_F} E \left[\frac{3N}{2E_F^{3/2}} E^{1/2}\right] dE = \frac{3N}{2E_F^{3/2}} \int_0^{E_F} E^{3/2}\, dE \tag{63}$$

Evaluating the integral:

$$\int_0^{E_F} E^{3/2}\, dE = \frac{2}{5} E_F^{5/2} \tag{64}$$

Substituting back:

$$U_0 = \frac{3N}{2E_F^{3/2}} \left(\frac{2}{5} E_F^{5/2}\right) = \frac{3}{5} N E_F \tag{65}$$

The average kinetic energy per electron in the ground state is therefore:

$$\boxed{\langle E \rangle_0 = \frac{U_0}{N} = \frac{3}{5} E_F} \tag{66}$$

Even at absolute zero, quantum confinement and Pauli exclusion force electrons to possess significant kinetic energy: for copper ($E_F = 7.0$ eV), the mean zero-point energy is $\langle E \rangle_0 = 4.2$ eV per electron, corresponding to an effective classical thermal temperature of $\frac{2}{3}\langle E \rangle_0 / k_B \approx 32{,}000$ K!

5.6 Thermodynamic Degeneracy Pressure and Electronic Bulk Modulus

From fundamental thermodynamics, the pressure exerted by a quantum gas at $T = 0$ K is defined by the volume derivative of its ground-state internal energy at constant particle number:

$$P_0 = -\left(\frac{\partial U_0}{\partial V}\right)_N \tag{67}$$

Using $U_0 = \frac{3}{5} N E_F$, the volume dependence enters entirely through E_F:

$$E_F = \frac{\hbar^2}{2m_e} \left(3\pi^2 \frac{N}{V}\right)^{2/3} \propto V^{-2/3} \tag{68}$$

Differentiating with respect to V:

$$\frac{dE_F}{dV} = -\frac{2}{3} \frac{E_F}{V} \tag{69}$$

Substituting this into equation (67):

$$P_0 = -\frac{3}{5} N \left(-\frac{2}{3} \frac{E_F}{V}\right) = \frac{2}{5} \left(\frac{N}{V}\right) E_F \tag{70}$$

We obtain the fundamental **electron degeneracy pressure**:

$$\boxed{P_0 = \frac{2}{5} n E_F} \tag{71}$$

Since $U_0/V = \frac{3}{5} n E_F$, the degeneracy pressure is related to the internal energy density by:

$$P_0 = \frac{2}{3} \left(\frac{U_0}{V}\right) \tag{72}$$

The mechanical resistance of the electron gas to isotropic compression is quantified by the **bulk modulus** B:

$$B \equiv -V \left(\frac{\partial P}{\partial V}\right)_N \tag{73}$$

Since $P_0 \propto nE_F \propto V^{-1}V^{-2/3} = V^{-5/3}$, differentiating gives:

$$\frac{\partial P_0}{\partial V} = -\frac{5}{3}\frac{P_0}{V} \tag{74}$$

Substituting into equation (73) yields:

$$\boxed{B = \frac{5}{3}P_0 = \frac{2}{3}nE_F} \tag{75}$$

Study Takeaway: Fermi Degeneracy Pressure and Stellar Stability

The degeneracy pressure $P_0 = \frac{2}{5}nE_F$ is a purely quantum kinetic phenomenon. It does not arise from electrostatic Coulomb repulsion, but from the Heisenberg uncertainty principle and Pauli exclusion: compressing the gas forces electrons into smaller spatial regions, increasing their quantum zero-point momenta and kinetic energies.
In astrophysics, this non-thermal electron degeneracy pressure is precisely the cosmic force that prevents **white dwarf stars** (such as Sirius B) from collapsing under their enormous gravitational self-attraction. When a star's core exhausts its nuclear fuel, electron degeneracy pressure supports the star against gravity up to the Chandrasekhar mass limit ($M \approx 1.44M_\odot$).

Worked Example: Degeneracy Pressure and Bulk Modulus of Metallic Copper

Problem Statement: For metallic copper at $T = 0$ K, the conduction electron density is $n = 8.47 \times 10^{28}$ m^{-3} and the Fermi energy is $E_F = 7.00$ eV.

(a) Calculate the zero-point internal energy density $u_0 = U_0/V$ in J/m^3.

(b) Determine the quantum degeneracy pressure P_0 in Pascals (N/m^2) and in atmospheres (1 atm $\approx 1.013 \times 10^5$ Pa).

(c) Compute the electronic bulk modulus B of copper, and compare it with the measured macroscopic bulk modulus of copper ($B_{\text{exp}} \approx 13.7 \times 10^{10}$ Pa).

Step-by-Step Solution:

(a) Zero-point energy density u_0: First convert E_F to Joules:

$$E_F = (7.00 \text{ eV})(1.6022 \times 10^{-19} \text{ J/eV}) = 1.1215 \times 10^{-18} \text{ J}$$

The energy density is:

$$\begin{aligned} u_0 &= \frac{3}{5}nE_F = \frac{3}{5}(8.47 \times 10^{28} \text{ m}^{-3})(1.1215 \times 10^{-18} \text{ J}) \\ &= (5.082 \times 10^{28})(1.1215 \times 10^{-18}) \approx \mathbf{5.70 \times 10^{10}} \text{ J/m}^3 \end{aligned}$$

(b) Quantum degeneracy pressure P_0: Using equation (71):

$$P_0 = \frac{2}{5}nE_F = \frac{2}{3}u_0 = \frac{2}{3}(5.70 \times 10^{10} \text{ J/m}^3) \approx \mathbf{3.80 \times 10^{10}} \text{ Pa}$$

Converting to atmospheres:

$$P_0 = \frac{3.80 \times 10^{10} \text{ Pa}}{1.01325 \times 10^5 \text{ Pa/atm}} \approx \mathbf{3.75 \times 10^5} \text{ atm} \approx \mathbf{375{,}000} \text{ atmospheres!}$$

(c) Electronic bulk modulus B: Using equation (75):

$$B = \frac{5}{3}P_0 = \frac{2}{3}nE_F = u_0 \approx \mathbf{5.70 \times 10^{10}} \text{ Pa} = 57.0 \text{ GPa}$$

Comparison: The experimental bulk modulus of copper is $B_{\text{exp}} \approx 137$ GPa. The free electron gas accounts for:

$$\frac{B_{\text{electronic}}}{B_{\text{exp}}} = \frac{57.0 \text{ GPa}}{137 \text{ GPa}} \approx \mathbf{41.6\%}$$

Over 40% of the macroscopic mechanical stiffness of metallic copper originates purely from quantum electron degeneracy pressure! The remainder arises from ion core Coulomb repulsions and exchange-correlation forces.

Worked Example: Thermally Active Fraction of Conduction Electrons at Room Temperature

Problem Statement: Consider conduction electrons in metallic copper ($E_F = 7.00$ eV) at room temperature ($T = 300$ K).

(a) Evaluate the thermal energy scale k_BT in electron-volts at 300 K.

(b) Estimate the fraction f_{active} of conduction electrons that lie within an energy window of width $\Delta E = \pm k_BT$ around the Fermi energy.

(c) Compute the number of thermally active electrons in 1.0 cm^3 of metallic copper.

Step-by-Step Solution:
(a) Thermal energy scale k_BT:

$$k_BT = (8.6173 \times 10^{-5} \text{ eV/K})(300 \text{ K}) \approx \mathbf{0.02585} \text{ eV} \approx \frac{1}{39} \text{ eV}$$

(b) Thermally active fraction f_{active}: The number of electrons per unit volume residing in an energy strip of width $\Delta E = 2k_BT$ centered at E_F is approximately:

$$\Delta n \approx D(E_F)\Delta E = \left(\frac{3n}{2E_F}\right)(2k_BT) = 3n\left(\frac{k_BT}{E_F}\right)$$

The active fraction is:

$$f_{\text{active}} \approx \frac{\Delta n}{n} = 3\left(\frac{k_BT}{E_F}\right) = 3\left(\frac{0.02585 \text{ eV}}{7.00 \text{ eV}}\right)$$
$$= 3(3.69 \times 10^{-3}) \approx \mathbf{1.11 \times 10^{-2}} \approx \mathbf{1.1\%}$$

(If using the single-sided interval k_BT/E_F, the fraction is 0.37%).
(c) Number of active electrons in 1 cm^3: For $V = 1.0$ cm^3 $= 1.0 \times 10^{-6}$ m^3, with $n = 8.47 \times 10^{28}$ m^{-3}:

$$N_{\text{total}} = nV = 8.47 \times 10^{22} \text{ electrons}$$

The number of thermally active electrons is:

$$N_{\text{active}} = f_{\text{active}} N_{\text{total}} \approx (0.0111)(8.47 \times 10^{22}) \approx \mathbf{9.4 \times 10^{20}} \text{ electrons}$$

Physical Meaning: Over 98.9% of the conduction electrons in copper at room temperature are quantum-mechanically locked in the interior of the Fermi sea and cannot participate in thermal absorption.

6 Quantum Statistics and Electronic Heat Capacity

6.1 The Fermi–Dirac Distribution Function

Because electrons are identical fermions possessing half-integer spin ($s = 1/2$), the probability $f(E)$ that a single-particle quantum state of energy E is occupied in thermal equilibrium at temperature T is governed by the **Fermi–Dirac distribution**:

$$\boxed{f(E) = \frac{1}{\exp\left(\frac{E-\mu}{k_B T}\right) + 1}} \tag{76}$$

where $\mu(T)$ is the chemical potential of the electron gas (the Gibbs free energy per particle).

Physical Properties of $f(E)$:

- **Zero-Temperature Limit** ($T \to 0$ K): $\mu(0) \equiv E_F$.

$$f(E) = \begin{cases} 1, & E < E_F \\ 0, & E > E_F \end{cases} \tag{77}$$

  The distribution is a pristine unit step function: every single quantum state below E_F is completely occupied, and every state above E_F is completely vacant.

- **Finite Temperatures** ($T > 0$ K): The sharp step at E_F becomes smoothed over an energy interval of characteristic width $\sim \pm k_B T$. At $E = \mu(T)$, the occupation probability is identically $1/2$ at all temperatures:

$$f(\mu) = \frac{1}{e^0 + 1} = \frac{1}{2} \tag{78}$$

- **Particle-Hole Symmetry**: The probability that an electron occupies a state an energy Δ above the chemical potential equals the probability that a state an energy Δ below the chemical potential is unoccupied (vacant hole):

$$f(\mu + \Delta) = 1 - f(\mu - \Delta) \iff f(\mu + \Delta) + f(\mu - \Delta) = 1 \tag{79}$$

6.2 Temperature Dependence of the Chemical Potential

The total electron density n in a metal is fixed by the positive ionic lattice charges and is independent of temperature. Particle conservation requires:

$$n = \int_0^\infty D(E) f(E)\, dE = \int_0^\infty \frac{3n}{2E_F^{3/2}} \frac{E^{1/2}}{\exp\left(\frac{E-\mu(T)}{k_B T}\right) + 1}\, dE \tag{80}$$

Because the 3D density of states $D(E) \propto E^{1/2}$ increases with energy, the density of states available above μ is slightly greater than that below μ. To maintain a constant particle count as states above μ become thermally populated, the chemical potential must shift slightly downward with increasing temperature.

Evaluating equation (80) via the Sommerfeld expansion yields the temperature dependence of $\mu(T)$:

$$\boxed{\mu(T) \approx E_F \left[1 - \frac{\pi^2}{12}\left(\frac{T}{T_F}\right)^2 + \mathcal{O}\left(\left(\frac{T}{T_F}\right)^4\right)\right]} \tag{81}$$

For metallic copper ($T_F \approx 81{,}200$ K) at room temperature ($T = 300$ K), $(T/T_F)^2 \approx (3.69 \times 10^{-3})^2 \approx 1.36 \times 10^{-5}$. The chemical potential drops by only 0.001%! In virtually all solid-state calculations, setting $\mu(T) \approx E_F$ is an extraordinarily accurate approximation.

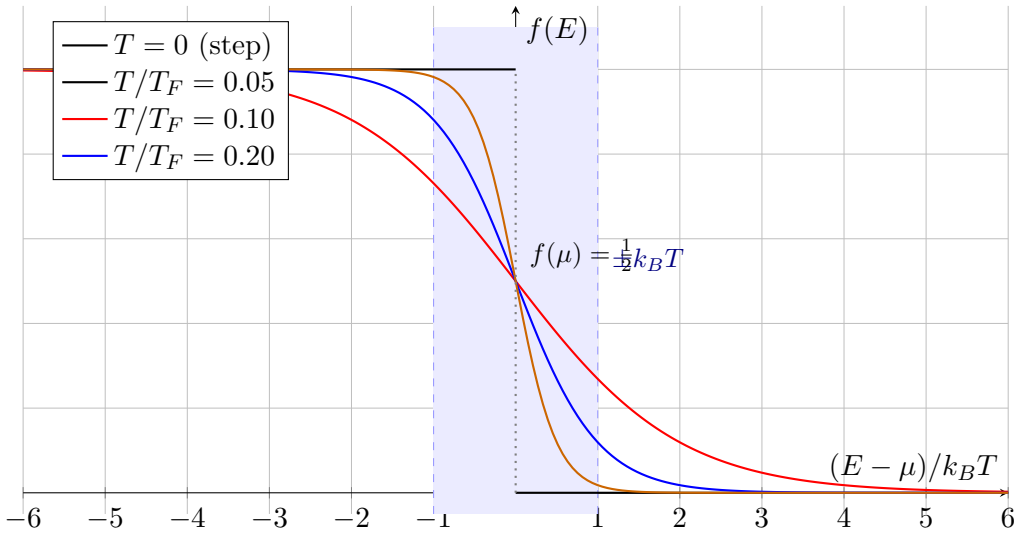


Figure 4: Fermi–Dirac distribution $f(E)$ at $T = 0$ (black step) and at finite temperatures, plotted versus $(E-\mu)/k_BT$. The shaded band marks the thermal smearing region of width $\sim \pm k_BT$. Even at $T/T_F \sim 0.1$ (typical room temperature for a metal), the distribution is still an almost perfect step.

6.3 The Classical Electronic Heat Capacity Failure

According to classical statistical mechanics (the equipartition theorem), each classical degree of freedom possesses an average thermal kinetic energy of $\frac{1}{2}k_BT$. In a three-dimensional gas, the internal thermal energy per mole of electrons should be:

$$U_{\text{classical}} = \frac{3}{2}N_A k_B T = \frac{3}{2}RT \tag{82}$$

predicting a constant molar electronic heat capacity:

$$C_{V,\text{classical}} = \frac{\partial U}{\partial T} = \frac{3}{2}R \approx 12.47 \text{ J/(mol} \cdot \text{K)} \tag{83}$$

However, 19th-century calorimetric measurements revealed that the total heat capacity of metals at room temperature follows the empirical Dulong–Petit law $C_V \approx 3R \approx 25$ J/(mol · K), which is fully accounted for by lattice vibrations (phonons) alone!

The conduction electrons appeared to contribute almost *nothing* to the specific heat. Classical physics was utterly powerless to explain why an electron gas capable of conducting massive electrical currents failed to absorb thermal energy.

6.4 Qualitative Quantum Resolution via the Pauli Principle

Quantum mechanics provides an elegant, physically transparent resolution:

1. When a metal is heated from 0 K to temperature T, thermal fluctuations can transfer energy packets of magnitude $\sim k_BT$ to electrons.

2. An electron lying at energy E can absorb this thermal quantum only if the target state at $E + k_BT$ is empty.

3. For an electron situated deep within the Fermi sea (where $E_F - E \gg k_BT$), all adjacent states up to $E + k_BT$ are already 100% occupied by other electrons. The Pauli exclusion principle strictly forbids the transition!

4. Therefore, **only electrons in a narrow surface shell** of thickness $\sim k_BT$ around the Fermi level can absorb thermal energy.

The fraction of thermally excited electrons is approximately:

$$f_{\text{excited}} \approx \frac{k_B T}{E_F} = \frac{T}{T_F} \sim 10^{-2} \tag{84}$$

Each excited electron absorbs an energy of order $k_B T$. The thermal energy per mole of electrons is:

$$\Delta U \approx N_A \left(\frac{T}{T_F}\right) k_B T = R\frac{T^2}{T_F} \tag{85}$$

Differentiating with respect to temperature gives the electronic heat capacity:

$$C_{el} = \frac{d(\Delta U)}{dT} \approx 2R\left(\frac{T}{T_F}\right) \propto T \tag{86}$$

Because $T/T_F \approx 10^{-2}$ at room temperature, the true electronic heat capacity is suppressed by two orders of magnitude relative to classical equipartition, completely solving the mystery!

Conceptual Checkpoint: Why is Electronic Heat Capacity Suppressed by 10^{-2}?

The suppression is a direct manifestation of Fermi–Dirac quantum statistics:

- In a classical Boltzmann gas, particles are distinguishable; all N particles absorb $\frac{3}{2}k_B T$ equally, producing a constant $C_V = \frac{3}{2}R$.
- In a quantum Fermi gas, particles are indistinguishable fermions; Pauli exclusion freezes over 99% of the carriers into inert, inaccessible quantum states.

6.5 Rigorous Sommerfeld Expansion and Electronic Heat Capacity

To evaluate the electronic internal energy rigorously, consider the general thermodynamic integral for any smooth function $\phi(E)$:

$$I = \int_0^\infty \phi(E) f(E)\, dE \tag{87}$$

Let $\Phi(E) \equiv \int_0^E \phi(E')\, dE'$. Integrating by parts:

$$I = \Big[\Phi(E) f(E)\Big]_0^\infty - \int_0^\infty \Phi(E)\frac{\partial f}{\partial E}\, dE = \int_0^\infty \Phi(E)\left(-\frac{\partial f}{\partial E}\right) dE \tag{88}$$

At low temperatures, $-\frac{\partial f}{\partial E}$ is a sharp symmetric bell curve centered at $E = \mu$ with width $\sim k_B T$. Expanding $\Phi(E)$ in a Taylor series about $E = \mu$:

$$\Phi(E) = \Phi(\mu) + \Phi'(\mu)(E-\mu) + \frac{1}{2}\Phi''(\mu)(E-\mu)^2 + \dots \tag{89}$$

Substituting this expansion into equation (88):

$$I = \Phi(\mu)\int_0^\infty \left(-\frac{\partial f}{\partial E}\right) dE + \phi(\mu)\int_0^\infty (E-\mu)\left(-\frac{\partial f}{\partial E}\right) dE + \frac{1}{2}\phi'(\mu)\int_0^\infty (E-\mu)^2\left(-\frac{\partial f}{\partial E}\right) dE + \dots \tag{90}$$

Using the dimensionless variable $x \equiv \frac{E-\mu}{k_B T}$:

$$-\frac{\partial f}{\partial E} = \frac{1}{k_B T}\frac{e^x}{(e^x+1)^2} \tag{91}$$

The three integrals evaluate as follows:

1. $\int_{-\infty}^{\infty} \frac{e^x}{(e^x+1)^2}\, dx = \left[-\frac{1}{e^x+1}\right]_{-\infty}^{\infty} = 1.$

2. $\int_{-\infty}^{\infty} x \frac{e^x}{(e^x+1)^2}\,dx = 0$ (integrand is strictly odd).

3. The second-order definite integral:

$$\int_{-\infty}^{\infty} x^2 \frac{e^x}{(e^x+1)^2}\,dx = 2\int_0^{\infty} x^2 \frac{e^{-x}}{(1+e^{-x})^2}\,dx \tag{92}$$

Using the binomial series $\frac{e^{-x}}{(1+e^{-x})^2} = \sum_{n=1}^{\infty}(-1)^{n-1} n e^{-nx}$:

$$\begin{aligned}\int_{-\infty}^{\infty} x^2 \frac{e^x}{(e^x+1)^2}\,dx &= 2\sum_{n=1}^{\infty}(-1)^{n-1} n \int_0^{\infty} x^2 e^{-nx}\,dx = 2\sum_{n=1}^{\infty}(-1)^{n-1} n \left(\frac{2!}{n^3}\right) \\ &= 4\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^2} = 4\left(\frac{\pi^2}{12}\right) = \frac{\pi^2}{3}\end{aligned} \tag{93}$$

This yields the celebrated **Sommerfeld Expansion Lemma**:

$$\boxed{\int_0^{\infty} \phi(E) f(E)\,dE = \int_0^{\mu} \phi(E)\,dE + \frac{\pi^2}{6}(k_B T)^2 \left.\frac{d\phi}{dE}\right|_{E=\mu} + \mathcal{O}\left((k_B T)^4\right)} \tag{94}$$

Application to Internal Energy $U(T)$: Setting $\phi(E) = E\,g(E)$:

$$U(T) = \int_0^{\mu} E\,g(E)\,dE + \frac{\pi^2}{6}(k_B T)^2 \left.\frac{d(E\,g(E))}{dE}\right|_{E=\mu} \tag{95}$$

Since $\mu(T) = E_F - \frac{\pi^2}{12}\frac{(k_B T)^2}{E_F}$, the integral $\int_0^{\mu} E g(E) dE$ evaluates to $U_0 - \frac{\pi^2}{12}(k_B T)^2 g(E_F)$. Combining terms:

$$U(T) = U_0 + \frac{\pi^2}{6} g(E_F)(k_B T)^2 \tag{96}$$

Differentiating with respect to temperature gives the exact **electronic specific heat**:

$$\boxed{C_{el} = \frac{\partial U}{\partial T} = \frac{\pi^2}{3} g(E_F) k_B^2 T} \tag{97}$$

Substituting $g(E_F) = \frac{3N}{2E_F} = \frac{3N}{2k_B T_F}$:

$$\boxed{C_{el} = \frac{\pi^2}{2} N k_B \left(\frac{T}{T_F}\right)} \tag{98}$$

For one mole of electrons ($N = N_A$, $N_A k_B = R$):

$$\boxed{C_{el} = \gamma T, \quad \text{where} \quad \gamma \equiv \frac{\pi^2}{2}\frac{R k_B}{E_F} = \frac{\pi^2}{3} D(E_F) k_B^2} \tag{99}$$

The constant γ is known as the **Sommerfeld coefficient** (or linear electronic specific heat coefficient).

6.6 Separation of Electronic and Phonon Heat Capacities

In a real metallic crystal, the total measured heat capacity at low temperatures ($T \ll \Theta_D$, where Θ_D is the Debye temperature) is the sum of conduction electron excitations and acoustic lattice vibrations (Debye phonons):

$$C(T) = C_{el} + C_{ph} = \gamma T + \beta T^3 \tag{100}$$

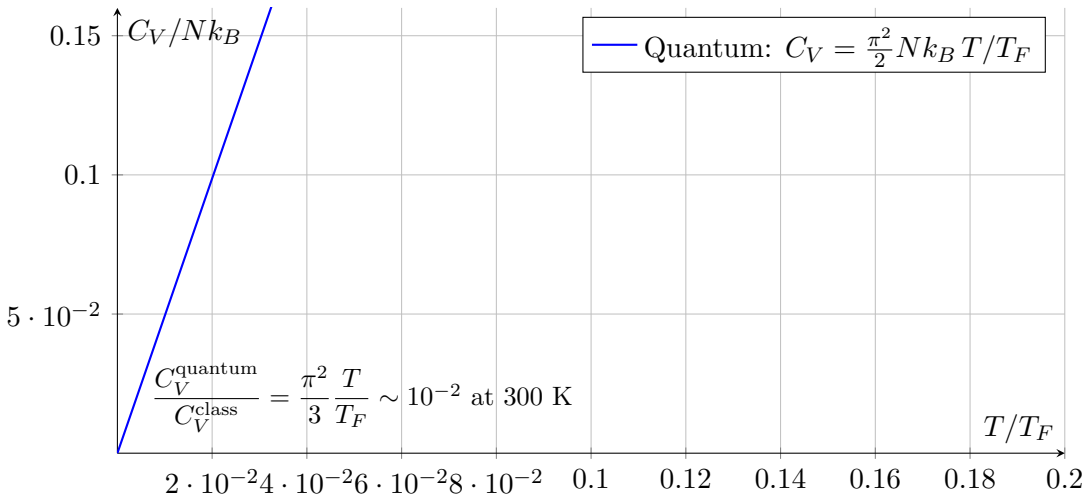


Figure 5: Linear-in-T electronic specific heat of a free-electron metal (blue) compared with the classical equipartition prediction (red dashed, off scale). The quantum result is smaller by a factor of order $T/T_F \sim 10^{-2}$ at room temperature.

where the Debye phonon coefficient β is:

$$\beta = \frac{12\pi^4}{5}\frac{N_A k_B}{\Theta_D^3} = \frac{12\pi^4 R}{5\Theta_D^3} \tag{101}$$

To disentangle the linear electronic contribution from the cubic phonon contribution experimentally, equation (100) is divided by T:

$$\boxed{\frac{C(T)}{T} = \gamma + \beta T^2} \tag{102}$$

A plot of experimental $C(T)/T$ versus T^2 yields a straight line:

- The **vertical intercept** at $T^2 = 0$ gives the Sommerfeld coefficient γ, directly measuring the electronic density of states at the Fermi energy $g(E_F)$!
- The **slope** of the line gives the Debye coefficient β, directly yielding the Debye temperature Θ_D.

The **crossover temperature** T^* where electronic and lattice heat capacities are equal ($C_{el} = C_{ph}$) is:

$$\gamma T^* = \beta (T^*)^3 \implies \boxed{T^* = \sqrt{\frac{\gamma}{\beta}}} \tag{103}$$

For typical metals, T^* lies between 1 K and 5 K. Below T^*, electronic specific heat dominates; above T^*, lattice phonons rapidly overwhelm the electronic contribution.

Problem-Solving Strategy: Extracting γ and Θ_D from Calorimetric Data

To extract physical parameters from experimental low-temperature specific heat data:

1. Tabulate experimental (T_i, C_i) points in the range 1 K $\leq T \leq$ 4 K.
2. Construct a linear regression of $Y = C/T$ against $X = T^2$.
3. Identify the intercept $Y_0 = \gamma$ in $\text{mJ}\,\text{mol}^{-1}\,\text{K}^{-2}$.
4. Extract the Fermi energy via $E_F = \frac{\pi^2 R k_B}{2\gamma}$.

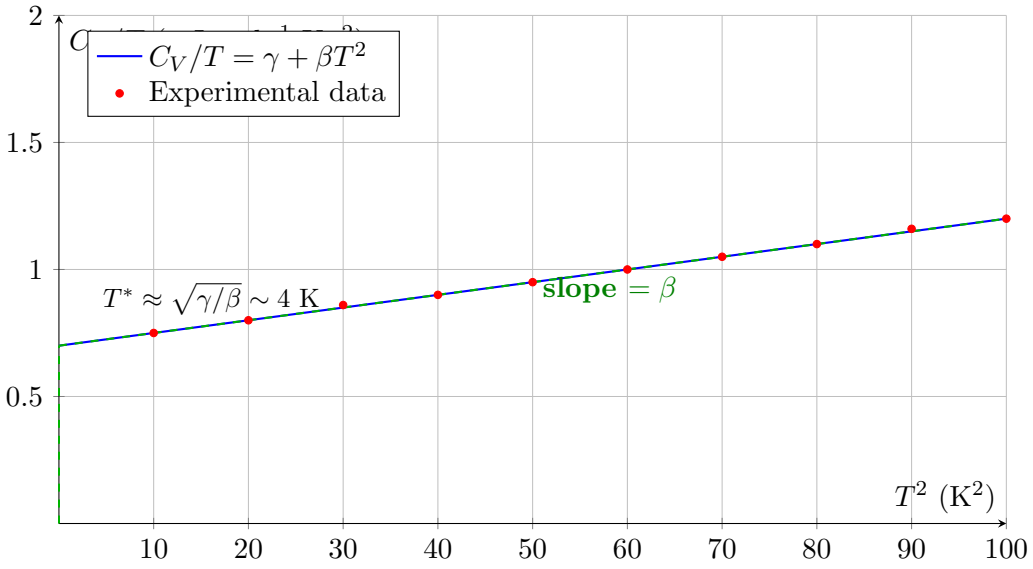


Figure 6: Plot of C_V/T versus T^2 for a metal such as copper. The intercept gives the Sommerfeld coefficient γ (electronic contribution), and the slope gives β (phonon contribution). Below the crossover temperature $T^* = \sqrt{\gamma/\beta}$, the linear electronic term dominates.

5. Identify the slope $m = \beta$ in $\mathrm{mJ\,mol^{-1}\,K^{-4}}$, and calculate the Debye temperature:

$$\Theta_D = \left(\frac{12\pi^4 R}{5\beta}\right)^{1/3}$$

6.7 Pauli Spin Paramagnetism of the Free Electron Gas

In classical statistical mechanics, magnetic moments of magnitude μ_B in an applied magnetic field B align according to the Langevin–Curie law:

$$\chi_{\text{Curie}} = \frac{\mu_0 n \mu_B^2}{k_B T} \propto \frac{1}{T} \tag{104}$$

However, non-ferromagnetic metals (such as Na, Al, Cu) display an extremely weak, nearly **temperature-independent paramagnetism**. Wolfgang Pauli (1927) explained this phenomenon as another fundamental consequence of Fermi–Dirac quantum statistics.

Each electron possesses an intrinsic magnetic dipole moment $\boldsymbol{\mu}_s = -g\mu_B \mathbf{S}/\hbar = \mp\mu_B$ directed antiparallel to its spin. In an external magnetic field $\mathbf{B} = B\hat{z}$, the Zeeman interaction energy is:

$$\Delta E = -\boldsymbol{\mu}_s \cdot \mathbf{B} = \pm \mu_B B \tag{105}$$

- Electrons with spin parallel to $\mathbf{B}$ (magnetic moment parallel, spin $\downarrow$) have their energy lowered: $E_\downarrow = E - \mu_B B$.
- Electrons with spin antiparallel to $\mathbf{B}$ (magnetic moment antiparallel, spin $\uparrow$) have their energy raised: $E_\uparrow = E + \mu_B B$.

Because electrons can flip their spin via spin-orbit interactions to reach thermal equilibrium, electrons near the Fermi boundary transfer from the higher-energy sub-band to the lower-energy sub-band until both spin manifolds share a common chemical potential μ.

The energy shift between the two sub-bands is $2\mu_B B$. The number of electrons transferred from spin-up to spin-down is the density of states per spin channel ($\frac{1}{2}g(E_F)$) multiplied by the Zeeman

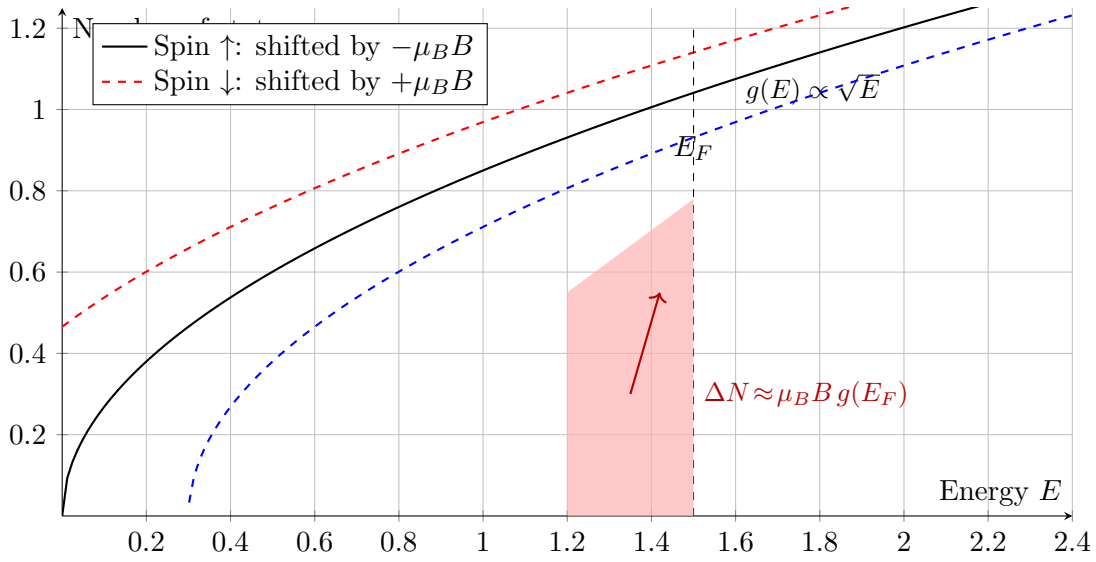


Figure 7: Pauli paramagnetism: in a magnetic field B, spin-up and spin-down DOS curves shift in opposite directions by $\pm\mu_B B$. Electrons near E_F flip to the lower-energy (spin-up) state, producing a net magnetization $M = \mu_B^2 B\, g(E_F)$. Only states within $\sim \mu_B B$ of E_F contribute, so $\chi \propto g(E_F)$ is essentially T-independent.

energy shift $\mu_B B$:

$$\Delta N = N_\downarrow - N_\uparrow \approx 2 \times \left[\frac{1}{2} g(E_F)\mu_B B\right] = g(E_F)\mu_B B \tag{106}$$

The resulting net macroscopic magnetization M is:

$$M = \mu_B \Delta N = \mu_B^2 g(E_F) B \tag{107}$$

The **Pauli paramagnetic volume susceptibility** $\chi_{\text{Pauli}} \equiv \mu_0 M/B$ is:

$$\boxed{\chi_{\text{Pauli}} = \mu_0 \mu_B^2 g(E_F) = \frac{3\mu_0 n \mu_B^2}{2E_F}} \tag{108}$$

Notice that χ_{Pauli} depends strictly on the density of states at the Fermi level $g(E_F)$ and is **completely independent of temperature** to leading order! Thermal corrections enter only at order $(T/T_F)^2 \sim 10^{-4}$, perfectly explaining the experimental observations.

Worked Example: Energy Levels for Given Fermi–Dirac Probabilities

Problem Statement: For metallic sodium at room temperature ($T = 300$ K, $k_B T \approx 0.02585$ eV), the chemical potential is $\mu = 3.130$ eV. Find the energy levels E for which the Fermi–Dirac occupation probability is: (a) $f(E) = 0.50$, (b) $f(E) = 0.75$, and (c) $f(E) = 0.25$.

Step-by-Step Solution:
Inverting the Fermi–Dirac distribution function $f(E) = \frac{1}{e^{(E-\mu)/k_B T}+1}$:

$$e^{(E-\mu)/k_B T} + 1 = \frac{1}{f} \implies e^{(E-\mu)/k_B T} = \frac{1-f}{f} \implies E = \mu + k_B T \ln\left(\frac{1-f}{f}\right)$$

(a) Case $f = 0.50$:

$$\frac{1-0.50}{0.50} = 1 \implies \ln(1) = 0 \implies E = \mu = \mathbf{3.130}\ \text{eV}$$

(b) Case $f = 0.75$:

$$\frac{1-0.75}{0.75} = \frac{0.25}{0.75} = \frac{1}{3}$$

$E = \mu + k_B T \ln(1/3) = \mu - k_B T \ln(3) = 3.130\text{ eV} - (0.02585\text{ eV})(1.0986) = 3.130 - 0.0284 \approx \mathbf{3.102}$ eV

(c) Case $f = 0.25$:

$$\frac{1-0.25}{0.25} = \frac{0.75}{0.25} = 3$$

$$E = \mu + k_B T \ln(3) = 3.130\text{ eV} + 0.0284\text{ eV} \approx \mathbf{3.158}\text{ eV}$$

Physical Insight: Notice that the energy shifts are perfectly symmetric: $E(0.75) = \mu - 1.10 k_B T$ and $E(0.25) = \mu + 1.10 k_B T$, illustrating exact particle-hole symmetry around the chemical potential.

Worked Example: Proof of Particle-Hole Symmetry of the Fermi Function

Problem Statement: Prove analytically that the Fermi–Dirac distribution function satisfies the exact particle-hole identity:

$$f(\mu + \Delta) + f(\mu - \Delta) \equiv 1 \quad \text{for all } \Delta \text{ and all } T > 0$$

Step-by-Step Proof:
Let $x \equiv \Delta/(k_B T)$. Then:

$$f(\mu + \Delta) = \frac{1}{e^x + 1}, \quad f(\mu - \Delta) = \frac{1}{e^{-x} + 1}$$

Multiply the numerator and denominator of $f(\mu - \Delta)$ by e^x:

$$f(\mu - \Delta) = \frac{1 \cdot e^x}{(e^{-x} + 1)e^x} = \frac{e^x}{1 + e^x}$$

Adding the two terms:

$$f(\mu + \Delta) + f(\mu - \Delta) = \frac{1}{e^x + 1} + \frac{e^x}{e^x + 1} = \frac{1 + e^x}{e^x + 1} \equiv \mathbf{1} \quad \textbf{Q.E.D.}$$

Worked Example: Sommerfeld Heat Capacity Coefficient of Aluminum

Problem Statement: Metallic aluminum has a Fermi energy $E_F = 11.7$ eV and valence $Z_v = 3$. Calculate the theoretical molar Sommerfeld coefficient γ in $\text{mJ mol}^{-1}\text{ K}^{-2}$, and determine the electronic heat capacity at $T = 4.2$ K.

Step-by-Step Solution:
(a) Molar Sommerfeld coefficient γ: First convert E_F to Joules:

$$E_F = (11.7\text{ eV})(1.6022 \times 10^{-19}\text{ J/eV}) \approx 1.875 \times 10^{-18}\text{ J}$$

For one mole of aluminum (N_A atoms), each atom contributes 3 conduction electrons, so $N = 3N_A$. The electronic heat capacity coefficient is:

$$\gamma = \frac{\pi^2}{2}\frac{N k_B^2}{E_F} = \frac{\pi^2}{2}\frac{(3N_A)k_B^2}{E_F} = \frac{3\pi^2 R k_B}{2E_F}$$

Evaluating with $R = 8.3145\ \text{J/(mol}\cdot\text{K)}$:

$$\gamma = \frac{3\pi^2(8.3145\ \text{J/(mol}\cdot\text{K)})(1.3807\times10^{-23}\ \text{J/K})}{2(1.875\times10^{-18}\ \text{J})}$$
$$= \frac{3.400\times10^{-21}}{3.750\times10^{-18}} \approx 9.07\times10^{-4}\ \text{J mol}^{-1}\,\text{K}^{-2} = \mathbf{0.907}\ \text{mJ mol}^{-1}\,\text{K}^{-2}$$

(If calculated per mole of electrons, $\gamma_e = \frac{\pi^2 R k_B}{2E_F} \approx 0.302\ \text{mJ mol}^{-1}\,\text{K}^{-2}$).

(b) Electronic heat capacity at $T = 4.2$ K:

$$C_{el}(4.2\ \text{K}) = \gamma T = (0.907\ \text{mJ mol}^{-1}\,\text{K}^{-2})(4.2\ \text{K}) \approx \mathbf{3.81}\ \text{mJ mol}^{-1}\,\text{K}^{-1}$$

Worked Example: Electronic vs. Lattice Specific Heat Crossover in Copper

Problem Statement: Low-temperature calorimetric measurements on copper yield a molar Sommerfeld coefficient $\gamma = 0.695\ \text{mJ mol}^{-1}\,\text{K}^{-2}$ and a Debye phonon coefficient $\beta = 0.048\ \text{mJ mol}^{-1}\,\text{K}^{-4}$.

(a) Determine the Debye characteristic temperature Θ_D of copper.

(b) Calculate the crossover temperature T^* where electronic and lattice heat capacities are equal.

(c) Compare the electronic and lattice contributions at $T = 1.0$ K and at $T = 300$ K.

Step-by-Step Solution:

(a) Debye temperature Θ_D: Using equation (101) with $\beta = 0.048\times10^{-3}\ \text{J mol}^{-1}\,\text{K}^{-4}$:

$$\Theta_D = \left(\frac{12\pi^4 R}{5\beta}\right)^{1/3} = \left(\frac{12\pi^4(8.3145)}{5(4.8\times10^{-5})}\right)^{1/3} = \left(\frac{9719}{2.4\times10^{-4}}\right)^{1/3} = (4.05\times10^{7})^{1/3} \approx \mathbf{343.4}\ \text{K}$$

(b) Crossover temperature T^*: Setting $C_{el} = C_{ph} \implies \gamma T^* = \beta (T^*)^3$:

$$T^* = \sqrt{\frac{\gamma}{\beta}} = \sqrt{\frac{0.695\ \text{mJ mol}^{-1}\,\text{K}^{-2}}{0.048\ \text{mJ mol}^{-1}\,\text{K}^{-4}}} = \sqrt{14.48} \approx \mathbf{3.81}\ \text{K}$$

(c) Contribution comparison:

- At $T = 1.0$ K:

  $$C_{el} = (0.695)(1.0) = 0.695\ \text{mJ/mol}\cdot\text{K}, \quad C_{ph} = (0.048)(1.0)^3 = 0.048\ \text{mJ/mol}\cdot\text{K}$$

  The electronic heat capacity is **14.5** times larger than the phonon contribution!

- At $T = 300$ K (room temperature):

  $$C_{el} = (0.695\times10^{-3}\ \text{J/mol}\cdot\text{K}^2)(300\ \text{K}) \approx \mathbf{0.208}\ \text{J/(mol}\cdot\text{K)}$$

  The high-temperature Dulong–Petit phonon heat capacity is $C_{ph} \approx 3R \approx \mathbf{24.94}$ J/(mol·K). The electronic fraction at room temperature is only:

  $$\frac{C_{el}}{C_{\text{total}}} \approx \frac{0.208}{25.15} \approx \mathbf{0.83}\%$$

7 Electrical and Thermal Transport in Metals

7.1 Semiclassical Dynamics of Conduction Electrons

In the quantum Sommerfeld model, conduction electrons are treated as localized wavepackets formed by a coherent superposition of plane waves centered at wavevector $\mathbf{k}$. The center-of-mass dynamics of an electron wavepacket in external macroscopic electromagnetic fields obeys the **semiclassical equations of motion**:

$$\mathbf{v}(\mathbf{k}) = \frac{1}{\hbar}\nabla_{\mathbf{k}} E(\mathbf{k}) \tag{109}$$

$$\hbar\frac{d\mathbf{k}}{dt} = \mathbf{F}_{\text{ext}} = -e\left(\mathbf{E} + \mathbf{v}\times\mathbf{B}\right) \tag{110}$$

where $\mathbf{v}(\mathbf{k})$ is the quantum group velocity. For a parabolic, isotropic free-electron dispersion $E(\mathbf{k}) = \frac{\hbar^2 k^2}{2m_e}$, the group velocity simplifies directly to the classical momentum relation:

$$\mathbf{v}(\mathbf{k}) = \frac{1}{\hbar}\nabla_{\mathbf{k}}\left(\frac{\hbar^2 k^2}{2m_e}\right) = \frac{\hbar\mathbf{k}}{m_e} \tag{111}$$

7.2 The Boltzmann Transport Equation in the Relaxation Time Approximation

To describe transport in non-equilibrium conditions, we define the distribution function $f(\mathbf{r}, \mathbf{k}, t)$ such that:

$$dn = 2 \times f(\mathbf{r}, \mathbf{k}, t)\frac{d^3r\, d^3k}{(2\pi)^3} \tag{112}$$

represents the number of electrons in the six-dimensional phase-space volume element $d^3r\, d^3k$ at time t (the factor of 2 accounts for spin).

According to Liouville's theorem, in the presence of external forces and microscopic scattering collisions, the total time rate of change of f is:

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \mathbf{v}\cdot\nabla_{\mathbf{r}} f + \dot{\mathbf{k}}\cdot\nabla_{\mathbf{k}} f = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \tag{113}$$

Under the **Relaxation Time Approximation** (RTA), the collision integral is modeled as an exponential decay toward the local Fermi–Dirac equilibrium distribution f_0:

$$\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = -\frac{f(\mathbf{r}, \mathbf{k}, t) - f_0(E)}{\tau(\mathbf{k})} \tag{114}$$

where $\tau(\mathbf{k})$ is the momentum relaxation time.

For steady-state ($\partial f/\partial t = 0$), spatially homogeneous ($\nabla_{\mathbf{r}} f = 0$) conduction driven by a uniform electric field $\mathbf{E}$, the wavevector acceleration is $\dot{\mathbf{k}} = -e\mathbf{E}/\hbar$. The Boltzmann equation (113) reduces to:

$$-\frac{e\mathbf{E}}{\hbar}\cdot\nabla_{\mathbf{k}} f = -\frac{f - f_0}{\tau} \tag{115}$$

7.3 Linearization of the Distribution Function

For ordinary electric fields ($E \sim$ 1–100 V/m), the perturbation away from equilibrium is exceedingly small. We linearize the distribution function:

$$f(\mathbf{k}) \approx f_0(E(\mathbf{k})) + f_1(\mathbf{k}), \quad |f_1| \ll f_0 \tag{116}$$

Substituting equation (116) into the left-hand side of equation (115) and approximating $\nabla_{\mathbf{k}} f \approx \nabla_{\mathbf{k}} f_0$:

$$\nabla_{\mathbf{k}} f_0 = \frac{\partial f_0}{\partial E}\nabla_{\mathbf{k}} E = \hbar\mathbf{v}\frac{\partial f_0}{\partial E} \tag{117}$$

Equation (115) becomes:

$$-\frac{e\mathbf{E}}{\hbar}\cdot\left(\hbar\mathbf{v}\frac{\partial f_0}{\partial E}\right)=-\frac{f_1}{\tau} \tag{118}$$

Solving directly for the non-equilibrium perturbation $f_1(\mathbf{k})$:

$$\boxed{f_1(\mathbf{k})=e\tau(\mathbf{v}\cdot\mathbf{E})\left(-\frac{\partial f_0}{\partial E}\right)} \tag{119}$$

Physical Interpretation (Shifted Fermi Sphere): Notice that for an electric field along $+x$, $\mathbf{v}\cdot\mathbf{E}=v_xE_x$. Since $\mathbf{v}=\frac{\hbar\mathbf{k}}{m_e}$:

$$f(\mathbf{k})\approx f_0(E)+\frac{e\tau E_x}{\hbar}\frac{\partial E}{\partial k_x}\left(-\frac{\partial f_0}{\partial E}\right)=f_0(E)-\Delta k_x\frac{\partial f_0}{\partial k_x}\approx f_0\left(k_x+\frac{eE_x\tau}{\hbar},k_y,k_z\right) \tag{120}$$

The application of an electric field simply shifts the entire Fermi sphere in reciprocal space rigidly opposite to the electric field by a displacement:

$$\boxed{\Delta\mathbf{k}=-\frac{e\mathbf{E}\tau}{\hbar}} \tag{121}$$

7.4 Rigorous Evaluation of Current Density and Conductivity Recovery

The macroscopic electric current density is obtained by integrating the velocity of all occupied states over k-space:

$$\mathbf{j}=-2e\int\frac{d^3k}{(2\pi)^3}\mathbf{v}(\mathbf{k})f(\mathbf{k})=-2e\int\frac{d^3k}{(2\pi)^3}\mathbf{v}(\mathbf{k})\left[f_0(E)+f_1(\mathbf{k})\right] \tag{122}$$

The integral splits naturally into two parts:

1. **Equilibrium Integral** (I_0):

$$\mathbf{j}_0=-2e\int\frac{d^3k}{(2\pi)^3}\mathbf{v}(\mathbf{k})f_0(E(\mathbf{k}))=\mathbf{0} \tag{123}$$

   The equilibrium current vanishes identically because $E(\mathbf{k})$ and $f_0(E(\mathbf{k}))$ are even functions of $\mathbf{k}$ ($f_0(-\mathbf{k})=f_0(\mathbf{k})$), whereas velocity $\mathbf{v}(\mathbf{k})=\hbar\mathbf{k}/m_e$ is strictly odd ($\mathbf{v}(-\mathbf{k})=-\mathbf{v}(\mathbf{k})$).

2. **Transport Integral** (I_1): For an electric field applied along $\hat{x}$ ($\mathbf{E}=E_x\hat{x}$):

$$j_x=-2e\int\frac{d^3k}{(2\pi)^3}v_x(\mathbf{k})\left[e\tau v_xE_x\left(-\frac{\partial f_0}{\partial E}\right)\right]=\frac{e^2\tau E_x}{4\pi^3}\int d^3k\,v_x^2\left(-\frac{\partial f_0}{\partial E}\right) \tag{124}$$

Coordinate Transformation to Iso-Energetic Surfaces: We transform the 3D volume element d^3k into an integral over constant-energy surfaces S_E in k-space:

$$d^3k=dS_Edk_\perp=dS_E\frac{dE}{|\nabla_{\mathbf{k}}E|}=\frac{dS_EdE}{\hbar v} \tag{125}$$

For an isotropic, spherical Fermi surface, directional averaging gives $\langle v_x^2\rangle=\frac{1}{3}v^2$. Thus:

$$v_x^2d^3k=\left(\frac{1}{3}v^2\right)\frac{dS_EdE}{\hbar v}=\frac{1}{3\hbar}v(E)dS_EdE \tag{126}$$

Substituting into equation (124):

$$j_x=\frac{e^2\tau E_x}{12\pi^3\hbar}\int_0^\infty dE\left(-\frac{\partial f_0}{\partial E}\right)v(E)\iint_{S_E}dS_E \tag{127}$$

For a sphere in k-space, $\iint_{S_E} dS_E = 4\pi k^2$.

At low temperatures ($T \ll T_F$), the derivative $-\frac{\partial f_0}{\partial E}$ collapses to a Dirac delta function centered at the Fermi energy:

$$\lim_{T\to 0}\left(-\frac{\partial f_0}{\partial E}\right) = \delta(E - E_F) \tag{128}$$

The energy integral collapses instantly to the Fermi surface:

$$j_x = \frac{e^2\tau E_x}{12\pi^3\hbar} v_F (4\pi k_F^2) = \frac{e^2\tau E_x}{3\pi^2\hbar} v_F k_F^2 \tag{129}$$

Using $v_F = \frac{\hbar k_F}{m_e}$:

$$j_x = \frac{e^2\tau E_x}{3\pi^2\hbar}\left(\frac{\hbar k_F}{m_e}\right) k_F^2 = \left(\frac{e^2\tau k_F^3}{3\pi^2 m_e}\right) E_x \tag{130}$$

Recalling the fundamental relation from Chapter 2, $k_F^3 = 3\pi^2 n$:

$$\boxed{j_x = \left(\frac{ne^2\tau}{m_e}\right) E_x \implies \sigma = \frac{ne^2\tau}{m_e}} \tag{131}$$

We have rigorously recovered Drude's classical conductivity formula from first-principles quantum Boltzmann transport theory!

Conceptual Checkpoint: Why Electrical Conduction Occurs Exclusively at the Fermi Surface

The appearance of the Dirac delta function $\delta(E - E_F)$ proves mathematically that **electrons deep inside the Fermi sea carry zero net current**!
Under an electric field, for every state (k_x, k_y, k_z) filled deep below E_F, the time-reversed state $(-k_x, k_y, k_z)$ is also completely filled. Their equal and opposite velocities cancel out identically. Only right at the Fermi sphere boundary does the field populate forward states while depopulating backward states, creating an uncompensated current carrying surface layer.

7.5 Microscopic Scattering Mechanisms and Matthiessen's Rule

In a pure metal, the carrier density n is temperature-independent. The observed temperature dependence of electrical resistivity $\rho(T) = \frac{m_e}{ne^2\tau(T)}$ is governed entirely by the microscopic collision time $\tau(T)$.

Electrons in a crystal experience two independent scattering processes:

1. **Impurity and Defect Scattering**: Electrons scatter elastically from stationary disruptions of lattice periodicity (chemical impurities, vacancies, interstitials, and grain boundaries). Because static defects do not freeze out at low temperatures, the impurity collision time τ_{imp} is independent of temperature, giving rise to a constant **residual resistivity** ρ_0:

$$\rho_0 = \frac{m_e}{ne^2\tau_{\text{imp}}} \tag{132}$$

2. **Thermal Phonon Scattering**: Electrons scatter inelastically from quantized lattice vibrations (phonons). At temperatures well above the Debye temperature ($T > \Theta_D$), the thermal phonon population is proportional to temperature ($n_{\text{ph}} \propto T$), leading to a scattering rate $\tau_{\text{ph}}^{-1} \propto T$ and a lattice resistivity $\rho_{\text{ph}}(T) \propto T$. At low temperatures ($T \ll \Theta_D$), small-angle scattering leads to the Bloch T^5 law ($\rho_{\text{ph}} \propto T^5$).

Because impurity and phonon scattering events are statistically independent, their scattering probabilities per unit time add linearly:

$$\boxed{\frac{1}{\tau(T)} = \frac{1}{\tau_{\text{imp}}} + \frac{1}{\tau_{\text{ph}}(T)}} \tag{133}$$

Multiplying by $\frac{m_e}{ne^2}$ yields **Matthiessen's Rule**:

$$\boxed{\rho(T) = \rho_0 + \rho_{\text{ph}}(T)} \tag{134}$$

Matthiessen's rule is widely utilized to quantify crystal purity via the **Residual Resistivity Ratio** (RRR $\equiv \rho(300\text{ K})/\rho(4.2\text{ K})$), and forms the physical operating principle of ultra-stable **Platinum Resistance Thermometers** (PRTs).

7.6 Quantum Thermal Conductivity and the Wiedemann–Franz Law

In metals, thermal energy is conducted predominantly by the mobile conduction electrons; the lattice phonon contribution to thermal conductivity is negligible by comparison ($K_{el} \gg K_{ph}$).

From kinetic gas theory, the thermal conductivity K of a gas of particles moving with characteristic speed v and mean free path $l = v\tau$ is:

$$K = \frac{1}{3} C_V v l = \frac{1}{3} C_{el} v_F^2 \tau \tag{135}$$

In the quantum Sommerfeld model, the carriers that transport thermal energy are the electrons near the Fermi boundary, which move at the Fermi velocity $v = v_F$.

Substituting the exact quantum electronic specific heat per unit volume $C_{el} = \frac{\pi^2}{2} n k_B \left(\frac{T}{T_F}\right) = \frac{\pi^2 n k_B^2 T}{m_e v_F^2}$ into equation (135):

$$K_{el} = \frac{1}{3}\left(\frac{\pi^2 n k_B^2 T}{m_e v_F^2}\right) v_F (v_F \tau) = \frac{\pi^2 n k_B^2 \tau T}{3 m_e} \tag{136}$$

Now, take the ratio of the electronic thermal conductivity K_{el} to the electrical conductivity $\sigma = \frac{ne^2\tau}{m_e}$:

$$\frac{K_{el}}{\sigma} = \frac{\frac{\pi^2 n k_B^2 \tau T}{3 m_e}}{\frac{ne^2\tau}{m_e}} = \frac{\pi^2 k_B^2}{3e^2} T \tag{137}$$

Dividing by absolute temperature T yields the universal **Wiedemann–Franz Law**:

$$\boxed{\frac{K}{\sigma T} = \frac{\pi^2}{3}\left(\frac{k_B}{e}\right)^2 \equiv L} \tag{138}$$

The theoretical **Sommerfeld Lorenz number** is:

$$L = \frac{\pi^2}{3}\left(\frac{1.3807 \times 10^{-23}\text{ J/K}}{1.6022 \times 10^{-19}\text{ C}}\right)^2 = \mathbf{2.443 \times 10^{-8}}\ \text{W}\,\Omega\,\text{K}^{-2} \tag{139}$$

Study Takeaway: The Universality of the Lorenz Number

The Sommerfeld Lorenz number $L = \frac{\pi^2}{3}(k_B/e)^2$ is a fundamental constant of nature. It contains:

- NO dependence on electron density n.
- NO dependence on electron effective mass m_e.

- NO dependence on the microscopic scattering relaxation time τ.

This explains why metals as diverse as copper, silver, gold, platinum, and lead all exhibit virtually identical Lorenz numbers at room temperature!

Metal	**Lorenz Number at** 0°C (10^{-8} W Ω K^{-2})	**Lorenz Number at** 100°C (10^{-8} W Ω K^{-2})
Silver (Ag)	2.31	2.37
Copper (Cu)	2.23	2.33
Gold (Au)	2.35	2.40
Platinum (Pt)	2.51	2.60
Lead (Pb)	2.47	2.56
Aluminum (Al)	2.14	2.19
Theoretical Value	$\mathbf{2.443 \times 10^{-8}}$ W Ω K^{-2}	

Table 4: Comparison of experimental Lorenz numbers for representative metals with the Sommerfeld quantum prediction.

Worked Example: Derivation of Electrical Conductivity from the Boltzmann Equation

Problem Statement: Starting from the steady-state Boltzmann Transport Equation $-\frac{e\mathbf{E}}{\hbar} \cdot \nabla_{\mathbf{k}} f = -\frac{f-f_0}{\tau}$, provide a complete mathematical derivation showing that the DC conductivity for an isotropic 3D parabolic band is $\sigma = \frac{ne^2\tau}{m_e}$.

Step-by-Step Derivation:

1. Linearization: Let $f(\mathbf{k}) = f_0(E) + f_1(\mathbf{k})$. Since f_0 depends on $\mathbf{k}$ through energy $E(\mathbf{k})$:

$$\nabla_{\mathbf{k}} f_0 = \frac{\partial f_0}{\partial E} \nabla_{\mathbf{k}} E = \hbar \mathbf{v} \frac{\partial f_0}{\partial E}$$

Substituting into the BTE gives:

$$-\frac{e\mathbf{E}}{\hbar} \cdot \left(\hbar \mathbf{v} \frac{\partial f_0}{\partial E} \right) = -\frac{f_1}{\tau} \implies f_1 = e\tau(\mathbf{v} \cdot \mathbf{E}) \left(-\frac{\partial f_0}{\partial E} \right)$$

2. Current Density Integral: Choosing $\mathbf{E} = E_x \hat{x}$, the current density along x is:

$$j_x = -2e \int \frac{d^3k}{(2\pi)^3} v_x [f_0(E) + f_1(\mathbf{k})] = -2e \int \frac{d^3k}{(2\pi)^3} v_x f_0(E) + 2e^2 \tau E_x \int \frac{d^3k}{(2\pi)^3} v_x^2 \left(-\frac{\partial f_0}{\partial E} \right)$$

The first integral vanishes because $v_x = \hbar k_x / m$ is odd in k_x while $f_0(E(k))$ is even.

3. Surface Integration: Transforming to spherical k-space coordinates: $d^3k = k^2 dk\, d\Omega = k^2 dk \sin\theta d\theta d\phi$. In spherical symmetry, $\langle v_x^2 \rangle = \frac{1}{3} v^2$:

$$j_x = \frac{2e^2 \tau E_x}{(2\pi)^3} \int_0^\infty k^2 dk \left(\frac{1}{3} v(k)^2 \right) \left(-\frac{\partial f_0}{\partial E} \right) \int d\Omega = \frac{2e^2 \tau E_x}{8\pi^3} (4\pi) \int_0^\infty \frac{1}{3} v(k)^2 k^2 dk \left(-\frac{\partial f_0}{\partial E} \right)$$

Using $E = \frac{\hbar^2 k^2}{2m} \implies dE = \frac{\hbar^2 k}{m} dk \implies k\, dk = \frac{m}{\hbar^2} dE$, and $v^2 = \frac{2E}{m}$:

$$k^2 v^2 dk = k(v^2)(k\, dk) = k \left(\frac{2E}{m} \right) \left(\frac{m}{\hbar^2} dE \right) = \frac{2kE}{\hbar^2} dE = \frac{2E}{\hbar^2} \left(\frac{\sqrt{2mE}}{\hbar} \right) dE = \frac{2\sqrt{2m}}{\hbar^3} E^{3/2} dE$$

As $T \to 0$, $-\frac{\partial f_0}{\partial E} \to \delta(E - E_F)$:

$$j_x = \frac{e^2 \tau E_x}{\pi^2} \frac{1}{3} \frac{2\sqrt{2m}}{\hbar^3} E_F^{3/2} = \frac{e^2 \tau E_x}{3\pi^2 m} \left(\frac{2mE_F}{\hbar^2} \right)^{3/2}$$

Recognizing from Chapter 2 that $k_F = \sqrt{2mE_F}/\hbar$ and $n = \frac{k_F^3}{3\pi^2} \implies 3\pi^2 n = \left(\frac{2mE_F}{\hbar^2}\right)^{3/2}$:

$$j_x = \frac{e^2 \tau E_x}{3\pi^2 m} (3\pi^2 n) = \left(\frac{ne^2\tau}{m} \right) E_x \implies \sigma = \boxed{\frac{\mathbf{ne^2\tau}}{\mathbf{m}}} \quad \textbf{Q.E.D.}$$

Worked Example: Thermal Conductivity of Copper and Wiedemann–Franz Verification

Problem Statement: At room temperature ($T = 300$ K), metallic copper has an electrical conductivity $\sigma = 5.90 \times 10^7\ \Omega^{-1}\text{m}^{-1}$.

(a) Using the quantum Wiedemann–Franz law with the theoretical Lorenz number $L = 2.443 \times 10^{-8}\ \text{W}\,\Omega\,\text{K}^{-2}$, calculate the electronic thermal conductivity K_{el} of copper.

(b) Compare this theoretical value with the experimental thermal conductivity of pure copper at 300 K ($K_{\text{exp}} \approx 401\ \text{W}/(\text{m}\cdot\text{K})$).

(c) What is the thermal conductivity predicted by Drude's classical model?

Step-by-Step Solution:

(a) Quantum Wiedemann–Franz thermal conductivity: From equation (138):

$$\begin{aligned} K_{el} &= L\sigma T = (2.443 \times 10^{-8}\ \text{W}\,\Omega\,\text{K}^{-2})(5.90 \times 10^7\ \Omega^{-1}\text{m}^{-1})(300\ \text{K}) \\ &= (2.443 \times 10^{-8})(1.770 \times 10^{10}) \approx \mathbf{432.4}\ \text{W}/(\text{m}\cdot\text{K}) \end{aligned}$$

(b) Comparison with experimental value: The measured thermal conductivity is $K_{\text{exp}} \approx 401\ \text{W}/(\text{m}\cdot\text{K})$. The theoretical prediction is within:

$$\frac{432.4 - 401}{401} \times 100\% \approx +\mathbf{7.8}\%$$

The small discrepancy arises from slight electron-phonon inelastic scattering at room temperature that relaxes thermal gradients slightly more effectively than electrical currents.

(c) Drude classical prediction: Using the classical Drude Lorenz number $L_{\text{Drude}} = \frac{3}{2}(k_B/e)^2 \approx 1.11 \times 10^{-8}\ \text{W}\,\Omega\,\text{K}^{-2}$:

$$K_{\text{Drude}} = L_{\text{Drude}}\sigma T = (1.11 \times 10^{-8})(1.770 \times 10^{10}) \approx \mathbf{196.5}\ \text{W}/(\text{m}\cdot\text{K})$$

Drude's classical calculation underestimates the experimental thermal conductivity by more than a factor of 2.

8 Thermoelectric Phenomena in Metals

8.1 Coupled Transport of Charge and Heat

In Chapters 1 and 5, electrical currents were driven by electric fields, and thermal currents were driven by temperature gradients. In real conductors, these two transport processes are fundamentally coupled: an electrical current inevitably carries thermal kinetic energy, and a thermal gradient inevitably drives a net diffusion of charge carriers.

The linear coupled transport equations connecting charge current density $\mathbf{j}$ and heat current density $\mathbf{j}_q$ to the generalized driving forces ($\mathbf{E}$ and ∇T) are:

$$\mathbf{j} = \sigma \mathbf{E} - \sigma S \nabla T \tag{140}$$

$$\mathbf{j}_q = \Pi \mathbf{j} - K_0 \nabla T \tag{141}$$

where σ is electrical conductivity, S is the Seebeck coefficient (thermopower), Π is the Peltier coefficient, and K_0 is the open-circuit thermal conductivity.

8.2 The Three Fundamental Thermoelectric Effects

Thermoelectric transport is characterized by three experimentally measurable phenomena:

8.2.1 1. The Seebeck Effect

Discovered by Thomas Johann Seebeck in 1821: when a temperature difference $\Delta T = T_{\text{hot}} - T_{\text{cold}}$ is maintained across an open-circuit conductor ($\mathbf{j} = \mathbf{0}$), mobile charge carriers diffuse from the hot end toward the cold end. This carrier migration establishes an internal electrostatic potential gradient (voltage difference) ΔV:

$$\mathbf{E} = S \nabla T \implies \Delta V = -\int_{T_1}^{T_2} S(T)\, dT \approx -S \Delta T \tag{142}$$

The proportionality constant $S \equiv -\frac{\Delta V}{\Delta T}$ is the **Seebeck coefficient** (or *thermopower*), measured in μV/K. In a closed thermocouple loop formed by two dissimilar conductors A and B with junctions held at temperatures T_1 and T_2, the net open-circuit electromotive force (EMF) is:

$$V_{\text{EMF}} = \int_{T_1}^{T_2} (S_B(T) - S_A(T))\, dT \approx (S_B - S_A)\Delta T \tag{143}$$

8.2.2 2. The Peltier Effect

Discovered by Jean Charles Athanase Peltier in 1834: when an external electric current I is driven across a junction between two dissimilar conductors A and B held at constant temperature, heat is reversibly absorbed or released at the junction at a rate $\dot{Q}$:

$$\dot{Q} = \Pi_{AB} I = (\Pi_B - \Pi_A) I \tag{144}$$

The proportionality constant Π is the **Peltier coefficient**, measured in Volts (V). Unlike irreversible Joule heating ($P_{\text{Joule}} = I^2 R \propto I^2$), the Peltier effect is strictly reversible: reversing the direction of current flips the junction from heating to cooling! This enables solid-state thermoelectric coolers and heat pumps.

8.2.3 3. The Thomson Effect

Predicted by William Thomson (Lord Kelvin) in 1851: when an electric current density $\mathbf{j}$ passes through a single homogeneous conductor along which a temperature gradient ∇T is maintained, heat is reversibly absorbed or liberated throughout the volume of the material at a rate per unit volume:

$$\dot{q}_{\text{Thomson}} = \mu_T\, \mathbf{j} \cdot \nabla T \tag{145}$$

where μ_T is the **Thomson coefficient** (measured in μV/K).

8.3 Kelvin (Onsager) Thermodynamic Relations

Using irreversible thermodynamics (and subsequently justified rigorously by Lars Onsager's reciprocal relations), Lord Kelvin proved that the three thermoelectric coefficients are interconnected by two exact thermodynamic laws:

$$\textbf{First Kelvin Relation:} \quad \boxed{\Pi = TS} \tag{146}$$

$$\textbf{Second Kelvin Relation:} \quad \boxed{\mu_T = T\frac{dS}{dT}} \tag{147}$$

Equation (146) is particularly profound: knowing the Seebeck coefficient S immediately yields the Peltier heat coefficient $\Pi = TS$ without needing a separate junction calculation!

8.4 Microscopic Derivation of the Peltier Heat Flux

To derive thermoelectric coefficients microscopically, consider an isothermal metal carrying an electric current density $\mathbf{j}_x$ along the x-axis.

Each moving electron carries an electric charge $-e$ and an energy relative to the chemical potential $(E_i - E_F)$. The total electric charge flux and thermal heat flux are:

$$j_x = -e\sum_i v_{x,i} = -2e\int \frac{d^3k}{(2\pi)^3} v_x f_1(\mathbf{k}) \tag{148}$$

$$j_{q,x} = \sum_i (E_i - E_F)v_{x,i} = 2\int \frac{d^3k}{(2\pi)^3}(E - E_F)v_x f_1(\mathbf{k}) \tag{149}$$

From equation (141), in the absence of a temperature gradient ($\nabla T = 0$), the heat flux is related to current density by $j_{q,x} = \Pi j_x$. Therefore, the Peltier coefficient is the microscopic ratio of heat flux to charge flux:

$$\Pi = \frac{j_{q,x}}{j_x} = -\frac{1}{e}\frac{\int \frac{d^3k}{(2\pi)^3}(E - E_F)v_x f_1(\mathbf{k})}{\int \frac{d^3k}{(2\pi)^3} v_x f_1(\mathbf{k})} \tag{150}$$

Substituting $f_1(\mathbf{k}) = e\tau E_x v_x \left(-\frac{\partial f_0}{\partial E}\right)$ and defining the **energy-dependent conductivity** $\sigma(E)$:

$$\sigma(E) \equiv \frac{1}{3}e^2 v(E)^2 \tau(E) D(E) \tag{151}$$

the integrals over iso-energetic surfaces in equation (150) become:

$$\Pi = -\frac{1}{e}\frac{\int_0^\infty (E - E_F)\sigma(E)\left(-\frac{\partial f_0}{\partial E}\right)dE}{\int_0^\infty \sigma(E)\left(-\frac{\partial f_0}{\partial E}\right)dE} \tag{152}$$

8.5 The Mott Formula for Thermopower

We now expand $\sigma(E)$ in a Taylor series about the Fermi energy E_F:

$$\sigma(E) \approx \sigma(E_F) + \left.\frac{d\sigma}{dE}\right|_{E_F}(E - E_F) + \ldots \tag{153}$$

1. The denominator of equation (152) is simply the macroscopic conductivity:

$$\int_0^\infty \sigma(E)\left(-\frac{\partial f_0}{\partial E}\right)dE \approx \sigma(E_F) \tag{154}$$

2. In the numerator:

$$\int_0^\infty (E-E_F)\sigma(E)\left(-\frac{\partial f_0}{\partial E}\right)dE \approx \sigma(E_F)\int_0^\infty (E-E_F)\left(-\frac{\partial f_0}{\partial E}\right)dE + \left.\frac{d\sigma}{dE}\right|_{E_F}\int_0^\infty (E-E_F)^2\left(-\frac{\partial f_0}{\partial E}\right)dE \tag{155}$$

The first integral vanishes by antisymmetry. The second integral was evaluated in equation (93) of Chapter 4:

$$\int_0^\infty (E-E_F)^2\left(-\frac{\partial f_0}{\partial E}\right)dE = (k_BT)^2\int_{-\infty}^\infty x^2\frac{e^x}{(e^x+1)^2}dx = \frac{\pi^2}{3}(k_BT)^2 \tag{156}$$

Substituting these results back into equation (152):

$$\Pi = -\frac{1}{e}\frac{\left.\frac{d\sigma}{dE}\right|_{E_F}\left[\frac{\pi^2}{3}(k_BT)^2\right]}{\sigma(E_F)} = -\frac{\pi^2k_B^2T^2}{3e}\left[\frac{1}{\sigma(E)}\frac{d\sigma(E)}{dE}\right]_{E_F} \tag{157}$$

Applying the First Kelvin Relation $\Pi = TS \implies S = \Pi/T$, we obtain the celebrated **Mott Formula for Thermopower**:

$$\boxed{S = -\frac{\pi^2k_B^2T}{3e}\left[\frac{d\ln\sigma(E)}{dE}\right]_{E_F}} \tag{158}$$

Evaluation for the 3D Free Electron Gas: For a 3D parabolic free electron band with energy-independent scattering time τ:

$$\sigma(E) \propto v^2(E)D(E) \propto E\cdot E^{1/2} = E^{3/2} \tag{159}$$

Taking the logarithmic derivative:

$$\frac{d\ln\sigma(E)}{dE} = \frac{d}{dE}\left(\frac{3}{2}\ln E\right) = \frac{3}{2E} \implies \left[\frac{d\ln\sigma}{dE}\right]_{E_F} = \frac{3}{2E_F} \tag{160}$$

Substituting equation (160) into the Mott formula (158):

$$S = -\frac{\pi^2k_B^2T}{3e}\left(\frac{3}{2E_F}\right) = -\frac{\pi^2k_B^2T}{2eE_F} \tag{161}$$

Expressing in terms of the Fermi temperature $T_F = E_F/k_B$:

$$\boxed{S \approx -\frac{\pi^2k_B}{2e}\left(\frac{T}{T_F}\right)} \tag{162}$$

Study Takeaway: Diagnostic Value of the Seebeck Coefficient

Equation (162) reveals the defining physical features of thermoelectricity in simple metals:

1. **Linear Temperature Dependence**: $S \propto T$ at low and room temperatures.
2. **Carrier Sign Sensitivity**: The factor $-1/e$ ensures that $S < 0$ for negatively charged electrons and $S > 0$ for positively charged holes! Measuring the sign of S provides an unambiguous test of carrier polarity.
3. **Extreme Quantum Suppression**: Because $k_B/e \approx 86.2\ \mu$V/K and $T/T_F \sim 10^{-2}$,

typical thermopowers in metals are tiny: $S \sim -1$–$10\ \mu$V/K (in stark contrast to semiconductors, where $S \sim 100$–$1000\ \mu$V/K).

Problem-Solving Strategy: Generalizing the Mott Formula to Real Band Structures

For real crystalline materials where the scattering relaxation time scales with energy as $\tau(E) \propto E^p$, and the density of states scales as $D(E) \propto E^q$:

1. Write the energy-dependent conductivity: $\sigma(E) \propto v^2(E)\tau(E)D(E) \propto E \cdot E^p \cdot E^q = E^{1+p+q}$.
2. Compute the logarithmic derivative: $\left[\frac{d\ln\sigma}{dE}\right]_{E_F} = \frac{1+p+q}{E_F}$.
3. Evaluate the thermopower:

$$S = -\frac{\pi^2 k_B^2 T}{3e}\left(\frac{1+p+q}{E_F}\right)$$

This formulation accounts for acoustic phonon scattering ($p = -1/2$) versus ionized impurity scattering ($p = +3/2$).

Worked Example: Fermi Velocity and Mean Free Path of Silver Conduction Electrons

Problem Statement: Metallic silver (Ag) has a Fermi energy $E_F = 5.49$ eV and a measured room-temperature electrical conductivity $\sigma = 6.21 \times 10^7\ \Omega^{-1}\text{m}^{-1}$. The conduction electron density is $n = 5.86 \times 10^{28}\ \text{m}^{-3}$.

(a) Calculate the quantum Fermi velocity v_F.

(b) Determine the momentum relaxation time τ.

(c) Compute the quantum mean free path $l = v_F\tau$ in nanometers and in Angstroms, and compare it with the lattice constant of silver ($a = 4.09$ Å).

Step-by-Step Solution:

(a) Fermi velocity v_F: First convert E_F to Joules:

$$E_F = (5.49\text{ eV})(1.6022 \times 10^{-19}\text{ J/eV}) \approx 8.796 \times 10^{-19}\text{ J}$$

$$v_F = \sqrt{\frac{2E_F}{m_e}} = \sqrt{\frac{2(8.796 \times 10^{-19}\text{ J})}{9.109 \times 10^{-31}\text{ kg}}} = \sqrt{1.931 \times 10^{12}} \approx \mathbf{1.39 \times 10^6}\text{ m/s}$$

(b) Relaxation time τ: From $\sigma = \frac{ne^2\tau}{m_e}$:

$$\tau = \frac{m_e\sigma}{ne^2} = \frac{(9.109 \times 10^{-31}\text{ kg})(6.21 \times 10^7\ \Omega^{-1}\text{m}^{-1})}{(5.86 \times 10^{28}\text{ m}^{-3})(1.6022 \times 10^{-19}\text{ C})^2}$$
$$= \frac{5.657 \times 10^{-23}}{5.86 \times 10^{28} \times 2.567 \times 10^{-38}} = \frac{5.657 \times 10^{-23}}{1.504 \times 10^{-9}} \approx \mathbf{3.76 \times 10^{-14}}\text{ s} \quad (\approx 38\text{ fs})$$

(c) Quantum mean free path l:

$$l = v_F\tau = (1.39 \times 10^6\text{ m/s})(3.76 \times 10^{-14}\text{ s}) \approx 5.23 \times 10^{-8}\text{ m}$$
$$= \mathbf{52.3}\text{ nm} = \mathbf{523}\text{ Å}$$

Comparison: Dividing by the silver lattice constant $a = 4.09$ Å:

$$\frac{l}{a} = \frac{523 \text{ Å}}{4.09 \text{ Å}} \approx \mathbf{128} \text{ lattice constants!}$$

Physical Discussion: In sharp contrast to Drude's classical prediction of $l \sim 3$ Å (scattering at every ion core), a conduction electron in room-temperature silver glides past **over 125 successive unit cells** without suffering a single collision! At liquid helium temperatures, this distance expands to over 10^6 Å (0.1 mm).

Worked Example: Mathematical Characterization of the 1D Displaced Fermi Sphere

Problem Statement: Consider a 1D electron gas at $T = 0$ K with equilibrium distribution $f_0(k) = \Theta(k_F - |k|)$. A steady electric field E_x is applied along $+x$.

(a) Derive the perturbed non-equilibrium distribution $f(k)$ in the relaxation time approximation.

(b) Sketch $f(k)$ and identify the rigid wavevector displacement Δk.

(c) Calculate the net number of uncompensated electrons per unit length carrying current.

Step-by-Step Solution:

(a) Perturbed distribution $f(k)$: Under steady electric field E_x, Newton's law with collisional damping gives the steady-state wavevector displacement:

$$\hbar \Delta k = -eE_x\tau \implies \Delta k = -\frac{eE_x\tau}{\hbar}$$

Each quantum state initially at k is shifted to $k + \Delta k$. The distribution is:

$$f(k) = f_0(k - \Delta k) = f_0\left(k + \frac{eE_x\tau}{\hbar}\right) = \Theta\left(k_F - \left|k + \frac{eE_x\tau}{\hbar}\right|\right)$$

Expanding in a Taylor series for small E_x:

$$f(k) \approx f_0(k) - \Delta k \frac{\partial f_0}{\partial k} = f_0(k) + \frac{eE_x\tau}{\hbar}\frac{\partial E}{\partial k}\left(-\frac{\partial f_0}{\partial E}\right) = f_0(k) + e\tau v(k)E_x\left(-\frac{\partial f_0}{\partial E}\right)$$

(b) Shifted Fermi surface boundary: The occupied interval shifts from $[-k_F, +k_F]$ to:

$$\left[-k_F - \frac{eE_x\tau}{\hbar},\ +k_F - \frac{eE_x\tau}{\hbar}\right]$$

States between $+k_F - \frac{eE_x\tau}{\hbar}$ and $+k_F$ are vacated, while states between $-k_F - \frac{eE_x\tau}{\hbar}$ and $-k_F$ are newly populated.

(c) Uncompensated electron density: The length of the uncompensated wavevector interval in k-space is $2|\Delta k| = 2\frac{eE_x\tau}{\hbar}$. The density of states in 1D k-space with spin is $2 \times \frac{L}{2\pi} = \frac{L}{\pi}$. The uncompensated linear carrier density is:

$$\Delta n = \frac{1}{L} \times \left(\frac{L}{\pi}|\Delta k|\right) = \frac{|\Delta k|}{\pi} = \frac{eE_x\tau}{\pi\hbar}$$

Each of these uncompensated electrons moves with velocity $-v_F = -\hbar k_F/m_e$. The resulting current is:

$$j = -e\Delta n(-v_F) = e\left(\frac{eE_x\tau}{\pi\hbar}\right)\left(\frac{\hbar k_F}{m_e}\right) = \frac{e^2\tau k_F}{\pi m_e}E_x$$

Since in 1D $n = 2k_F/\pi \implies k_F = \frac{\pi n}{2}$:

$$j = \left(\frac{ne^2\tau}{m_e}\right) E_x \implies \sigma_{1\mathrm{D}} = \frac{ne^2\tau}{m_e}$$

confirming the universal Drude conductivity form in one dimension.

9 Limitations of Free Electron Theory and Transition to Band Theory

9.1 Scorecard of the Free Electron Model

The Sommerfeld quantum free electron model represents one of the most remarkable triumphs of early solid-state physics. By treating conduction electrons as a quantum gas obeying Fermi–Dirac statistics, Sommerfeld achieved striking quantitative successes. Yet, as experimental techniques advanced, equally striking contradictions emerged.

9.1.1 Six Major Successes of the Free Electron Model

1. **Electrical and Thermal Conduction**: Naturally explains high electrical ($\sigma \sim 10^7\ \Omega^{-1}\mathrm{m}^{-1}$) and thermal ($K \sim 100$–$400\ \mathrm{W/(m \cdot K)}$) conductivities.

2. **Wiedemann–Franz Law**: Correctly derives the universal Sommerfeld Lorenz number $L = \frac{\pi^2}{3}(k_B/e)^2 \approx 2.44 \times 10^{-8}\ \mathrm{W\,\Omega\,K^{-2}}$, resolving Drude's factor-of-two error.

3. **Electronic Specific Heat**: Resolves the classical Dulong–Petit failure by proving that $C_{el} = \gamma T$ is suppressed by $T/T_F \sim 10^{-2}$.

4. **Pauli Paramagnetism**: Accurately accounts for the weak, temperature-independent spin susceptibility $\chi = \mu_0 \mu_B^2 g(E_F)$.

5. **Realistic Fermi Scales**: Correctly predicts that conduction electron densities $n \sim 10^{28}$–$10^{29}\ \mathrm{m}^{-3}$ lead to Fermi energies of several electron-volts and Fermi velocities of order 10^6 m/s.

6. **Thermionic and Photoelectric Emission**: Successfully describes electron emission over the surface work function barrier Φ.

9.1.2 Six Critical Failures Demanding a Periodic Potential

1. **The Conductor / Semiconductor / Insulator Distinction**: The free electron model assumes that all valence electrons delocalize into conduction electrons. It completely fails to explain why diamond (carbon, 4 valence electrons) is a transparent insulator ($E_g \approx 5.5$ eV), silicon (4 valence electrons) is a semiconductor ($E_g \approx 1.1$ eV), and lead (4 valence electrons) is a brilliant metal!

2. **Positive Hall Coefficients**: As derived in equation (26), free electron theory strictly demands $R_H = -1/(ne) < 0$. Experimentally, metals such as zinc (Zn), cadmium (Cd), beryllium (Be), and aluminum (Al) exhibit **positive Hall coefficients** ($R_H > 0$), as if current were transported by positive charges!

3. **Divalent Metal Conduction**: Divalent metals (e.g., alkaline earths Be, Mg, Ca, Ba) have two valence electrons per atom. As shown in Chapter 8, a completely filled band accommodates exactly two electrons per unit cell. Free electron theory cannot explain why divalent elements are good conductors rather than band insulators.

4. **Temperature Dependence of Resistivity**: Drude and Sommerfeld predict $\rho \propto T$ at high temperatures, but fail at low temperatures where high-purity metals follow the **Bloch T^5 law** ($\rho(T) \propto T^5$) due to electron-phonon scattering geometry.

5. **Magnetoresistance Anomalies**: Semiclassical transport in an isotropic single-band model predicts that transverse magnetoresistance vanishes identically ($\Delta\rho/\rho = 0$). In real metals, magnetoresistance is large, highly anisotropic, and often does not saturate even in extreme magnetic fields ($B > 30$ T).

6. **Anisotropic and Non-Spherical Fermi Surfaces**: Free electron theory mandates that the Fermi surface is an exact, isotropic sphere ($E = \hbar^2 k^2/2m_e$). Experimental probes (such as the de Haas–van Alphen effect, Chapter 10) reveal that real Fermi surfaces possess intricate necks, bellies, rosettes, and open orbits that reflect the underlying crystalline lattice symmetry.

9.2 The Physical Necessity of a Periodic Lattice Potential

The fundamental deficiency of the free electron model is its total neglect of the spatial periodicity of the crystal. In a real solid, the positive ion cores are not smeared into a uniform background; they form an ultra-precise, periodic Bravais lattice with translation vectors:

$$\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3, \quad n_1, n_2, n_3 \in \mathbb{Z} \tag{163}$$

Consequently, the potential energy $V(\mathbf{r})$ experienced by a conduction electron is strictly lattice-periodic:

$$\boxed{V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})} \tag{164}$$

Near each nucleus, $V(\mathbf{r}) \sim -Ze^2/(4\pi\epsilon_0 r)$ possesses deep, Coulombic potential wells. The interaction of electron de Broglie waves with this periodic array produces **constructive and destructive Bragg diffraction**, completely restructuring the continuous free-electron parabola into discrete, energy-separated **energy bands**.

9.3 Intuitive Model of Band Formation: Coupled Oscillators

To understand intuitively how a periodic lattice creates energy bands, consider the quantum analogy of coupled identical harmonic oscillators:

- **Single Isolated Atom**: An isolated atom has discrete, sharp quantum energy levels ($1s, 2s, 2p, \ldots$).
- **Two Interacting Atoms** (Diatomic Molecule): When two identical atoms are brought close together, their valence wavefunctions overlap. The single atomic level splits into two distinct molecular states: a symmetric bonding orbital (lower energy) and an antisymmetric antibonding orbital (higher energy).
- N_c **Interacting Atoms** (Macroscopic Crystal): When $N_c \sim 10^{23}$ identical atoms condense into a crystal lattice, each atomic energy level splits into N_c distinct, closely spaced quantum states.

Because N_c is macroscopic, the energy spacing between adjacent states ($\Delta E \sim$ few eV$/10^{23} \sim 10^{-22}$ eV) is infinitesimal. The N_c states merge into a continuous band of allowed energy levels, separated from adjacent bands by **forbidden energy band gaps** (E_g).

Conceptual Checkpoint: From Coupled Oscillators to Energy Bands

Consider two identical pendula of frequency ω_0 coupled by a light spring of stiffness κ. When uncoupled ($\kappa = 0$), both oscillate independently at ω_0. When coupled ($\kappa > 0$), the degenerate frequency splits into two normal modes:

- Symmetric mode: both pendula swing in phase ($\omega_1 = \omega_0$, spring unstretched).
- Antisymmetric mode: pendula swing in opposition ($\omega_2 = \sqrt{\omega_0^2 + 2\kappa/m} > \omega_0$, spring compressed/extended).

For a chain of N coupled pendula, the single frequency ω_0 broadens into a continuous band of N normal mode frequencies spanning $[\omega_{\min}, \omega_{\max}]$. In exact mathematical equivalence, N overlapping atomic orbitals in a crystal broaden into an energy band spanning a bandwidth $W \sim$ 2–10 eV!

Common Misconception & Pitfall: Free Electron Theory is an Approximation, Not an Error

It is crucial not to view the free electron model as "incorrect." Rather, it is the zero-order baseline upon which modern solid-state physics is constructed.
For monovalent alkali metals (Li, Na, K, Rb, Cs), the single valence electron fills only half of the first Brillouin zone. The Fermi sphere is small ($k_F \approx 0.78(\pi/a)$) and never comes into contact with the zone boundaries where Bragg reflection occurs. In these metals, conduction electrons behave almost exactly like pure Sommerfeld free electrons! Band theory becomes essential precisely when the Fermi sphere expands to touch or cross the Brillouin zone boundaries.

Worked Example: Hall Coefficient and Carrier Type Anomalies in Divalent Metals

Problem Statement: Beryllium (Be, $Z = 4$) and Zinc (Zn, $Z = 30$) are divalent metals (two valence electrons per atom). Free electron theory predicts that: (1) Their Hall coefficients should be negative: $R_H = -1/(ne) < 0$. (2) Their electron concentrations should be $n = 2N_{\text{atoms}}/V$. Yet, high-purity zinc crystals exhibit a positive Hall coefficient ($R_H \approx +3.3 \times 10^{-11}$ m^3/C) at room temperature! Explain qualitatively how the presence of a periodic lattice potential resolves both the conductive nature of divalent metals and the origin of positive Hall coefficients.

Step-by-Step Resolution:

1. The Divalent Insulator Paradox: In a crystal with N_c primitive unit cells, each energy band has a capacity of exactly $2N_c$ electrons (including spin). Because divalent atoms contribute 2 electrons per cell, $2N_c$ valence electrons are available. If the periodic potential opened a complete energy gap across the entire Brillouin zone, the first band would be 100% full and the second band 100% empty, making all divalent elements insulators!

2. Resolution via Band Overlap: In three dimensions, the energy gap varies with wavevector direction $\hat{\mathbf{k}}$. In divalent metals like Be and Zn, the lattice potential is sufficiently weak that the minimum energy of the second band (the conduction band bottom, occurring at zone corners) drops below the maximum energy of the first band (the valence band top, occurring at zone faces):

$$E_{2,\min} < E_{1,\max}$$

Electrons spill over from the top of the first band into the bottom of the second band. As a result:

- The first band is slightly unfilled, leaving **vacant states (hole pockets)** near the band top.
- The second band is slightly populated, creating **electron pockets** near the band bottom.

Both bands are partially filled, guaranteeing metallic conductivity! Metals with this electronic structure are known as **semimetals**.

3. Origin of the Positive Hall Coefficient ($R_H > 0$): In a two-band system with electrons (density n_e, mobility μ_e) and holes (density n_h, mobility μ_h), the low-field Hall coefficient is:

$$R_H = \frac{n_h\mu_h^2 - n_e\mu_e^2}{e(n_e\mu_e + n_h\mu_h)^2}$$

In zinc, the curvature of the valence band top is very sharp, giving hole-like carriers a small effective mass and an exceptionally high mobility ($\mu_h \gg \mu_e$). Consequently, the positive

term $n_h\mu_h^2$ overwhelms the negative electron term, resulting in an overall **positive Hall coefficient** ($R_H > 0$).

10 Electrons in a Periodic Potential and Bloch's Theorem

10.1 The Periodic Crystal Potential and Translation Operators

In a crystalline solid, the ions arrange themselves into a regular Bravais lattice characterized by primitive translation vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. Any lattice site can be expressed as:

$$\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3, \quad n_1, n_2, n_3 \in \mathbb{Z} \tag{165}$$

Because the arrangement of atomic nuclei is invariant under translation by any lattice vector $\mathbf{R}$, the single-electron potential energy $V(\mathbf{r})$ is strictly lattice-periodic:

$$V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r}) \tag{166}$$

The single-particle Hamiltonian for an electron moving in this periodic potential is:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla^2 + V(\mathbf{r}) \tag{167}$$

We define the **lattice translation operator** $\hat{T}_{\mathbf{R}}$ as the quantum operator that translates any spatial coordinate by $\mathbf{R}$:

$$\hat{T}_{\mathbf{R}}\psi(\mathbf{r}) \equiv \psi(\mathbf{r} + \mathbf{R}) \tag{168}$$

Acting with $\hat{T}_{\mathbf{R}}$ on the Hamiltonian:

$$\hat{T}_{\mathbf{R}}\hat{H}\psi(\mathbf{r}) = \hat{H}(\mathbf{r} + \mathbf{R})\psi(\mathbf{r} + \mathbf{R}) = \left[-\frac{\hbar^2}{2m_e}\nabla^2 + V(\mathbf{r} + \mathbf{R})\right]\psi(\mathbf{r} + \mathbf{R}) = \hat{H}\hat{T}_{\mathbf{R}}\psi(\mathbf{r}) \tag{169}$$

Because this holds for any state $\psi(\mathbf{r})$, the translation operator commutes with the crystal Hamiltonian:

$$\boxed{[\hat{H}, \hat{T}_{\mathbf{R}}] = 0 \quad \text{for all lattice vectors } \mathbf{R}} \tag{170}$$

Furthermore, any two translation operators commute with each other:

$$\hat{T}_{\mathbf{R}}\hat{T}_{\mathbf{R}'} = \hat{T}_{\mathbf{R}'}\hat{T}_{\mathbf{R}} = \hat{T}_{\mathbf{R}+\mathbf{R}'} \tag{171}$$

10.2 Rigorous Proof of Bloch's Theorem

From fundamental quantum mechanics, because $\hat{H}$ and the family of translation operators $\{\hat{T}_{\mathbf{R}}\}$ are commuting normal operators, they possess a complete set of **simultaneous eigenstates**.

Let $\psi(\mathbf{r})$ be an energy eigenstate with eigenvalue E. Because $\psi(\mathbf{r})$ is simultaneously an eigenstate of $\hat{T}_{\mathbf{R}}$, it must satisfy:

$$\hat{T}_{\mathbf{R}}\psi(\mathbf{r}) = \psi(\mathbf{r} + \mathbf{R}) = C(\mathbf{R})\psi(\mathbf{r}) \tag{172}$$

where $C(\mathbf{R})$ is the eigenvalue of the translation operator.

Determination of the Eigenvalues $C(\mathbf{R})$: Applying the composition property (171) to an eigenstate $\psi(\mathbf{r})$:

$$\hat{T}_{\mathbf{R}}\hat{T}_{\mathbf{R}'}\psi(\mathbf{r}) = C(\mathbf{R})C(\mathbf{R}')\psi(\mathbf{r}) = \hat{T}_{\mathbf{R}+\mathbf{R}'}\psi(\mathbf{r}) = C(\mathbf{R} + \mathbf{R}')\psi(\mathbf{r}) \tag{173}$$

Therefore, the eigenvalues must satisfy the multiplicative group property:

$$C(\mathbf{R})C(\mathbf{R}') = C(\mathbf{R} + \mathbf{R}') \tag{174}$$

The most general mathematical function that maps addition to multiplication is an exponential:

$$C(\mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{R}} \tag{175}$$

where $\mathbf{k}$ is a wavevector.

To ensure that the electron wavefunction remains bounded and normalizable throughout an infinite crystal, the exponent must be purely imaginary ($|C(\mathbf{R})| = 1$), which requires $\mathbf{k}$ to be a purely real vector.

Substituting equation (175) back into equation (172) yields the first statement of **Bloch's Theorem**:

$$\boxed{\psi_{\mathbf{k}}(\mathbf{r}+\mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{R}}\psi_{\mathbf{k}}(\mathbf{r})} \tag{176}$$

Equivalent Formulation (Modulated Plane Wave): Now define an auxiliary function $u_{\mathbf{k}}(\mathbf{r}) \equiv e^{-i\mathbf{k}\cdot\mathbf{r}}\psi_{\mathbf{k}}(\mathbf{r})$, so that $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{\mathbf{k}}(\mathbf{r})$. Testing the periodicity of $u_{\mathbf{k}}(\mathbf{r})$ under lattice translation:

$$u_{\mathbf{k}}(\mathbf{r}+\mathbf{R}) = e^{-i\mathbf{k}\cdot(\mathbf{r}+\mathbf{R})}\psi_{\mathbf{k}}(\mathbf{r}+\mathbf{R}) = e^{-i\mathbf{k}\cdot\mathbf{r}}e^{-i\mathbf{k}\cdot\mathbf{R}}\left[e^{i\mathbf{k}\cdot\mathbf{R}}\psi_{\mathbf{k}}(\mathbf{r})\right] = e^{-i\mathbf{k}\cdot\mathbf{r}}\psi_{\mathbf{k}}(\mathbf{r}) = u_{\mathbf{k}}(\mathbf{r}) \tag{177}$$

We arrive at the standard alternative statement of **Bloch's Theorem**:

$$\boxed{\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{k}}(\mathbf{r}), \quad \text{where} \quad u_{n\mathbf{k}}(\mathbf{r}+\mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r})} \tag{178}$$

where n is a discrete **band index**.

Physical Meaning: An electron moving through a perfectly periodic potential is described by a plane wave $e^{i\mathbf{k}\cdot\mathbf{r}}$ whose amplitude is periodically modulated by a function $u_{n\mathbf{k}}(\mathbf{r})$ that possesses the exact spatial periodicity of the crystal lattice!

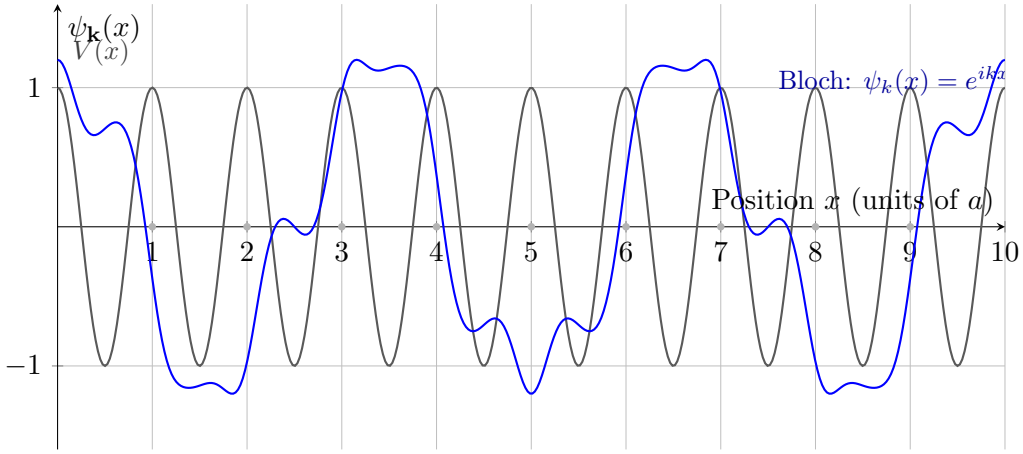


Figure 8: Real part of a 1D Bloch wave (blue) in a weak periodic potential $V(x)$ (gray). The wave is a plane wave e^{ikx} modulated by a lattice-periodic function $u_k(x)$ with the same period a as the potential. This is the general form of eigenstates in a crystal.

10.3 Physical Nature of Crystal Momentum $\hbar\mathbf{k}$

In free space, an electron with wavefunction $\psi = e^{i\mathbf{k}\cdot\mathbf{r}}$ has a definite kinematic momentum $\hat{\mathbf{p}}\psi = -i\hbar\nabla\psi = \hbar\mathbf{k}\psi$.

In a crystalline potential, however, a Bloch wave is NOT an eigenstate of true momentum $\hat{\mathbf{p}}$:

$$-i\hbar\nabla\psi_{n\mathbf{k}}(\mathbf{r}) = -i\hbar\nabla\left[e^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{k}}(\mathbf{r})\right] = \hbar\mathbf{k}\psi_{n\mathbf{k}}(\mathbf{r}) + e^{i\mathbf{k}\cdot\mathbf{r}}(-i\hbar\nabla u_{n\mathbf{k}}(\mathbf{r})) \neq \text{const} \times \psi_{n\mathbf{k}}(\mathbf{r}) \tag{179}$$

The electron continuously exchanges linear momentum with the periodic array of ion cores. The vector $\hbar\mathbf{k}$ is called the **crystal momentum**. It acts as an effective quantum number governing the translational symmetry of the wavepacket.

Conservation Modulo Reciprocal Lattice Vectors: Let $\mathbf{G}$ be any reciprocal lattice vector satisfying $e^{i\mathbf{G}\cdot\mathbf{R}} = 1$ for all direct lattice vectors $\mathbf{R}$. Under lattice translation:

$$\psi_{\mathbf{k}+\mathbf{G}}(\mathbf{r}+\mathbf{R}) = e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{R}}\psi_{\mathbf{k}+\mathbf{G}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{R}}e^{i\mathbf{G}\cdot\mathbf{R}}\psi_{\mathbf{k}+\mathbf{G}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{R}}\psi_{\mathbf{k}+\mathbf{G}}(\mathbf{r}) \tag{180}$$

The phase factor $e^{i\mathbf{k}\cdot\mathbf{R}}$ is completely unchanged if $\mathbf{k}$ is replaced by $\mathbf{k}+\mathbf{G}$!

Therefore, wavevectors that differ by a reciprocal lattice vector are physically equivalent:

$$\boxed{\mathbf{k} \equiv \mathbf{k}+\mathbf{G}} \tag{181}$$

In all quantum scattering events in crystals (such as electron–phonon or electron–photon collisions), total crystal momentum is conserved **modulo G**:

$$\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4 + \mathbf{G} \tag{182}$$

When $\mathbf{G} = \mathbf{0}$, the event is a *Normal process* (N-process). When $\mathbf{G} \neq \mathbf{0}$, the event is an *Umklapp process* (U-process), where the crystal lattice as a whole recoils by momentum $\hbar\mathbf{G}$.

10.4 Born–von Karman Boundary Conditions and State Counting

To enumerate allowed $\mathbf{k}$-states, we apply Born–von Karman periodic boundary conditions across a macroscopic crystal containing $N_1 \times N_2 \times N_3 = N_c$ unit cells:

$$\psi(\mathbf{r}+N_i\mathbf{a}_i) = \psi(\mathbf{r}) \implies e^{iN_i\mathbf{k}\cdot\mathbf{a}_i} = 1 \implies \mathbf{k}\cdot\mathbf{a}_i = \frac{2\pi m_i}{N_i}, \quad m_i \in \mathbb{Z} \tag{183}$$

Expanding $\mathbf{k}$ in the primitive reciprocal basis vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ (where $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$):

$$\mathbf{k} = \frac{m_1}{N_1}\mathbf{b}_1 + \frac{m_2}{N_2}\mathbf{b}_2 + \frac{m_3}{N_3}\mathbf{b}_3 \tag{184}$$

The volume occupied by a single $\mathbf{k}$-state in reciprocal space is:

$$\Delta V_k = \frac{\mathbf{b}_1 \cdot (\mathbf{b}_2 \times \mathbf{b}_3)}{N_1 N_2 N_3} = \frac{V_{\text{BZ}}}{N_c} \tag{185}$$

where $V_{\text{BZ}} = \frac{(2\pi)^3}{V_{\text{cell}}}$ is the total volume of the **first Brillouin zone**.

Consequently, the total number of distinct allowed wavevector states within the first Brillouin zone is:

$$\frac{V_{\text{BZ}}}{\Delta V_k} = N_c \tag{186}$$

Each orbital state accommodates two electrons of opposite spin ($g_s = 2$). We arrive at a central theorem of solid-state physics:

$$\boxed{\text{Each non-degenerate energy band can accommodate at most } 2N_c \text{ electrons}} \tag{187}$$

10.5 The Three Zone Schemes

Because $E_n(\mathbf{k}+\mathbf{G}) = E_n(\mathbf{k})$, the band dispersion $E(\mathbf{k})$ can be graphically represented in three standard schemes:

1. **Extended Zone Scheme**: Different energy bands are plotted in different Brillouin zones. Band 1 is plotted in the 1st BZ ($|k| \leq \pi/a$), Band 2 in the 2nd BZ ($\pi/a < |k| \leq 2\pi/a$), Band 3 in the 3rd BZ, and so on. In the limit of zero crystal potential ($V \to 0$), the dispersion forms a continuous free-electron parabola $E = \hbar^2k^2/2m_e$.

2. **Reduced Zone Scheme**: All energy bands are mapped back into the first Brillouin zone ($-\pi/a \leq k \leq \pi/a$) by translating wavevectors by appropriate reciprocal lattice vectors $\mathbf{G}$. In this scheme, the energy dispersion is multi-valued: for each $\mathbf{k}$ in the 1st BZ, there exists an infinite ladder of discrete energy eigenvalues $E_n(\mathbf{k})$ labeled by band index $n = 1, 2, 3, \ldots$.

3. **Periodic (Repeated) Zone Scheme**: The energy bands in the first Brillouin zone are repeated periodically throughout all reciprocal space by adding **G** in all directions. This scheme is particularly useful for visualizing electron orbits and cyclotron trajectories across multiple zones in magnetic fields.

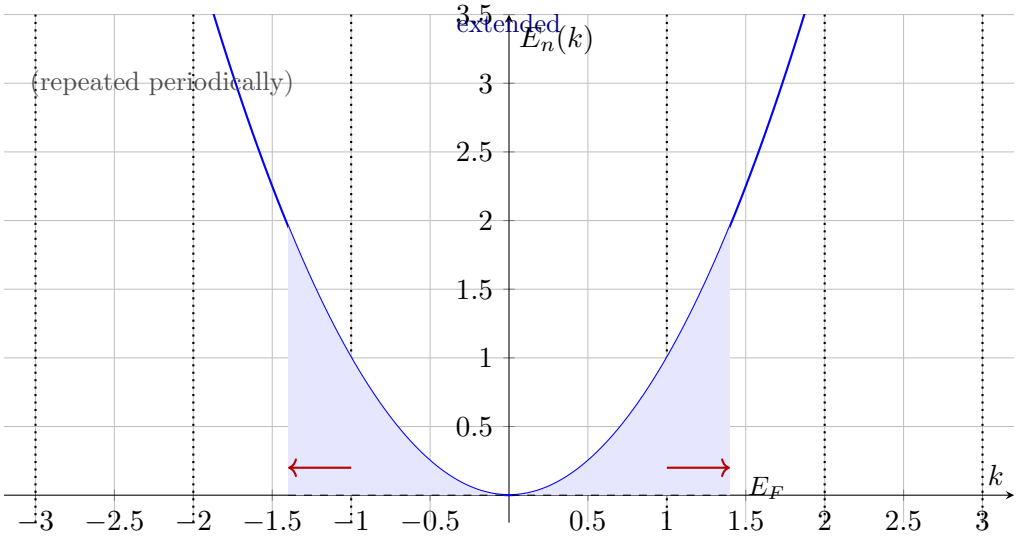


Figure 9: Extended-zone scheme: the free-electron parabola $E = \hbar^2 k^2/2m_e$ is plotted for all k. Allowed k values from Born–von Karman BCs, the first Brillouin zone, and the filled states up to E_F are shown. The same dispersion can be folded into the first BZ (reduced zone) or copied periodically (repeated zone).

Conceptual Checkpoint: Why Crystal Momentum $\hbar$k is Not True Momentum

True momentum is a continuous translation generator; crystal momentum is a discrete translation generator.
When an external force $\mathbf{F}_{\text{ext}}$ acts on a Bloch electron, the semiclassical equation of motion is $\hbar\frac{d\mathbf{k}}{dt} = \mathbf{F}_{\text{ext}}$. Although this equation mirrors Newton's second law, $\hbar\mathbf{k}$ is NOT the mechanical momentum of the electron. The actual physical momentum of the electron is continuously modulated by the periodic forces exerted by the lattice ions. When an electron is accelerated to a Brillouin zone boundary $k = \pi/a$, it undergoes Bragg reflection back to $-\pi/a$, transferring an impulse of $\hbar G = 2\pi\hbar/a$ directly to the entire crystal lattice.

Study Takeaway: State Counting in Brillouin Zones

The fact that each energy band accommodates exactly $2N_c$ electrons (with spin) leads directly to the fundamental classification of solids:

- **Odd Valency**: If a crystal has an *odd* number of valence electrons per primitive unit cell (e.g., alkali metals Li, Na, K with 1, or noble metals Cu, Ag, Au with 1), the highest occupied band can be at most half-filled. The material **must be a metal**!
- **Even Valency**: If a crystal has an *even* number of valence electrons per unit cell (e.g., diamond with 8 electrons per primitive cell), bands can be completely filled. If a finite energy gap E_g separates the highest filled band (valence band) from the lowest empty band (conduction band), the material is an **insulator** or **semiconductor**.

Worked Example: Free Electron Plane Wave as a Trivial Bloch State

Problem Statement: Show that a standard free-electron plane wave $\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}}$ satisfies Bloch's theorem for an arbitrary periodic lattice with translation vector $\mathbf{R}$. Identify the cell-periodic function $u_{\mathbf{k}}(\mathbf{r})$ and verify that it possesses the required lattice periodicity.

Step-by-Step Solution:

1. Verification of the First Bloch Formulation: Evaluate $\psi_{\mathbf{k}}(\mathbf{r}+\mathbf{R})$:

$$\psi_{\mathbf{k}}(\mathbf{r}+\mathbf{R}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot(\mathbf{r}+\mathbf{R})} = e^{i\mathbf{k}\cdot\mathbf{R}} \left[\frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}}\right] = e^{i\mathbf{k}\cdot\mathbf{R}} \psi_{\mathbf{k}}(\mathbf{r})$$

This satisfies equation (176) identically!

2. Identification of $u_{\mathbf{k}}(\mathbf{r})$: Comparing $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r})$ with the plane wave:

$$e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}} \implies u_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{V}} = \text{constant}$$

3. Verification of Cell Periodicity: Since $u_{\mathbf{k}}(\mathbf{r})$ is a spatially uniform constant:

$$u_{\mathbf{k}}(\mathbf{r}+\mathbf{R}) = \frac{1}{\sqrt{V}} = u_{\mathbf{k}}(\mathbf{r})$$

A spatially constant function trivially satisfies the lattice periodicity requirement $u(\mathbf{r}+\mathbf{R}) = u(\mathbf{r})$ for any arbitrary lattice vector $\mathbf{R}$. Thus, free electrons are merely the special limiting case of Bloch electrons in a vanishing crystal potential ($V(\mathbf{r}) \to 0$).

11 Nearly Free Electron Model and Energy Gaps

11.1 Perturbation Theory for a Weak Periodic Potential

In the **Nearly Free Electron** (NFE) model, we treat the lattice-periodic potential $V(\mathbf{r})$ as a weak perturbation on an otherwise free electron gas:

$$\hat{H} = \hat{H}_0 + V(\mathbf{r}) = -\frac{\hbar^2}{2m_e}\nabla^2 + V(\mathbf{r}) \tag{188}$$

Because $V(\mathbf{r}+\mathbf{R}) = V(\mathbf{r})$, it can be expanded in a three-dimensional complex Fourier series over all reciprocal lattice vectors $\mathbf{G}$:

$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}} \tag{189}$$

where the Fourier coefficients $V_{\mathbf{G}}$ are given by the unit cell integral:

$$V_{\mathbf{G}} = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} V(\mathbf{r}) e^{-i\mathbf{G}\cdot\mathbf{r}}\, d^3r \tag{190}$$

Because $V(\mathbf{r})$ is real ($V^*(\mathbf{r}) = V(\mathbf{r})$), the Fourier amplitudes satisfy $V_{-\mathbf{G}} = V_{\mathbf{G}}^*$. We can choose the arbitrary reference zero of potential energy such that the spatial average vanishes: $V_0 = 0$.

11.2 Matrix Elements and Selection Rules

Using normalized free-electron plane waves $|\mathbf{k}\rangle = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}}$ as the unperturbed basis:

$$\begin{aligned}\langle \mathbf{k}'|V|\mathbf{k}\rangle &= \frac{1}{V}\int_V e^{-i\mathbf{k}'\cdot\mathbf{r}} \left[\sum_{\mathbf{G}} V_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}\right] e^{i\mathbf{k}\cdot\mathbf{r}}\, d^3r \\ &= \sum_{\mathbf{G}} V_{\mathbf{G}} \left[\frac{1}{V}\int_V e^{i(\mathbf{k}-\mathbf{k}'+\mathbf{G})\cdot\mathbf{r}}\, d^3r\right] = \sum_{\mathbf{G}} V_{\mathbf{G}}\,\delta_{\mathbf{k}',\mathbf{k}+\mathbf{G}}\end{aligned} \tag{191}$$

This is a powerful selection rule: **the periodic potential couples only plane-wave states whose wavevectors differ by a reciprocal lattice vector G**.

11.3 Non-Degenerate Perturbation Theory and its Breakdown

In standard Rayleigh–Schrödinger non-degenerate perturbation theory, the energy of state $|\mathbf{k}\rangle$ to second order is:

$$E(\mathbf{k}) = E^{(0)}(\mathbf{k}) + \langle \mathbf{k}|V|\mathbf{k}\rangle + \sum_{\mathbf{k}'\neq\mathbf{k}} \frac{|\langle \mathbf{k}'|V|\mathbf{k}\rangle|^2}{E^{(0)}(\mathbf{k}) - E^{(0)}(\mathbf{k}')} \tag{192}$$

Applying the selection rule (191) and setting $\langle \mathbf{k}|V|\mathbf{k}\rangle = V_0 = 0$:

$$E(\mathbf{k}) = E^{(0)}(\mathbf{k}) + \sum_{\mathbf{G}\neq\mathbf{0}} \frac{|V_{\mathbf{G}}|^2}{E^{(0)}(\mathbf{k}) - E^{(0)}(\mathbf{k}+\mathbf{G})} \tag{193}$$

where $E^{(0)}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m_e}$.

Breakdown at Zone Boundaries: Equation (193) is valid as long as the denominator is large: $|E^{(0)}(\mathbf{k}) - E^{(0)}(\mathbf{k}+\mathbf{G})| \gg |V_{\mathbf{G}}|$.

However, whenever the unperturbed energies of $|\mathbf{k}\rangle$ and $|\mathbf{k}+\mathbf{G}\rangle$ become equal:

$$E^{(0)}(\mathbf{k}) = E^{(0)}(\mathbf{k}+\mathbf{G}) \iff k^2 = |\mathbf{k}+\mathbf{G}|^2 = k^2 + 2\mathbf{k}\cdot\mathbf{G} + G^2 \tag{194}$$

the denominator vanishes ($E^{(0)}(\mathbf{k}) - E^{(0)}(\mathbf{k}+\mathbf{G}) = 0$) and the energy correction diverges to infinity!

Canceling k^2 gives the geometric condition for degeneracy:

$$2\mathbf{k}\cdot\mathbf{G}+G^2=0 \iff \boxed{\mathbf{k}\cdot\left(\frac{1}{2}\mathbf{G}\right)=-\frac{1}{4}G^2 \iff \mathbf{G}\cdot\left(\mathbf{k}+\frac{1}{2}\mathbf{G}\right)=0} \tag{195}$$

Equation (195) is the exact mathematical definition of the **Brillouin zone boundary**! It is precisely the Laue condition for the constructive Bragg reflection of an electron de Broglie wave by the crystal planes perpendicular to $\mathbf{G}$.

11.4 Degenerate Perturbation Theory and the 2×2 Secular Equation

Near a Brillouin zone boundary where $|\mathbf{k}\rangle$ and $|\mathbf{k}+\mathbf{G}\rangle$ are nearly degenerate, non-degenerate perturbation theory must be replaced by **degenerate perturbation theory**. We construct a trial Bloch wavefunction as a linear superposition of the two strongly coupled states:

$$\psi(\mathbf{r})=c_1|\mathbf{k}\rangle+c_2|\mathbf{k}+\mathbf{G}\rangle=c_1\frac{1}{\sqrt{V}}e^{i\mathbf{k}\cdot\mathbf{r}}+c_2\frac{1}{\sqrt{V}}e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} \tag{196}$$

Substituting this trial function into the Schrödinger equation $\hat{H}\psi=E\psi$:

$$(\hat{H}_0+V)\left(c_1|\mathbf{k}\rangle+c_2|\mathbf{k}+\mathbf{G}\rangle\right)=E\left(c_1|\mathbf{k}\rangle+c_2|\mathbf{k}+\mathbf{G}\rangle\right) \tag{197}$$

Projecting successively onto $\langle\mathbf{k}|$ and $\langle\mathbf{k}+\mathbf{G}|$:

$$\left(E^{(0)}(\mathbf{k})-E\right)c_1+V_{\mathbf{G}}^{*}c_2=0 \tag{198}$$

$$V_{\mathbf{G}}c_1+\left(E^{(0)}(\mathbf{k}+\mathbf{G})-E\right)c_2=0 \tag{199}$$

For non-trivial coefficients $(c_1,c_2)\neq(0,0)$, the determinant of the coefficients must vanish, yielding the fundamental 2×2 **Secular Equation**:

$$\begin{vmatrix} E^{(0)}(\mathbf{k})-E & V_{\mathbf{G}}^{*} \\ V_{\mathbf{G}} & E^{(0)}(\mathbf{k}+\mathbf{G})-E \end{vmatrix}=0 \tag{200}$$

Expanding the determinant:

$$\left(E^{(0)}(\mathbf{k})-E\right)\left(E^{(0)}(\mathbf{k}+\mathbf{G})-E\right)-|V_{\mathbf{G}}|^2=0 \tag{201}$$

Solving this quadratic equation for the two energy eigenvalues $E_{\pm}$:

$$E_{\pm}(\mathbf{k})=\frac{E^{(0)}(\mathbf{k})+E^{(0)}(\mathbf{k}+\mathbf{G})}{2}\pm\sqrt{\left(\frac{E^{(0)}(\mathbf{k})-E^{(0)}(\mathbf{k}+\mathbf{G})}{2}\right)^2+|V_{\mathbf{G}}|^2} \tag{202}$$

11.5 Energy Band Gap at the Brillouin Zone Boundary

Directly at the Brillouin zone boundary, the two unperturbed plane-wave energies are degenerate:

$$E^{(0)}(\mathbf{k})=E^{(0)}(\mathbf{k}+\mathbf{G})\equiv E_0=\frac{\hbar^2k_0^2}{2m_e}=\frac{\hbar^2(G/2)^2}{2m_e} \tag{203}$$

The square root in equation (202) simplifies cleanly:

$$E_{\pm}=E_0\pm\sqrt{0+|V_{\mathbf{G}}|^2}=E_0\pm|V_{\mathbf{G}}| \tag{204}$$

The single unperturbed energy level E_0 splits into two separate energy levels:

- Lower band energy: $E_-=E_0-|V_{\mathbf{G}}|$
- Upper band energy: $E_+=E_0+|V_{\mathbf{G}}|$

The forbidden **energy band gap** $\Delta E=E_g$ opened by the periodic potential is:

$$\boxed{E_g=\Delta E=E_+-E_-=2|V_{\mathbf{G}}|} \tag{205}$$

The magnitude of the energy band gap at any Brillouin zone boundary is **exactly twice the absolute value of the corresponding Fourier component of the crystal potential**!

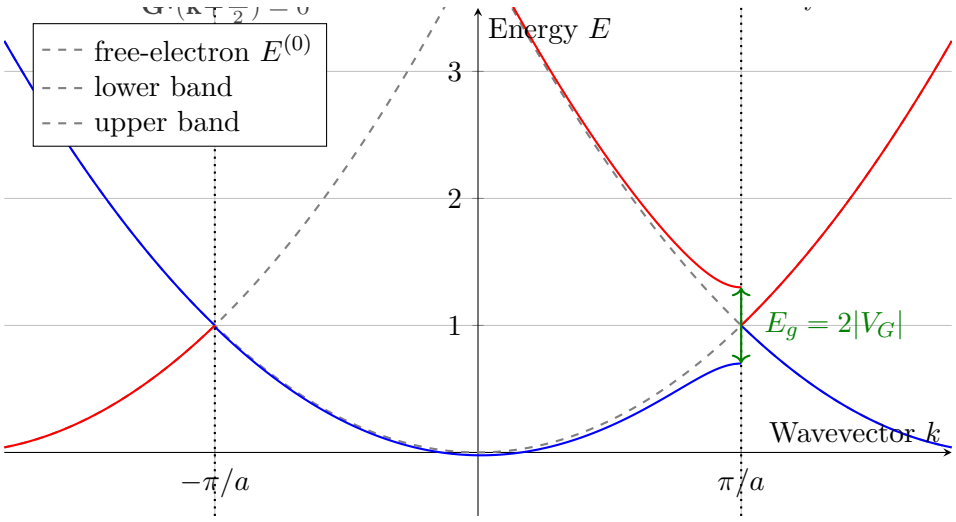


Figure 10: Nearly free electron band structure near the first Brillouin-zone boundary ($k = \pm\pi/a$). The dashed gray lines show the free-electron parabolas $E^{(0)}$, and the solid blue/red curves show the avoided crossing that opens an energy gap $E_g = 2|V_G|$ at the Bragg condition $\mathbf{G} \cdot (\mathbf{k} + \mathbf{G}/2) = 0$.

11.6 Physical Origin of the Band Gap: Standing Wave Electrostatics

Why does a weak periodic potential split the energy spectrum into allowed bands and forbidden band gaps?

At the zone boundary in 1D ($k = \pm\pi/a$ with $G = -2\pi/a$), the forward-traveling wave $e^{i\pi x/a}$ and backward-traveling wave $e^{-i\pi x/a}$ have identical amplitudes. Instead of traveling waves, Bragg reflection locks them into two orthogonal **stationary standing waves**:

$$\psi_+(x) = \frac{1}{\sqrt{2}}\left(e^{i\pi x/a} + e^{-i\pi x/a}\right) = \sqrt{2}\cos\left(\frac{\pi x}{a}\right) \tag{206}$$

$$\psi_-(x) = \frac{1}{\sqrt{2}i}\left(e^{i\pi x/a} - e^{-i\pi x/a}\right) = \sqrt{2}\sin\left(\frac{\pi x}{a}\right) \tag{207}$$

Now examine the spatial probability densities of these two standing waves:

$$\rho_+(x) = |\psi_+(x)|^2 = 2\cos^2\left(\frac{\pi x}{a}\right) = 1 + \cos\left(\frac{2\pi x}{a}\right) \tag{208}$$

$$\rho_-(x) = |\psi_-(x)|^2 = 2\sin^2\left(\frac{\pi x}{a}\right) = 1 - \cos\left(\frac{2\pi x}{a}\right) \tag{209}$$

Assume the positive ion cores are located at lattice sites $x = 0, \pm a, \pm 2a, \ldots$, where the attractive potential energy $V(x) \approx -V_0\cos(2\pi x/a)$ is at its most negative:

1. **Wavefunction** ψ_+: The charge density $\rho_+(x)$ peaks at $x = 0, \pm a, \ldots$, concentrating the negative electron charge **directly on top of the positive ion cores**. The attractive Coulomb interaction is maximized, **lowering the potential energy**:

$$\langle V\rangle_+ = \int_0^a \rho_+(x)V(x)\,dx = \int_0^a \left[1 + \cos\left(\frac{2\pi x}{a}\right)\right]\left[-V_0\cos\left(\frac{2\pi x}{a}\right)\right]dx = -\frac{V_0}{2} \tag{210}$$

2. **Wavefunction** ψ_-: The charge density $\rho_-(x)$ vanishes at the ion cores ($x = 0$) and peaks midway between the cores ($x = \pm a/2$), piling electron charge in the regions of weakest attraction, **raising the potential energy**:

$$\langle V\rangle_- = \int_0^a \rho_-(x)V(x)\,dx = \int_0^a \left[1 - \cos\left(\frac{2\pi x}{a}\right)\right]\left[-V_0\cos\left(\frac{2\pi x}{a}\right)\right]dx = +\frac{V_0}{2} \tag{211}$$

The difference in electrostatic potential energy between these two charge distributions is:

$$\Delta E = \langle V \rangle_- - \langle V \rangle_+ = +\frac{V_0}{2} - \left(-\frac{V_0}{2}\right) = V_0 = 2|V_G| \tag{212}$$

The energy band gap is not an abstract mathematical artifact: **it is the electrostatic potential energy difference between an electron cloud that clusters on positive ion cores versus an electron cloud that clusters between them!**

Conceptual Checkpoint: Physical Intuition for Energy Gap Formation

At any wavevector k far from zone boundaries, an electron wavepacket travels smoothly through the crystal with uniform probability density $|\psi|^2 = 1/V$, sampling the attractive and repulsive regions equally.
At the zone boundary, Bragg reflection creates stationary nodes. If the electron puts its charge antinodes on the positive ions, its energy drops; if it puts its antinodes between the ions, its energy rises. No stationary quantum state can exist with an energy intermediate between these two configurations: hence, an energy gap opens!

Problem-Solving Strategy: Setting Up the Secular Determinant at Zone Boundaries

For any weak crystal potential $V(\mathbf{r})$:

1. Identify the Brillouin zone boundary of interest (e.g., $k = \pm\pi/a$ with $G = 2\pi/a$).
2. Compute the Fourier transform of the crystal potential for that reciprocal lattice vector:
$$V_{\mathbf{G}} = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} V(\mathbf{r}) e^{-i\mathbf{G}\cdot\mathbf{r}}\, d^3r$$
3. The band gap opened at that specific zone boundary is immediately given by $\Delta E = 2|V_{\mathbf{G}}|$.
4. The unperturbed kinetic energy at the boundary is $E_0 = \frac{\hbar^2(G/2)^2}{2m_e}$.
5. The band edges are $E_{\text{lower}} = E_0 - |V_{\mathbf{G}}|$ and $E_{\text{upper}} = E_0 + |V_{\mathbf{G}}|$.

Worked Example: Exact Solution of the 2×2 Zone-Boundary Secular Equation

Problem Statement: Consider a 1D lattice with potential $V(x) = V_0 \cos(2\pi x/a)$ where $V_0 = 1.20$ eV and lattice constant $a = 3.00$ Å.

(a) Find the Fourier components V_G for $G = \pm 2\pi/a$.

(b) Calculate the unperturbed kinetic energy E_0 at the zone boundary $k = \pi/a$.

(c) Determine the upper and lower band edge energies and the band gap ΔE.

Step-by-Step Solution:
(a) Fourier components: Using Euler's identity:

$$V(x) = V_0 \cos\left(\frac{2\pi x}{a}\right) = \frac{V_0}{2} e^{i(2\pi/a)x} + \frac{V_0}{2} e^{-i(2\pi/a)x}$$

Comparing with $V(x) = \sum_G V_G e^{iGx}$, the non-zero Fourier components are:

$$V_{2\pi/a} = V_{-2\pi/a} = \frac{V_0}{2} = \frac{1.20 \text{ eV}}{2} = \mathbf{0.60} \text{ eV}$$

(b) Unperturbed energy E_0 at $k = \pi/a$:

$$k_0 = \frac{\pi}{a} = \frac{\pi}{3.00 \times 10^{-10}\ \text{m}} \approx 1.047 \times 10^{10}\ \text{m}^{-1}$$

$$E_0 = \frac{\hbar^2 k_0^2}{2m_e} = \frac{(1.0546 \times 10^{-34}\ \text{J}\cdot\text{s})^2 (1.047 \times 10^{10}\ \text{m}^{-1})^2}{2(9.109 \times 10^{-31}\ \text{kg})}$$
$$= \frac{1.220 \times 10^{-48}}{1.822 \times 10^{-30}} \approx 6.697 \times 10^{-19}\ \text{J} = \frac{6.697 \times 10^{-19}\ \text{J}}{1.6022 \times 10^{-19}\ \text{J/eV}} \approx \mathbf{4.18}\ \text{eV}$$

(c) Band edge energies and band gap:

$$E_- = E_0 - |V_G| = 4.18\ \text{eV} - 0.60\ \text{eV} = \mathbf{3.58}\ \text{eV}$$
$$E_+ = E_0 + |V_G| = 4.18\ \text{eV} + 0.60\ \text{eV} = \mathbf{4.78}\ \text{eV}$$
$$\Delta E = 2|V_G| = 2(0.60\ \text{eV}) = \mathbf{1.20}\ \text{eV}$$

Worked Example: Coulomb Energy Difference of Zone-Boundary Standing Waves

Problem Statement: For a 1D lattice potential $V(x) = -V_0 \cos(2\pi x/a)$ with $V_0 > 0$, calculate the first-order perturbation energy shift $\Delta E_\pm = \langle \psi_\pm | V | \psi_\pm \rangle$ for the two normalized standing waves:

$$\psi_+(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{a}\right), \quad \psi_-(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{a}\right)$$

and verify that the resulting splitting reproduces the band gap $\Delta E = 2|V_G|$.

Step-by-Step Solution:
The expectation value of the potential energy is:

$$\langle V \rangle_+ = \int_0^L |\psi_+(x)|^2 V(x)\, dx = \frac{2}{L} \int_0^L \cos^2\left(\frac{\pi x}{a}\right) \left[-V_0 \cos\left(\frac{2\pi x}{a}\right)\right] dx$$

Using the trigonometric identity $\cos^2\theta = \frac{1+\cos(2\theta)}{2}$:

$$\langle V \rangle_+ = -\frac{V_0}{L} \int_0^L \left[1 + \cos\left(\frac{2\pi x}{a}\right)\right] \cos\left(\frac{2\pi x}{a}\right) dx$$
$$= -\frac{V_0}{L} \left[\int_0^L \cos\left(\frac{2\pi x}{a}\right) dx + \int_0^L \cos^2\left(\frac{2\pi x}{a}\right) dx\right]$$

Over an integer number of unit cells ($L = Na$), $\int_0^L \cos(2\pi x/a)dx = 0$, and $\int_0^L \cos^2(2\pi x/a)dx = L/2$. Thus:

$$\langle V \rangle_+ = -\frac{V_0}{L}\left(0 + \frac{L}{2}\right) = -\frac{\mathbf{V_0}}{\mathbf{2}}$$

Similarly for $\psi_-(x)$, using $\sin^2\theta = \frac{1-\cos(2\theta)}{2}$:

$$\langle V \rangle_- = \frac{2}{L} \int_0^L \sin^2\left(\frac{\pi x}{a}\right) \left[-V_0 \cos\left(\frac{2\pi x}{a}\right)\right] dx$$
$$= -\frac{V_0}{L} \int_0^L \left[1 - \cos\left(\frac{2\pi x}{a}\right)\right] \cos\left(\frac{2\pi x}{a}\right) dx = -\frac{V_0}{L}\left(0 - \frac{L}{2}\right) = +\frac{\mathbf{V_0}}{\mathbf{2}}$$

The total energy splitting between the two standing waves is:

$$\Delta E = \langle V \rangle_- - \langle V \rangle_+ = +\frac{V_0}{2} - \left(-\frac{V_0}{2}\right) = V_0$$

Since the Fourier expansion of $V(x) = -V_0 \cos(2\pi x/a)$ gives $|V_G| = V_0/2$, this exactly yields $\Delta E = 2|V_G|$, confirming the electrostatic standing-wave origin of the energy gap.

12 Fermi Surfaces and Experimental Probes

12.1 Geometry of the Fermi Surface in Brillouin Zones

The **Fermi surface** is the constant-energy surface in reciprocal space that separates occupied electronic states from unoccupied states at absolute zero:

$$E_n(\mathbf{k}) = E_F \tag{213}$$

In a pure free-electron gas, the Fermi surface is an exact sphere of radius $k_F = (3\pi^2 n)^{1/3}$. In a crystalline lattice, however, the presence of Brillouin zone boundaries distort this sphere.

Harrison's Free-Electron Construction: To visualize the Fermi surface in a crystalline lattice within the nearly free electron approximation, Walter Harrison developed a simple geometric recipe:

1. Draw reciprocal lattice points **G** and construct the Brillouin zone boundaries (the perpendicular bisector Bragg planes).
2. Center a free-electron sphere of radius k_F at every reciprocal lattice point.
3. Points in k-space that lie inside at least one sphere belong to the **first band**; points inside at least two spheres belong to the **second band**; points inside at least three spheres belong to the **third band**, and so forth.
4. Translate the segments of each band back into the first Brillouin zone using reciprocal lattice vectors **G** to obtain the Fermi surface in the **reduced zone scheme**.

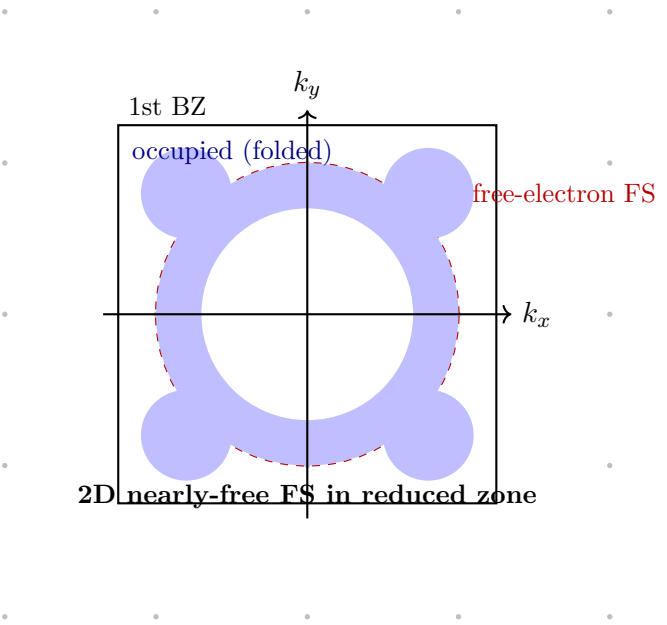


Figure 11: Construction of the Fermi surface in the reduced zone scheme for a 2D square lattice. The free-electron circle (red dashed) is folded into the first Brillouin zone by reciprocal-lattice translations: the central disk becomes the first-band Fermi surface, while the four corner pieces map to four hole pockets (blue) in the second band.

Electron vs. Hole Pockets:

- If a band is mostly empty with states occupied near the band bottom, the Fermi surface encloses occupied states, forming an **electron pocket**.
- If a band is mostly full with unoccupied states near the band top, the Fermi surface encloses empty states, forming a **hole pocket**.

12.2 Semiclassical Electron Dynamics in a Magnetic Field

When a uniform magnetic field $\mathbf{B} = B\hat{z}$ is applied to a metal, the Lorentz force governs the time evolution of the wavevector:

$$\hbar\frac{d\mathbf{k}}{dt} = -e\,(\mathbf{v}\times\mathbf{B}) \tag{214}$$

This semiclassical equation leads to two fundamental conservation laws:

1. **Energy Conservation**: The rate of work done by the magnetic force is:

$$\frac{dE}{dt} = \nabla_{\mathbf{k}}E\cdot\frac{d\mathbf{k}}{dt} = (\hbar\mathbf{v})\cdot\left[-\frac{e}{\hbar}(\mathbf{v}\times\mathbf{B})\right] = -e\,\mathbf{v}\cdot(\mathbf{v}\times\mathbf{B}) \equiv 0 \tag{215}$$

   The electron energy $E(\mathbf{k}) = \text{constant}$ is strictly conserved throughout the trajectory.

2. **Parallel Wavevector Conservation**: Taking the dot product with $\mathbf{B}$:

$$\hbar\frac{d(\mathbf{k}\cdot\mathbf{B})}{dt} = -e(\mathbf{v}\times\mathbf{B})\cdot\mathbf{B} \equiv 0 \implies k_z = \text{constant} \tag{216}$$

Geometric Trajectory: In reciprocal space, electrons traverse closed orbits formed by the **intersection of a constant-energy surface $E(\mathbf{k}) = E_F$ with a plane of constant k_z perpendicular to B**!

Cyclotron Frequency and Effective Mass: The time required for an electron to complete one full revolution around the orbit is the cyclotron period $T_c = 2\pi/\omega_c$. Integrating $dt = \frac{\hbar\,dk_\perp}{eBv_\perp}$:

$$T_c = \oint dt = \frac{\hbar}{eB}\oint\frac{dk_\perp}{v_\perp} \tag{217}$$

Recognizing that the area element between two adjacent energy contours in the k_xk_y-plane is $dA = dk_\perp dk_\parallel = dk_\perp\left(\frac{dE}{\hbar v_\perp}\right)$:

$$\oint\frac{dk_\perp}{v_\perp} = \hbar\frac{\partial}{\partial E}\oint dk_\perp dk_\parallel = \hbar\frac{\partial A(E,k_z)}{\partial E} \tag{218}$$

Therefore:

$$T_c = \frac{\hbar^2}{eB}\frac{\partial A}{\partial E} \implies \omega_c = \frac{2\pi}{T_c} = \frac{2\pi eB}{\hbar^2\left(\frac{\partial A}{\partial E}\right)} \tag{219}$$

We define the **cyclotron effective mass** m_c as:

$$\boxed{m_c \equiv \frac{\hbar^2}{2\pi}\left.\frac{\partial A(E,k_z)}{\partial E}\right|_{E=E_F}} \iff \omega_c = \frac{eB}{m_c} \tag{220}$$

For a free-electron circle $A(E) = \pi k^2 = \pi\frac{2m_eE}{\hbar^2} \implies \frac{\partial A}{\partial E} = \frac{2\pi m_e}{\hbar^2}$, which recovers $m_c = m_e$.

12.3 Landau Level Quantization

In a quantum mechanical treatment, a perpendicular magnetic field $\mathbf{B} = B\hat{z}$ breaks the continuous k_xk_y continuum into discrete, highly degenerate harmonic oscillator levels known as **Landau levels**:

$$E_n(k_z) = \left(n+\frac{1}{2}\right)\hbar\omega_c + \frac{\hbar^2k_z^2}{2m^*} \tag{221}$$

In reciprocal space, the allowed electronic states condense onto a family of concentric, coaxial **Landau cylinders** (or Landau tubes) aligned along the k_z-axis.

The k-space cross-sectional area enclosed by the n-th Landau cylinder is given by the Bohr–Sommerfeld semiclassical **Onsager relation**:

$$\boxed{A_n = \frac{2\pi eB}{\hbar}\left(n + \frac{1}{2}\right)} \tag{222}$$

The area spacing between adjacent concentric cylinders in k-space is:

$$\Delta A = A_{n+1} - A_n = \frac{2\pi eB}{\hbar} \tag{223}$$

Notice that as the magnetic field B increases, the cylinders expand in radius, and their area spacing $\Delta A \propto B$ increases proportionally!

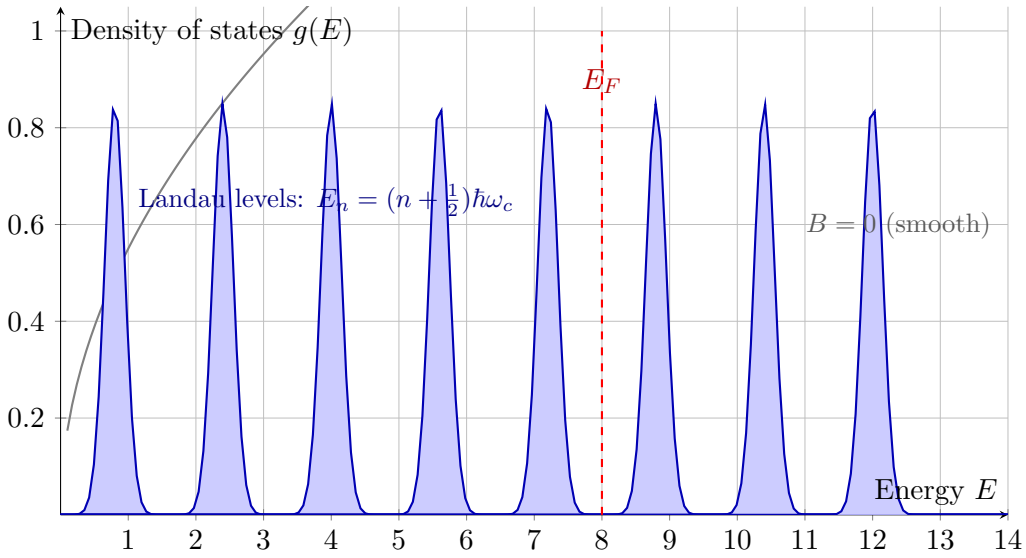


Figure 12: Landau levels in a magnetic field: the 3D free-electron $g(E) \propto \sqrt{E}$ (gray) splits into equally spaced, highly degenerate Landau levels at $E_n = (n + \frac{1}{2})\hbar\omega_c$ (blue). As B changes, these peaks sweep through E_F, causing the oscillatory density of states at the Fermi level that drives the de Haas–van Alphen signal.

12.4 The de Haas–van Alphen (dHvA) Effect

As the magnetic field B is continuously increased, the Landau cylinders expand outwards through the fixed Fermi surface.

Whenever the outermost occupied Landau cylinder expands to touch the extremal cross-section of the Fermi surface and pops through it into the unoccupied region above E_F, all the electrons on that cylinder suddenly depopulate, dropping into lower Landau levels. This sudden, periodic depopulation causes the total thermodynamic free energy $F(B)$, magnetization $M(B)$, and magnetic susceptibility $\chi(B)$ to oscillate as a function of the magnetic field!

This celebrated quantum phenomenon is known as the **de Haas–van Alphen (dHvA) effect** (discovered in 1930 by Wander Johannes de Haas and Pieter M. van Alphen).

Periodicity in Inverse Field $1/B$: A Landau level depopulates whenever its cross-sectional area matches the **extremal cross-sectional area** A_{ext} of the Fermi surface:

$$A_{\text{ext}} = \frac{2\pi eB_n}{\hbar}(n + \gamma) \implies \frac{1}{B_n} = \frac{2\pi e}{\hbar A_{\text{ext}}}(n + \gamma) \tag{224}$$

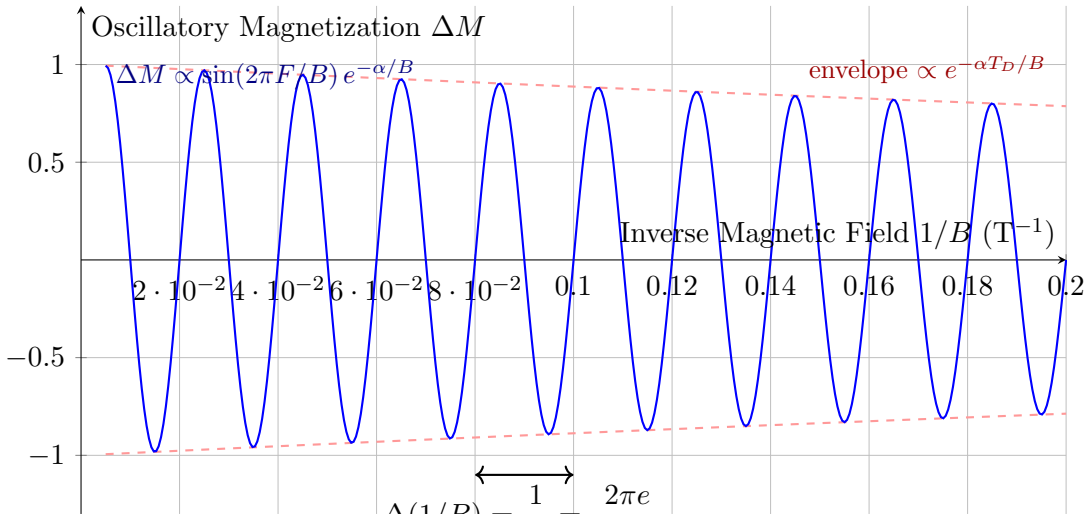


Figure 13: Schematic of the de Haas–van Alphen effect, showing oscillations in magnetization periodic in $1/B$. The period is inversely proportional to the extremal cross-sectional area A_{ext} of the Fermi surface. The Dingle factor $e^{-\alpha T_D/B}$ encodes impurity scattering and finite temperature, damping the oscillations at small B (large $1/B$).

The condition between consecutive oscillation peaks ($n \to n+1$) reveals that dHvA oscillations are strictly **periodic in inverse magnetic field** $1/B$:

$$\boxed{\Delta\left(\frac{1}{B}\right) = \frac{2\pi e}{\hbar A_{\text{ext}}} = \frac{1}{F}} \tag{225}$$

where F is the **dHvA oscillation frequency** (measured in Tesla):

$$\boxed{F = \frac{\hbar}{2\pi e} A_{\text{ext}}} \tag{226}$$

Study Takeaway: Why Quantum Oscillations Measure Extremal Cross-Sections

In three dimensions, the total magnetization involves an integral over all cross-sections along the k_z-axis: $M \sim \int dk_z \sin\left(\frac{\hbar A(k_z)}{eB}\right)$.
For general values of k_z, the phase oscillates rapidly, and the contributions from neighboring slices destructively interfere to zero. By the **principle of stationary phase**, constructive interference occurs exclusively at slices where the cross-sectional area is stationary with respect to k_z:

$$\frac{\partial A(k_z)}{\partial k_z} = 0$$

These stationary points are precisely the **extremal cross-sections** (maximal "belly" orbits or minimal "neck" orbits). Therefore, measuring the dHvA frequency F provides a direct, sub-percent caliper measurement of the extremal cross-sectional dimensions of the Fermi surface!

Common Misconception & Pitfall: Thermal Washing-Out of Quantum Oscillations

To observe de Haas–van Alphen oscillations experimentally, two strict physical conditions must be satisfied:

1. **Low Temperature Criterion**: $k_B T \ll \hbar\omega_c$. If the thermal smearing window $k_B T$ is comparable to the Landau level spacing $\hbar\omega_c$, electrons are thermally blurred across

multiple levels, completely damping the oscillations. For $B \approx 10$ T, $\hbar\omega_c \approx 1$ meV $\sim$ 10 K, necessitating cryogenic liquid helium temperatures ($T \leq 4.2$ K).

2. **High Purity Criterion**: $\omega_c\tau \gg 1$. Electrons must complete at least one full cyclotron orbit around the Fermi surface before being scattered by impurities ($\tau > T_c$). This requires ultra-high-purity single crystals (RRR > 1000).

Worked Example: Extremal Area of Copper from dHvA Oscillations

Problem Statement: In a de Haas–van Alphen experiment on a high-purity single crystal of copper with magnetic field aligned along the [111] crystallographic direction, magnetization oscillations are observed with an inverse field period:

$$\Delta\left(\frac{1}{B}\right) = 2.18 \times 10^{-5}\ \text{T}^{-1}$$

Calculate:

(a) The extremal cross-sectional area A_{ext} of the copper Fermi surface in m^{-2} and in Å^{-2}.

(b) The fundamental dHvA frequency F in Tesla.

(c) Assuming a spherical Fermi surface, deduce the effective Fermi wavevector k_F and the corresponding Fermi energy E_F in eV.

Step-by-Step Solution:

(a) Extremal cross-sectional area A_{ext}: From equation (225):

$$A_{\text{ext}} = \frac{2\pi e}{\hbar\Delta(1/B)} = \frac{2\pi(1.6022 \times 10^{-19}\ \text{C})}{(1.0546 \times 10^{-34}\ \text{J}\cdot\text{s})(2.18 \times 10^{-5}\ \text{T}^{-1})}$$
$$= \frac{1.0067 \times 10^{-18}}{2.299 \times 10^{-39}} \approx \mathbf{4.38 \times 10^{20}}\ \text{m}^{-2} = \mathbf{4.38}\ \text{Å}^{\mathbf{-2}}$$

(b) dHvA frequency F:

$$F = \frac{1}{\Delta(1/B)} = \frac{1}{2.18 \times 10^{-5}\ \text{T}^{-1}} \approx \mathbf{45{,}870}\ \text{T} \approx 4.59 \times 10^4\ \text{T}$$

(c) Effective Fermi wavevector and Fermi energy: For a circular cross-section, $A_{\text{ext}} = \pi k_F^2$:

$$k_F = \sqrt{\frac{A_{\text{ext}}}{\pi}} = \sqrt{\frac{4.38 \times 10^{20}\ \text{m}^{-2}}{\pi}} = \sqrt{1.394 \times 10^{20}} \approx \mathbf{1.18 \times 10^{10}}\ \text{m}^{-1} = 1.18\ \text{Å}^{-1}$$

The corresponding Fermi energy is:

$$E_F = \frac{\hbar^2 k_F^2}{2m_e} = \frac{(1.0546 \times 10^{-34}\ \text{J}\cdot\text{s})^2(1.181 \times 10^{10}\ \text{m}^{-1})^2}{2(9.109 \times 10^{-31}\ \text{kg})}$$
$$= \frac{(1.112 \times 10^{-68})(1.394 \times 10^{20})}{1.822 \times 10^{-30}} \approx 8.51 \times 10^{-19}\ \text{J} = \frac{8.51 \times 10^{-19}\ \text{J}}{1.6022 \times 10^{-19}\ \text{J/eV}} \approx \mathbf{5.31}\ \text{eV}$$

(The slight deviation from the belly value $E_F \approx 7.0$ eV reflects the fact that the [111] orbit in copper measures the narrower neck that contacts the Brillouin zone face!).

Worked Example: Extraction of Fermi Parameters from dHvA Frequency

Problem Statement: A metal displays a dominant dHvA oscillation frequency of $F = 5000$ T.

(a) Calculate the extremal area A_{ext} of the Fermi surface in m^{-2}.

(b) Assuming an isotropic spherical Fermi surface, determine the Fermi wavevector k_F, the electron density n, and the Fermi energy E_F in eV.

Step-by-Step Solution:

(a) Extremal cross-sectional area: From equation (226):

$$\begin{aligned} A_{\text{ext}} &= \frac{2\pi e}{\hbar} F = \frac{2\pi(1.6022 \times 10^{-19}\ \text{C})}{1.0546 \times 10^{-34}\ \text{J}\cdot\text{s}}(5000\ \text{T}) \\ &= (9.546 \times 10^{15}\ \text{m}^{-2}\text{T}^{-1})(5000\ \text{T}) \approx \mathbf{4.77 \times 10^{19}}\ \text{m}^{-2} = 0.477\ \text{Å}^{-2} \end{aligned}$$

(b) Fermi wavevector, electron density, and Fermi energy: For a sphere, $A_{\text{ext}} = \pi k_F^2$:

$$k_F = \sqrt{\frac{A_{\text{ext}}}{\pi}} = \sqrt{\frac{4.773 \times 10^{19}\ \text{m}^{-2}}{\pi}} \approx \mathbf{3.90 \times 10^{9}}\ \text{m}^{-1} = 0.390\ \text{Å}^{-1}$$

The conduction electron density is:

$$n = \frac{k_F^3}{3\pi^2} = \frac{(3.898 \times 10^{9}\ \text{m}^{-1})^3}{3\pi^2} = \frac{5.923 \times 10^{28}}{29.609} \approx \mathbf{2.00 \times 10^{27}}\ \text{m}^{-3}$$

The Fermi energy is:

$$E_F = \frac{\hbar^2 k_F^2}{2m_e} = \frac{(1.0546 \times 10^{-34})^2(3.898 \times 10^{9})^2}{2(9.109 \times 10^{-31})} = \frac{1.690 \times 10^{-49}}{1.822 \times 10^{-30}} \approx 9.276 \times 10^{-20}\ \text{J}$$

$$E_F = \frac{9.276 \times 10^{-20}\ \text{J}}{1.6022 \times 10^{-19}\ \text{J/eV}} \approx \mathbf{0.579}\ \text{eV} \approx \mathbf{0.58}\ \text{eV}$$

13 Comprehensive Solved Problems and Tutorial Solutions

This chapter provides complete, multi-part, step-by-step solutions to advanced tutorial problems that synthesize thermodynamics, transport, band theory, and quantum oscillations across Module 3.

13.1 Problem 11.1: Complete Electronic Ground State of Monovalent Noble Metals

Problem Statement: Silver (Ag) crystallizes in a face-centered cubic (FCC) structure with a conventional lattice parameter $a = 4.09$ Å. Silver has atomic mass $A = 107.87$ g/mol, mass density $\rho_m = 10.50 \times 10^3$ kg/m^3, and each atom contributes exactly one valence electron ($Z_v = 1$).

(a) Calculate the conduction electron density n independently via: (i) the crystallographic FCC unit cell volume, and (ii) macroscopic mass density. Compare the two results.

(b) Determine the Fermi wavevector k_F, the radius of the Fermi sphere, and verify that the Fermi sphere does not intersect the first Brillouin zone boundary along the [100] direction ($k_{\text{BZ}} = 2\pi/a$).

(c) Calculate the Fermi energy E_F (in eV), the Fermi velocity v_F, and the Fermi temperature T_F.

(d) Compute the zero-point ground-state internal energy U_0 and the degeneracy pressure P_0 exerted by the conduction electrons.

Step-by-Step Solution:

(a) Conduction electron concentration n:

(i) *From crystallographic unit cell*: In an FCC Bravais lattice, there are 4 atoms per cubic unit cell ($8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$). Since each atom donates 1 electron:

$$n = \frac{4Z_v}{a^3} = \frac{4 \times 1}{(4.09 \times 10^{-10}\ \text{m})^3} = \frac{4}{6.842 \times 10^{-29}\ \text{m}^3} \approx \mathbf{5.846 \times 10^{28}}\ \text{m}^{-3}$$

(ii) *From macroscopic mass density*:

$$n = \frac{Z_v N_A \rho_m}{A} = \frac{(1)(6.0221 \times 10^{23}\ \text{mol}^{-1})(10.50 \times 10^3\ \text{kg/m}^3)}{107.87 \times 10^{-3}\ \text{kg/mol}} = \frac{6.323 \times 10^{27}}{0.10787} \approx \mathbf{5.862 \times 10^{28}}\ \text{m}^{-3}$$

The two independent methods agree to within 0.27%! We adopt $n = 5.86 \times 10^{28}$ m^{-3}.

(b) Fermi wavevector and Brillouin zone geometry:

$$k_F = (3\pi^2 n)^{1/3} = \left[3\pi^2 (5.86 \times 10^{28}\ \text{m}^{-3})\right]^{1/3} = (1.735 \times 10^{30})^{1/3} \approx \mathbf{1.202 \times 10^{10}}\ \text{m}^{-1} = 1.202\ \text{Å}^{-1}$$

For an FCC lattice, the shortest distance from the zone center Γ to the square face of the first Brillouin zone (X-point along [100]) is:

$$k_X = \frac{2\pi}{a} = \frac{2\pi}{4.09 \times 10^{-10}\ \text{m}} \approx \mathbf{1.536 \times 10^{10}}\ \text{m}^{-1} = 1.536\ \text{Å}^{-1}$$

Comparing the radii:

$$\frac{k_F}{k_X} = \frac{1.202}{1.536} \approx 0.783 < 1$$

The Fermi sphere lies well inside the first Brillouin zone along [100]. (Along [111], the distance to the hexagonal face is $k_L = \sqrt{3}\pi/a \approx 1.330$ Å^{-1}, where the neck of the silver Fermi surface pulls out and contacts the zone face).

(c) Fermi energy, velocity, and temperature:

$$E_F = \frac{\hbar^2 k_F^2}{2m_e} = \frac{(1.0546 \times 10^{-34}\ \text{J}\cdot\text{s})^2 (1.202 \times 10^{10}\ \text{m}^{-1})^2}{2(9.1094 \times 10^{-31}\ \text{kg})} \approx 8.816 \times 10^{-19}\ \text{J} = \mathbf{5.50}\ \text{eV}$$

$$v_F = \frac{\hbar k_F}{m_e} = \frac{(1.0546 \times 10^{-34})(1.202 \times 10^{10})}{9.1094 \times 10^{-31}} \approx \mathbf{1.39 \times 10^6}\ \text{m/s}$$

$$T_F = \frac{E_F}{k_B} = \frac{8.816 \times 10^{-19}\ \text{J}}{1.3807 \times 10^{-23}\ \text{J/K}} \approx \mathbf{6.38 \times 10^4}\ \text{K} \approx 63{,}800\ \text{K}$$

(d) Ground-state energy density and degeneracy pressure:

$$u_0 = \frac{U_0}{V} = \frac{3}{5} n E_F = \frac{3}{5}(5.86 \times 10^{28}\ \text{m}^{-3})(8.816 \times 10^{-19}\ \text{J}) \approx \mathbf{3.10 \times 10^{10}}\ \text{J/m}^3$$

$$P_0 = \frac{2}{3} u_0 = \frac{2}{3}(3.10 \times 10^{10}\ \text{J/m}^3) \approx \mathbf{2.07 \times 10^{10}}\ \text{Pa} \approx \mathbf{204{,}000}\ \text{atm}$$

13.2 Problem 11.2: Quantum Degeneracy Pressure and White Dwarf Collapse

Problem Statement: In a white dwarf star, matter is completely ionized into helium or carbon/oxygen nuclei and a dense, degenerate electron gas. Consider a white dwarf of mass M and radius R with uniform mass density $\rho_m = \frac{M}{\frac{4}{3}\pi R^3}$. Let the number of nucleons per electron be $\mu_e = 2$ (e.g., ^{12}C has 6 protons, 6 neutrons, and 6 electrons, so $\mu_e = 12/6 = 2$).

(a) Express the electron number density n in terms of mass density ρ_m and proton mass m_p.

(b) Derive the non-relativistic degeneracy pressure P_{NR} as a function of ρ_m.

(c) In the ultra-relativistic limit ($E \approx pc \gg m_e c^2$), show that the degeneracy pressure scales as $P_{\text{UR}} \propto \rho_m^{4/3}$.

(d) Explain physically why this transition from $P \propto \rho^{5/3}$ to $P \propto \rho^{4/3}$ leads directly to the Chandrasekhar mass limit ($M_{\text{Ch}} \approx 1.44 M_\odot$).

Step-by-Step Solution:

(a) Electron concentration n: The mass of the star is overwhelmingly carried by the nucleons: $M \approx N_{\text{nucleons}} m_p$. Because there is 1 electron for every μ_e nucleons:

$$n = \frac{N_e}{V} = \frac{N_{\text{nucleons}}}{\mu_e V} = \frac{\rho_m}{\mu_e m_p}$$

(b) Non-relativistic degeneracy pressure: From equation (71), $P_0 = \frac{2}{5} n E_F = \frac{\hbar^2}{5m_e}(3\pi^2)^{2/3} n^{5/3}$. Substituting $n = \rho_m/(\mu_e m_p)$:

$$P_{\text{NR}} = \frac{\hbar^2}{5m_e}(3\pi^2)^{2/3}\left(\frac{\rho_m}{\mu_e m_p}\right)^{5/3} \propto \rho_m^{5/3} \propto \left(\frac{M}{R^3}\right)^{5/3} = \frac{M^{5/3}}{R^5}$$

(c) Ultra-relativistic degeneracy pressure: When density becomes extreme ($\rho_m \gtrsim 10^9\ \text{kg/m}^3$), $p_F \gg m_e c$, and electron dispersion becomes ultra-relativistic: $E(p) \approx pc = \hbar c k$. The total ground-state internal energy is:

$$U_{\text{UR}} = 2\sum_{|\mathbf{k}| \le k_F} \hbar c k = 2\frac{V}{(2\pi)^3}\int_0^{k_F} (\hbar c k)(4\pi k^2 dk) = \frac{V\hbar c}{\pi^2}\int_0^{k_F} k^3 dk = \frac{V\hbar c}{4\pi^2} k_F^4$$

Since $k_F = (3\pi^2 n)^{1/3}$:

$$U_{\text{UR}} = \frac{V\hbar c}{4\pi^2}(3\pi^2 n)^{4/3} = \frac{3}{4}\hbar c(3\pi^2)^{1/3} n^{4/3} V = \frac{3}{4}NE_F$$

The relativistic degeneracy pressure is:

$$P_{\text{UR}} = -\frac{\partial U_{\text{UR}}}{\partial V} = \frac{1}{3}\frac{U_{\text{UR}}}{V} = \frac{1}{4}\hbar c(3\pi^2)^{1/3} n^{4/3}$$

Substituting $n = \rho_m/(\mu_e m_p)$:

$$\boxed{P_{\text{UR}} = \frac{\hbar c}{4}(3\pi^2)^{1/3}\left(\frac{\rho_m}{\mu_e m_p}\right)^{4/3} \propto \rho_m^{4/3} \propto \frac{M^{4/3}}{R^4}}$$

(d) Physical origin of the Chandrasekhar Limit: Hydrostatic equilibrium requires the internal pressure gradient force to balance gravitational inward pull:

$$F_{\text{pressure}} \sim PR^2, \quad F_{\text{gravity}} \sim \frac{GM^2}{R^2}$$

- In the non-relativistic regime:

$$P_{\text{NR}} R^2 \sim \frac{M^{5/3}}{R^5}R^2 = \frac{M^{5/3}}{R^3} \sim \frac{GM^2}{R^2} \implies R \propto M^{-1/3}$$

  As mass increases, the star contracts to a smaller, stable radius R.

- In the ultra-relativistic regime:

$$P_{\text{UR}} R^2 \sim \frac{M^{4/3}}{R^4}R^2 = \frac{M^{4/3}}{R^2} \sim \frac{GM^2}{R^2}$$

  The radius R cancels out completely from both sides! Gravitational force and degeneracy pressure exhibit the **identical** $1/R^2$ **scaling**.

Equating the two yields an absolute critical maximum mass:

$$M_{\text{Ch}} \approx \frac{\omega_3^0}{m_p^2}\left(\frac{\hbar c}{G}\right)^{3/2} \approx 1.44 M_\odot$$

If a stellar remnant exceeds $1.44M_\odot$, relativistic electron degeneracy pressure is mathematically incapable of halting gravitational contraction; the star must collapse into a neutron star or black hole.

13.3 Problem 11.3: Determination of Fermi Energy from Specific Heat

Problem Statement: Low-temperature heat capacity measurements on an unknown monovalent metallic element yield an electronic coefficient $C_{el} = 2.50 \times 10^{-4} T$ J mol^{-1} K^{-1}.

(a) Calculate the experimental Fermi energy E_F in Joules and in electron-volts.

(b) Determine the density of states at the Fermi energy per mole, $g(E_F)$.

(c) Compute the theoretical Fermi wavevector k_F and the conduction electron concentration n.

Step-by-Step Solution:

(a) Fermi energy E_F: The experimental Sommerfeld coefficient is $\gamma = 2.50 \times 10^{-4}\ \text{J mol}^{-1}\,\text{K}^{-2}$. Using equation (99) for one mole:

$$\gamma = \frac{\pi^2 R k_B}{2E_F} \implies E_F = \frac{\pi^2 R k_B}{2\gamma}$$

Substituting $R = 8.3145\ \text{J/(mol}\cdot\text{K)}$ and $k_B = 1.3807 \times 10^{-23}\ \text{J/K}$:

$$E_F = \frac{\pi^2(8.3145)(1.3807 \times 10^{-23})}{2(2.50 \times 10^{-4})} = \frac{1.133 \times 10^{-21}}{5.00 \times 10^{-4}} \approx \mathbf{2.266 \times 10^{-18}}\ \text{J}$$

$$E_F = \frac{2.266 \times 10^{-18}\ \text{J}}{1.6022 \times 10^{-19}\ \text{J/eV}} \approx \mathbf{14.14}\ \text{eV}$$

(b) Density of states at E_F: From equation (99), $\gamma = \frac{\pi^2}{3} k_B^2 g(E_F)$:

$$g(E_F) = \frac{3\gamma}{\pi^2 k_B^2} = \frac{3(2.50 \times 10^{-4}\ \text{J mol}^{-1}\,\text{K}^{-2})}{\pi^2(1.3807 \times 10^{-23}\ \text{J/K})^2} = \frac{7.50 \times 10^{-4}}{\pi^2(1.906 \times 10^{-45})}$$
$$\approx \mathbf{3.99 \times 10^{41}}\ \text{states J}^{-1}\,\text{mol}^{-1} = \mathbf{6.39 \times 10^{22}}\ \text{states eV}^{-1}\,\text{mol}^{-1}$$

(c) Fermi wavevector and electron density: From $E_F = \frac{\hbar^2 k_F^2}{2m_e}$:

$$k_F = \sqrt{\frac{2m_e E_F}{\hbar^2}} = \sqrt{\frac{2(9.1094 \times 10^{-31}\ \text{kg})(2.266 \times 10^{-18}\ \text{J})}{(1.0546 \times 10^{-34}\ \text{J}\cdot\text{s})^2}} = \sqrt{\frac{4.128 \times 10^{-48}}{1.112 \times 10^{-68}}}$$
$$= \sqrt{3.712 \times 10^{20}} \approx \mathbf{1.927 \times 10^{10}}\ \text{m}^{-1} = 1.927\ \text{Å}^{-1}$$

The electron concentration is:

$$n = \frac{k_F^3}{3\pi^2} = \frac{(1.927 \times 10^{10}\ \text{m}^{-1})^3}{3\pi^2} = \frac{7.155 \times 10^{30}}{29.609} \approx \mathbf{2.42 \times 10^{29}}\ \text{m}^{-3}$$

13.4 Problem 11.4: Band Gaps in a 1D Superlattice Potential

Problem Statement: An artificial 1D semiconductor superlattice of period $a = 5.00$ Å has a crystal potential represented by $V(x) = V_0 \cos(2\pi x/a)$ with $V_0 = 1.00$ eV.

(a) Determine the non-zero Fourier coefficients V_G.

(b) Calculate the energy band gap ΔE_1 at the first Brillouin zone boundary $k = \pi/a$.

(c) Does this potential open an energy gap at the second zone boundary $k = 2\pi/a$ to first order in perturbation theory? What higher-order process opens a gap at $k = 2\pi/a$?

Step-by-Step Solution:

(a) Fourier coefficients: Expanding $V(x) = \frac{V_0}{2} e^{i(2\pi/a)x} + \frac{V_0}{2} e^{-i(2\pi/a)x}$:

$$V_{\pm 2\pi/a} = \frac{V_0}{2} = \frac{1.00\ \text{eV}}{2} = \mathbf{0.50}\ \text{eV}$$

All other Fourier components V_G for $G \neq \pm 2\pi/a$ are identically zero.

(b) Band gap at the first zone boundary ($k = \pi/a$): At $k = \pi/a$, the state $|k\rangle = |\pi/a\rangle$ is coupled to $|k - G\rangle = |\pi/a - 2\pi/a\rangle = |-\pi/a\rangle$ by $G = 2\pi/a$. The band gap is:

$$\Delta E_1 = 2|V_{2\pi/a}| = 2(0.50\ \text{eV}) = \mathbf{1.00}\ \text{eV}$$

(c) Second zone boundary ($k = 2\pi/a$): At the second zone boundary $k = 2\pi/a$, coupling to $|k'\rangle = |-2\pi/a\rangle$ requires a reciprocal lattice vector $G' = 4\pi/a$. Because $V(x)$ is a pure cosine, the Fourier amplitude $V_{4\pi/a} = 0$. Therefore, to **first order** in perturbation theory, the band gap at $k = 2\pi/a$ vanishes: $\Delta E_2^{(1)} = 2|V_{4\pi/a}| = \mathbf{0}$.

However, a non-zero band gap is opened to **second order** via an intermediate virtual state. The state $|2\pi/a\rangle$ couples to $|0\rangle$ via $V_{-2\pi/a}$, which subsequently couples to $|-2\pi/a\rangle$ via $V_{-2\pi/a}$. The second-order effective matrix element is:

$$V_{\text{eff}}^{(2)} \sim \frac{|V_{2\pi/a}|^2}{E^{(0)}(2\pi/a) - E^{(0)}(0)} \implies \Delta E_2^{(2)} \approx \frac{2|V_{2\pi/a}|^2}{E^{(0)}(2\pi/a)}$$

This produces a much narrower second-order band gap at $k = 2\pi/a$.

13.5 Problem 11.5: Multi-Orbit Fermi Surface Reconstruction in Divalent Metals

Problem Statement: In a de Haas–van Alphen oscillation experiment on a divalent hexagonal metal, the magnetic field is oriented along the hexagonal c-axis. Two distinct oscillation frequencies are observed in the Fourier spectrum of the magnetization:

$$F_1 = 1200 \text{ T} \quad \text{and} \quad F_2 = 5600 \text{ T}$$

(a) Calculate the extremal cross-sectional areas A_1 and A_2 of the two orbits in m^{-2} and in Å^{-2}.

(b) Explain physically how a single metal can exhibit two distinct dHvA frequencies for the same field orientation.

(c) Assuming both orbits are circular, compute their effective radii in k-space (k_1 and k_2).

Step-by-Step Solution:

(a) Extremal cross-sectional areas: Using the Onsager frequency relation $A_{\text{ext}} = \frac{2\pi e}{\hbar} F$:

$$A_1 = \frac{2\pi(1.6022 \times 10^{-19} \text{ C})}{1.0546 \times 10^{-34} \text{ J} \cdot \text{s}}(1200 \text{ T}) = (9.546 \times 10^{15})(1200) \approx \mathbf{1.146 \times 10^{19}} \text{ m}^{-2} = \mathbf{0.115} \text{ Å}^{\mathbf{-2}}$$

$$A_2 = (9.546 \times 10^{15})(5600 \text{ T}) \approx \mathbf{5.346 \times 10^{19}} \text{ m}^{-2} = \mathbf{0.535} \text{ Å}^{\mathbf{-2}}$$

(b) Physical origin of multiple frequencies: Multiple dHvA frequencies for a single magnetic field orientation arise from two distinct physical mechanisms:

1. **Multiple Extremal Slices of a Single Surface**: A non-spherical Fermi surface (such as a dumbbell or corrugated cylinder) possesses multiple cross-sections where $\partial A/\partial k_z = 0$. For example, a dumbbell has a maximal cross-section (the wide "belly") and a minimal cross-section (the narrow "neck").

2. **Multiple Separate Fermi Surface Sheets**: In a multiband metal (such as a divalent metal with overlapping bands), the Fermi surface consists of multiple disconnected pieces in different Brillouin zones: small hole pockets in one band (yielding small frequency F_1) and large electron pockets in another band (yielding large frequency F_2).

(c) Effective orbital radii: Assuming circular cross-sections ($A = \pi k^2$):

$$k_1 = \sqrt{\frac{A_1}{\pi}} = \sqrt{\frac{1.146 \times 10^{19}}{\pi}} \approx \mathbf{1.91 \times 10^{9}} \text{ m}^{-1} = 0.191 \text{ Å}^{-1}$$

$$k_2 = \sqrt{\frac{A_2}{\pi}} = \sqrt{\frac{5.346 \times 10^{19}}{\pi}} \approx \mathbf{4.13 \times 10^{9}} \text{ m}^{-1} = 0.413 \text{ Å}^{-1}$$

A Appendix A: Python Demonstration Scripts

The following standalone Python scripts provide interactive computational implementations of the central theoretical concepts in Module 3. Students are encouraged to execute, modify, and explore these scripts to deepen their physical and numerical intuition.

A.1 Fermi Energy Calculator

```python
import numpy as np

# Physical constants
hbar = 1.055e-34    # J s
m_e = 9.11e-31      # kg
eV = 1.602e-19      # J

# Typical electron densities in m^-3
metals = {
    "Cu": 8.5e28,
    "Ag": 5.86e28,
    "Al": 18.1e28,
    "Na": 2.65e28,
    "Fe": 17.0e28,
    "Au": 5.90e28
}

def fermi_energy(n):
    """Return Fermi energy in eV."""
    kF = (3.0 * np.pi**2 * n)**(1.0/3.0)
    EF = hbar**2 * kF**2 / (2.0 * m_e)
    return EF / eV

def fermi_velocity(n):
    """Return Fermi velocity in m/s."""
    kF = (3.0 * np.pi**2 * n)**(1.0/3.0)
    return hbar * kF / m_e

print(f"{'Metal':<6} {'n (1e28/m^3)':<16} {'EF (eV)':<10} {'vF (1e6 m/s)':<15}")
print("-" * 55)

for metal, n in metals.items():
    EF = fermi_energy(n)
    vF = fermi_velocity(n) / 1.0e6
    print(f"{metal:<6} {n/1.0e28:<16.2f} {EF:<10.2f} {vF:<15.2f}")
```

A.2 Fermi–Dirac Distribution Visualization

```python
import numpy as np
import matplotlib.pyplot as plt

def fermi_dirac(E, mu, T):
    """Fermi-Dirac distribution. E and mu in eV, T in K."""
    kB = 8.617e-5  # eV/K
    if T == 0:
        return np.where(E < mu, 1.0, 0.0)
    return 1.0 / (np.exp((E - mu) / (kB * T)) + 1.0)

E = np.linspace(-0.5, 0.5, 1000)
```

```
mu = 0.0
T_values = [0, 100, 300, 1000, 3000]

plt.figure(figsize=(7, 4))

for T in T_values:
    f = fermi_dirac(E, mu, T)
    plt.plot(E, f, label=f"T = {T} K")

plt.axvline(0, color="gray", linestyle="--", alpha=0.5)
plt.xlabel("E - EF (eV)")
plt.ylabel("f(E)")
plt.title("Fermi-Dirac Distribution")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

A.3 Electronic and Phonon Specific Heat

```
import numpy as np
import matplotlib.pyplot as plt

# Approximate copper parameters
gamma = 0.695   # mJ/mol/K^2
beta = 0.048    # mJ/mol/K^4

T = np.linspace(0.1, 15, 500)

C_el = gamma * T
C_ph = beta * T**3
C_total = C_el + C_ph

plt.figure(figsize=(7, 4))
plt.plot(T, C_el, label="Electronic, gamma*T")
plt.plot(T, C_ph, label="Phonon, beta*T^3")
plt.plot(T, C_total, "k--", label="Total")

plt.xlabel("T (K)")
plt.ylabel("C_V (mJ/mol/K)")
plt.title("Low-T Specific Heat of Copper")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

# Temperature where electronic and phonon parts are equal
T_eq = np.sqrt(gamma / beta)
print(f"Crossover temperature = {T_eq:.2f} K")
```

A.4 Nearly Free Electron Gap Illustration

```
import numpy as np
import matplotlib.pyplot as plt

# 1D nearly-free-electron illustration
a = 1.0
```

```
G = 2.0 * np.pi / a
VG = 0.3

k = np.linspace(-1.8, 1.8, 2000)
E_free = k**2

# Two-band avoided-crossing model near k = +/- 1 (which is +/- pi/a for a
    =2)
# Using the exact secular equation solution
E_lower = (k**2 + (k-2)**2)/2 - np.sqrt(((k**2 - (k-2)**2)/2)**2 + VG**2)
E_upper = (k**2 + (k-2)**2)/2 + np.sqrt(((k**2 - (k-2)**2)/2)**2 + VG**2)

plt.figure(figsize=(8, 4.5))
plt.plot(k, E_free, "k--", alpha=0.5, label="Free electron")
plt.plot(k, (k-2)**2, "k--", alpha=0.5)
plt.plot(k, E_lower, "b", linewidth=2, label="Lower band")
plt.plot(k, E_upper, "r", linewidth=2, label="Upper band")

plt.axvline(1, color="gray", linestyle=":")
plt.axvline(-1, color="gray", linestyle=":")

plt.xlabel("k")
plt.ylabel("E (arbitrary units)")
plt.title("Nearly Free Electron Band Gap at Zone Boundary")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

A.5 de Haas–van Alphen Oscillation Model

```
import numpy as np
import matplotlib.pyplot as plt

B = np.linspace(1.0, 20.0, 2000)
F = 50.0        # dHvA frequency factor
T = 1.0         # temperature in K

# Simplified oscillation with damping
invB = 1.0 / B
chi = np.sin(2.0 * np.pi * F * invB) * np.exp(-2.0 * invB)

fig, axes = plt.subplots(1, 2, figsize=(11, 4))

axes[0].plot(B, chi)
axes[0].set_xlabel("B (T)")
axes[0].set_ylabel("Oscillatory susceptibility")
axes[0].set_title("dHvA oscillations vs B")
axes[0].grid(True, alpha=0.3)

axes[1].plot(invB, chi)
axes[1].set_xlabel("1/B (1/T)")
axes[1].set_ylabel("Oscillatory susceptibility")
axes[1].set_title("Periodic in 1/B")
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```

B Appendix B: Master Formula Reference Table

A comprehensive analytical reference guide summarizing the 26 central formulas across Module 3:

Physical Quantity	Mathematical Formula	Chapter Reference
Drude DC Conductivity	$\sigma_0 = \dfrac{ne^2\tau}{m_e}$	Chapter 3
Electron Mobility	$\mu = \dfrac{\lvert\mathbf{v}_d\rvert}{E} = \dfrac{e\tau}{m_e}$	Chapter 3
AC Optical Conductivity	$\sigma(\omega) = \dfrac{\sigma_0}{1 - i\omega\tau}$	Chapter 3
Electron Plasma Frequency	$\omega_p = \sqrt{\dfrac{ne^2}{\epsilon_0 m_e}}$	Chapter 3
Classical Hall Coefficient	$R_H = \dfrac{E_y}{j_x B} = -\dfrac{1}{ne}$	Chapter 3
Empirical Electron Density	$n = \dfrac{Z_v N_A \rho_m}{A} = \dfrac{Z_v N_c}{a^3}$	Chapter 4
Fermi Wavevector	$k_F = (3\pi^2 n)^{1/3}$	Chapter 4
Fermi Momentum	$p_F = \hbar k_F = \hbar(3\pi^2 n)^{1/3}$	Chapter 4
Fermi Energy	$E_F = \dfrac{\hbar^2 k_F^2}{2m_e} = \dfrac{\hbar^2}{2m_e}(3\pi^2 n)^{2/3}$	Chapter 4
Fermi Velocity	$v_F = \dfrac{\hbar k_F}{m_e} = \sqrt{\dfrac{2E_F}{m_e}}$	Chapter 4
Fermi Temperature	$T_F = \dfrac{E_F}{k_B}$	Chapter 4
3D Density of States	$g(E) = \dfrac{V}{2\pi^2}\left(\dfrac{2m_e}{\hbar^2}\right)^{3/2} E^{1/2}$	Chapter 5
Normalized 3D DOS	$D(E) = \dfrac{3n}{2E_F}\left(\dfrac{E}{E_F}\right)^{1/2}$	Chapter 5
DOS at Fermi Level	$D(E_F) = \dfrac{3n}{2E_F}, \quad g(E_F) = \dfrac{3N}{2E_F}$	Chapter 5
2D Density of States	$g_{2D}(E) = \dfrac{Am_e}{\pi\hbar^2} = \text{constant}$	Chapter 5
Ground-State Energy	$U_0 = \dfrac{3}{5}NE_F, \quad \langle E\rangle_0 = \dfrac{3}{5}E_F$	Chapter 5
Fermi Degeneracy Pressure	$P_0 = \dfrac{2}{5}nE_F = \dfrac{2}{3}\left(\dfrac{U_0}{V}\right)$	Chapter 5
Electronic Bulk Modulus	$B = \dfrac{5}{3}P_0 = \dfrac{2}{3}nE_F$	Chapter 5
Fermi–Dirac Distribution	$f(E) = \dfrac{1}{\exp\left(\dfrac{E-\mu}{k_B T}\right) + 1}$	Chapter 6
Sommerfeld Expansion	$\int_0^\infty \phi(E) f(E) dE = \int_0^\mu \phi dE + \dfrac{\pi^2}{6}(k_B T)^2 \phi'(\mu)$	Chapter 6
Chemical Potential $\mu(T)$	$\mu(T) \approx E_F\left[1 - \dfrac{\pi^2}{12}\left(\dfrac{T}{T_F}\right)^2\right]$	Chapter 6
Electronic Specific Heat	$C_{el} = \dfrac{\pi^2}{3}g(E_F)k_B^2 T = \dfrac{\pi^2}{2}Nk_B\left(\dfrac{T}{T_F}\right) = \gamma T$	Chapter 6
Total Metal Heat Capacity	$C = \gamma T + \beta T^3 \implies \dfrac{C}{T} = \gamma + \beta T^2$	Chapter 6
Pauli Paramagnetism	$\chi_{\text{Pauli}} = \mu_0 \mu_B^2 g(E_F) = \dfrac{3\mu_0 n \mu_B^2}{2E_F}$	Chapter 6

Physical Quantity	Mathematical Formula	Chapter Reference
BTE Current Perturbation	$f_1(\mathbf{k}) = e\tau(\mathbf{v}\cdot\mathbf{E})\left(-\dfrac{\partial f_0}{\partial E}\right)$	Chapter 7
Matthiessen's Rule	$\rho(T) = \rho_0 + \rho_{\text{ph}}(T)$	Chapter 7
Electronic Thermal Cond.	$K_{el} = \dfrac{1}{3}C_{el}v_F^2\tau = \dfrac{\pi^2 n k_B^2 \tau T}{3m_e}$	Chapter 7
Wiedemann–Franz Law	$\dfrac{K}{\sigma T} = \dfrac{\pi^2}{3}\left(\dfrac{k_B}{e}\right)^2 \equiv L \approx 2.44 \times 10^{-8}\ \text{W}\,\Omega\,\text{K}^{-2}$	Chapter 7
Mott Thermopower Formula	$S = -\dfrac{\pi^2 k_B^2 T}{3e}\left[\dfrac{d\ln\sigma}{dE}\right]_{E_F} \approx -\dfrac{\pi^2 k_B}{2e}\left(\dfrac{T}{T_F}\right)$	Chapter 8
First Kelvin Relation	$\Pi = TS$	Chapter 8
Bloch Wavefunction	$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{k}}(\mathbf{r}), \quad u(\mathbf{r}+\mathbf{R}) = u(\mathbf{r})$	Chapter 10
Bragg Zone Boundary	$\mathbf{G}\cdot\left(\mathbf{k}+\dfrac{1}{2}\mathbf{G}\right) = 0$	Chapter 11
NFE Energy Band Gap	$\Delta E = E_g = 2\lvert V_{\mathbf{G}}\rvert$	Chapter 11
Semiclassical Onsager Rel.	$A_n = \dfrac{2\pi e B}{\hbar}\left(n+\dfrac{1}{2}\right)$	Chapter 12
dHvA Oscillation Period	$\Delta\left(\dfrac{1}{B}\right) = \dfrac{2\pi e}{\hbar A_{\text{ext}}} = \dfrac{1}{F}$	Chapter 12
Cyclotron Effective Mass	$m_c = \dfrac{\hbar^2}{2\pi}\left.\dfrac{\partial A}{\partial E}\right\rvert_{E_F}$	Chapter 12

Module 3 Summary: Key Takeaways & Bridge to Band Theory

As we conclude Module 3, it is vital to synthesize the central pedagogical journey that spans classical mechanics, quantum statistics, and modern solid-state band theory:

1. **The Drude Baseline**: Drude's classical model established that electrical and thermal transport in metals are driven by mobile valence electrons. However, classical Maxwell–Boltzmann equipartition led to disastrous failures in specific heat ($C_V = \frac{3}{2}R$) and low-temperature mean free paths ($l \sim 3$ Å).

2. **The Quantum Sommerfeld Revolution**: By treating electrons as indistinguishable fermions obeying the Pauli exclusion principle and Fermi–Dirac statistics, Arnold Sommerfeld demonstrated that only electrons in an active thermal skin of width $\sim k_B T$ around the Fermi energy can participate in thermal absorption ($C_{el} = \gamma T$) and magnetic response ($\chi_{\text{Pauli}} = \mu_0 \mu_B^2 g(E_F)$).

3. **Transport Reconciled**: Semiclassical Boltzmann transport theory rigorously resolved the velocity hierarchy paradox: although only a tiny fraction ($T/T_F \sim 10^{-2}$) of electrons carry net current, they move at the immense quantum Fermi velocity ($v_F \sim 10^6$ m/s). This reconciled Drude's empirical conductivity with quantum mechanics and yielded the exact universal Lorenz number $L = \frac{\pi^2}{3}(k_B/e)^2$.

4. **Thermoelectric Probes**: The coupling of thermal and electrical currents gives rise to the Seebeck, Peltier, and Thomson effects, unified by Kelvin's relations and Mott's formula. The sign of the Seebeck coefficient provides an indispensable diagnostic tool for carrier polarity.

5. **The Need for Band Theory**: The failure of free electron theory to account for positive Hall coefficients ($R_H > 0$), divalent metal conduction, and the fundamental distinction between metals, semiconductors, and insulators proved that the periodic lattice potential cannot be ignored.

6. **Bloch Waves and Energy Gaps**: In a periodic crystal potential $V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$, electron wavepackets take the form of modulated Bloch waves $\psi = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r})$ characterized by conserved crystal momentum $\hbar\mathbf{k}$. Constructive Bragg reflection at Brillouin zone boundaries creates standing waves whose electrostatic potential energy difference produces forbidden energy band gaps $\Delta E = 2|V_{\mathbf{G}}|$.

7. **Fermi Surface Topology**: The geometry of the Fermi surface in reciprocal space governs all electronic properties of metals, and can be measured with extraordinary sub-percent precision through quantum oscillations such as the de Haas–van Alphen effect.

Looking Ahead to Module 4: With the foundations of Bloch's theorem, crystal momentum, and band gap formation firmly established, we are now fully prepared to investigate the detailed electronic structure of solids. In Module 4, we will develop the **Tight-Binding Approximation** (LCAO method), Kronig–Penney model solutions, energy band calculations in semiconductors, and the dynamics of electrons and holes in applied fields.

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